DevOrSysAdminScripts

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Abstract

Some "small" useful scripts for enhancing development or system-administrators tasks

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Code lines: 11036 total lines, 10186 not empty lines, 850 empty lines.

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1 Files tree

```
.
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_dependencies_notes.exec.php
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spelling_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── repository_data.libr.sh
│   ├── shell_checks_complement.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── dependencies_data.json
│   ├── dependencies_notes.txt
│   ├── DevOrSysAdminScripts.html.tpl
│   ├── DevOrSysAdminScripts.tex
```

```
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_names_listing.txt
│   ├── .gitignore
│   ├── licenses_notes.txt
│   ├── phpmd_baseline.xml
│   ├── pylintrc.toml
│   ├── repository_data.txt
│   ├── repository_data.txt.tpl
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── IMAP_transfert.py
├── LiensUtiles.txt
├── miscellaneous
│   ├──
│   1234567890123456789012345678901234567890123456789012345678901234567890123456789
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│   012345
│   ├──
│   1234567890123456789012345678901234567890123456789012345678901234567890123456789
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│   ├──
│   1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   0123456
│   ├──
│   1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   0123456\
│   ├──
│   a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   an_see_that_it_works.txt
│   ├── funny.sh
│   ├── spampty_data.txt
```

```
│    ├── string_escaping.html
│    ├── string_escaping.xhtml
│    └──
│      yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│      that_you_can_see_that_it_works.txt
├── README.html
├── README.md
├── README.md.tpl
├── string_escaping.libr.php
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

6 directories, 80 files

[ 12K Mar 24 20:38]  ./
├── [4.0K Mar 24 20:38]  build_and_checks_dependencies/
│    ├── [4.3K Mar 24 19:32]  build_and_checks.exec.sh.tpl*
│    ├── [6.3K Mar 24 20:34]  build_dependencies_notes.exec.php*
│    ├── [4.2K Mar 23 08:55]  build_md_from_printable_md.exec.sh*
│    ├── [3.2K Mar 23 09:48]  check_shell_scripts_beginnings.libr.sh
│    ├── [5.3K Mar 23 09:47]  check_shell_scripts_indentation.libr.sh
│    ├── [3.6K Mar 23 10:11]  check_URLs.libr.sh
│    ├── [ 30K Mar 24 20:38]  common_build_and_checks.exec.sh*
│    ├── [ 26K Mar 23 18:46]  common_build_and_checks.exec.sh.tpl
│    ├── [ 16K Mar 23 15:17]  common_options.libr.sh
│    ├── [6.1K Mar 22 16:03]  comparisons.libr.sh
│    ├── [ 21K Mar 24 20:37]  create_PDF.exec.sh*
│    ├── [3.7K Mar 22 16:36]  generate_from_template.libr.sh
│    ├── [2.0K Mar  5 21:22]  get_common_text_glob_patterns.libr.sh
│    ├── [3.3K Mar 23 10:18]  grammar_and_spelling_check.libr.sh
│    ├── [4.0K Mar 23 18:40]  licenses_templates/
│    │    ├── [ 14K Mar 23 10:24]  build_licenses_templates.exec.sh*
│    │    ├── [ 728 Feb 28 20:38]  license_file_header_AGPLv3+.tpl
│    │    ├── [ 707 Feb 28 20:38]  license_file_header_GPLv3+.tpl
│    │    └── [ 728 Feb 28 20:38]  license_file_header_LGPLv3+.tpl
│    ├── [6.2K Mar 23 15:40]  lines_counts.libr.sh
│    ├── [ 14K Mar 22 18:05]  lines_filters.libr.sh
│    ├── [2.7K Feb 19 16:16]  lines_maps.libr.sh
│    ├── [2.7K Mar 22 18:33]  overwrite_if_not_equal.libr.sh
│    ├── [2.5K Mar 23 15:29]  python_black_complement.libr.sh
│    ├── [3.0K Mar 23 15:30]  python_isort_complement.libr.sh
│    ├── [2.1K Mar 23 10:14]  repository_data.libr.sh
│    ├── [3.4K Mar 23 15:30]  shell_checks_complement.libr.sh
│    ├── [1.6K Feb  3 17:57]  split_score.exec.php
│    ├── [1.9K Mar 22 19:01]  split_score.libr.php
```

```
│   ├── [ 48K Mar 23 10:16]  strings_functions.libr.sh
│   ├── [4.5K Mar 24 20:26]  too_long_code_lines.libr.sh
│   ├── [4.2K Mar 23 17:15]  update_common_build_and_checks.exec.sh*
│   └── [6.1K Mar 23 17:56]  update_or_check_files_names_listing.exec.sh*
├── [4.5K Mar 24 20:38]  build_and_checks.exec.sh*
├── [4.0K Mar 24 20:38]  build_and_checks_variables/
│   ├── [  15 Jan  5 20:00]  bandit_baseline.json
│   ├── [  25 Jan  5 18:49]  bandit.ini
│   ├── [  30 Feb 28 20:30]  black.toml
│   ├── [4.6K Mar 23 16:44]  dependencies_data.json
│   ├── [3.8K Mar 23 16:44]  dependencies_notes.txt
│   ├── [5.7K Mar 23 17:43]  DevOrSysAdminScripts.html.tpl
│   ├── [ 34K Mar 24 20:35]  DevOrSysAdminScripts.tex
│   ├── [4.2K Mar 23 17:40]  DevOrSysAdminScripts.tex.tpl
│   ├── [5.3K Mar 24 19:44]  files_names_listing.txt
│   ├── [  23 Mar 24 19:33]  .gitignore
│   ├── [1.4K Mar 23 16:11]  licenses_notes.txt
│   ├── [2.2K Mar 23 17:31]  phpmd_baseline.xml
│   ├── [ 678 Feb 24 22:38]  pylintrc.toml
│   ├── [4.5K Mar 22 15:51]  repository_data.txt
│   ├── [ 786 Mar 22 15:39]  repository_data.txt.tpl
│   ├── [1.4K Feb 24 23:25]  ruff.toml
│   ├── [ 166 Mar 24 19:33]  shellcheck.ini
│   └── [3.5K Mar  9 14:46]  shellcheck_notes.txt
├── [ 34K May  6  2024]  COPYING
├── [7.5K Dec 27 21:48]  COPYING.LESSER
├── [441K Mar 24 20:35]  DevOrSysAdminScripts.html
├── [387K Mar 24 20:35]  DevOrSysAdminScripts.pdf
├── [4.0K Mar 23 10:34]  git_helpers/
│   ├── [3.8K Mar 22 21:06]  git_checks.libr.sh
│   ├── [2.1K Mar 22 21:09]  post-commit*
│   ├── [ 15K Mar 23 10:34]  post_commit_renaming_comment.libr.sh
│   └── [1.8K Mar 22 21:11]  pre-commit*
├── [8.3K Feb 28 20:31]  IMAP_transfert.py
├── [ 830 Dec 29 19:10]  LiensUtiles.txt
├── [4.0K Mar 18 17:27]  miscellaneous/
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123\
│   ├── [   0 Mar  4 10:03]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 01234\
│   ├── [   0 Mar  4 10:01]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345
│   ├── [   0 Mar  4 10:37]
```

```
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345\
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456\
│   ├── [   0 Mar  4 09:13]
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │ an_see_that_it_works.txt
│   ├── [1.3K Mar  9 15:15]  funny.sh
│   ├── [  94 Mar  3 19:56]  spampty_data.txt
│   ├── [ 442 Mar 18 16:50]  string_escaping.html
│   ├── [ 654 Mar 18 17:27]  string_escaping.xhtml
│   └── [   0 Mar  4 09:13]
│     yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│     that_you_can_see_that_it_works.txt
├── [8.1K Mar 22 21:29]  README.html
├── [4.3K Mar 22 21:29]  README.md
├── [4.3K Mar 22 21:29]  README.md.tpl
├── [3.2K Mar 22 21:36]  string_escaping.libr.php
├── [ 13K Mar 24 19:32]  unicode.libr.php
├── [1.8K Mar 23 18:45]  upgrade_build_and_checks.exec.sh*
└── [2.5K Mar 23 18:45]  wget_sha512.libr.sh

6 directories, 80 files
```

2 Notes on licenses and dependencies

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:
  basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;

- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;

- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
  bandit;
- The following dependencies are BSD-3-Clause licensed:
  phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD license.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
`https://www.gnu.org/software/bash/`,
`https://tiswww.case.edu/php/chet/bash/bashtop.html`,
`https://savannah.gnu.org/projects/bash/`.

GNU coreutils can be found at the following URLs:
`https://www.gnu.org/software/coreutils/`,
`https://savannah.gnu.org/projects/coreutils/`.

uutils coreutils can be found at the following URLs:
`https://uutils.github.io/coreutils/`,
`https://github.com/uutils/coreutils`.

diff (GNU diffutils) can be found at the following URLs:
`https://www.gnu.org/software/diffutils/`,
`https://savannah.gnu.org/projects/diffutils/`.

find, xargs (GNU findutils) can be found at the following URLs:
`https://www.gnu.org/software/findutils/`,
`https://savannah.gnu.org/projects/findutils/`.

grep (GNU grep) can be found at the following URLs:
`https://www.gnu.org/software/grep/`,
`https://savannah.gnu.org/projects/grep/`.

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/,
https://savannah.gnu.org/projects/sed/.

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/,
https://github.com/koalaman/shellcheck.

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/,
https://savannah.gnu.org/projects/wget/.

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/,
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/,
https://github.com/util-linux/util-linux official?.

pandoc can be found at the following URLs:
https://pandoc.org/,
https://github.com/jgm/pandoc.

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/,
https://tug.org/texlive/svn/.

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree,
https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.

git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.

pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.

black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.

isort can be found at the following URLs:
https://pycqa.github.io/isort/,
https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.

mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,

https://github.com/python/mypy,
https://pypi.org/project/mypy/.

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/,
https://github.com/pypa/pip,
https://pypi.org/project/pip/ (egg and chicken ;)).

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.

phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.

php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,
https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.

python3 can be found at the following URLs:
https://www.python.org/,
https://docs.python.org/3/.

If you see something that I missed regarding my dependencies, please tell me/email me :).

3 Listing of files

3.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
```

```
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download* ]]; then
```

```
    LFBFL_check_download="--check-download"
  fi
  readonly LFBFL_check_download

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir -p "${LFBFL_subdir}/licenses_templates"
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  mkdir -p "${LFBFL_variables_directory}/temp"
  if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    printf "temp/\nupgrade_venvs_ts\n"\
      > "${LFBFL_variables_directory}/.gitignore"
  fi

  # LFBFL_dependencies_raw_content_download_URL
  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  # LFBFL_script_download_URL
  declare -r\
    LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
  declare -r\
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  @sha512_common_build_and_checks.exec.sh@
  wget_sha512 "${LFBFL_file_path}"\
    "${LFBFL_script}"\
    "${LFBFL_correct_sha512}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
  chmod +x "./${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
```

```
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

3.2 build_and_checks_dependencies/build_dependencies_notes.exec.php

```
#!/usr/bin/php
<?php
/*
This file is part of DevOrSysAdminScripts library.

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```

```
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*/

require_once('string_escaping.libr.php');


function build_dependencies_notes(string $s_working_directory) : void {
  $s_subdir1 = $s_working_directory.'/build_and_checks_variables';
  $s_subdir2 = $s_subdir1.'/temp';
  $s_data_file_path = $s_subdir1.'/dependencies_data.json';
  $s_data = file_get_contents($s_data_file_path);
  $arr_data = json_decode($s_data, true);
  $s_result_content1 = '';
  $s_result_content2 = '';
  $s_result_content3 = '';

  foreach($arr_data as $arr_dependency){
    $s_URL_word = "URL";
    if(count($arr_dependency["URLs"]) > 1){
      $s_URL_word .= "s";
    }

    $s_result_content3 .= "<p>\n";

    $s_result_content1 .= (
      $arr_dependency["dependency_name"]
      ." can be found at the following "
      .$s_URL_word
      ":\n"
    );
    $s_result_content2 .= (
      $arr_dependency["dependency_name"]
      ." can be found at the following "
      .$s_URL_word
      ":\\newline\n"
    );
    $s_result_content3 .= (
      $arr_dependency["dependency_name"]
      ." can be found at the following "
```

```
        .$s_URL_word
        ."<br>\n"
      );

      for(
        $i = 0, $i_max = count($arr_dependency["URLs"]) - 1;
        $i <= $i_max;
        ++$i
      ){
        $mixed_URL = $arr_dependency["URLs"][$i];
        $s_comment = '';
        $s_URL = '';
        if(is_array($mixed_URL)){
          $s_URL = $mixed_URL["URL"];
          $s_comment = ' '.$mixed_URL["comment"];
        }
        else{
          $s_URL = $mixed_URL;
        }

        $s_result_content1 .= $s_URL.$s_comment."\n";

        if(strlen($s_URL) > 60){
          $s_result_content2 .= (
            "{\\small\n"
            .'  \url{'.$s_URL."}\n"
            .'}'
          );
        }
        else{
          $s_result_content2 .= '\url{'.$s_URL.'}';
        }
        if($i === $i_max){
          $s_result_content2 .= $s_comment.".\n";
        }
        else{
          $s_result_content2 .= $s_comment.",\\newline\n";
        }

        $s_result_content3 .= (
          "<a\n"
          .'href="'.string_escaping\HTML\escape_attribute($s_URL)."\"\n"
          .'><span style="font-family:monospace"'."\n"
          .'>'.string_escaping\HTML\escape_text($s_URL)."</span\n"
          .'></a>'
        );
```

```
      if($i === $i_max){
        $s_result_content3 .= $s_comment.".\n";
      }
      else{
        $s_result_content3 .= $s_comment.",<br>\n";
      }
    }

    $s_result_content1 .= "\n";
    $s_result_content2 .= "\n";
    $s_result_content3 .= "</p>\n\n";
  }

  // Add a "footer".
  $s_result_content1 .= (
    "If you see something that I missed regarding my dependencies,\n"
    ."please tell me/email me :).\n"
  );
  $s_result_content2 .= (
    "If you see something that I missed regarding my dependencies,\n"
    ."please tell me/email me :).\n"
  );
  $s_result_content3 .= (
    "<p>\n"
    ."If you see something that I missed regarding my dependencies,\n"
    ."please tell me/email me :).\n"
    ."</p>\n"
  );

  // We overwrite results if needed.
  $s_result_path1 = $s_subdir1.'/dependencies_notes.txt';
  $s_existing_result_content1 = '';
  if(file_exists($s_result_path1)){
    $s_existing_result_content1 = file_get_contents($s_result_path1);
  }
  if($s_result_content1 !== $s_existing_result_content1){
    file_put_contents($s_result_path1, $s_result_content1);
  }
  /*
  Example of generated content of dependencies_notes.txt:
  ---------------------------------------------------------------------------
  more of util-linux can be found at the following URLs:
  https://www.kernel.org/pub/linux/utils/util-linux/
  https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
  https://github.com/util-linux/util-linux official?
  ---------------------------------------------------------------------------
```

```
*/
$s_result_path2 = $s_subdir2.'/dependencies_notes.tex.sub';
$s_existing_result_content2 = '';
if(file_exists($s_result_path2)){
  $s_existing_result_content2 = file_get_contents($s_result_path2);
}
if($s_result_content2 !== $s_existing_result_content2){
  file_put_contents($s_result_path2, $s_result_content2);
}
/*
Example of generated content of dependencies_notes.tex.sub:
---------------------------------------------------------------------------
more of util-linux can be found at the following URLs:\newline
\url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
{\small
  \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
},\newline
\url{https://github.com/util-linux/util-linux} official?.
---------------------------------------------------------------------------
*/
$s_result_path3 = $s_subdir2.'/dependencies_notes.html.sub';
$s_existing_result_content3 = '';
if(file_exists($s_result_path3)){
  $s_existing_result_content3 = file_get_contents($s_result_path3);
}
if($s_result_content3 !== $s_existing_result_content3){
  file_put_contents($s_result_path3, $s_result_content3);
}
/*
Example of generated content of dependencies_notes.html.sub:
<p>
more of util-linux can be found at the following URLs:<br>
<a
href="https://www.kernel.org/pub/linux/utils/util-linux/"
><span style="font-family:monospace"
>https://www.kernel.org/pub/linux/utils/util-linux/</span
></a>,<br>
<a
href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
><span style="font-family:monospace"
>https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
></a>,<br>
<a
href="https://github.com/util-linux/util-linux"
><span style="font-family:monospace"
>https://github.com/util-linux/util-linux</span
```

```
  ></a> official?.
  </p>
  */
}

build_dependencies_notes($argv[1]);
```

3.3 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "build_md_from_printable_md.sh"
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
```

```
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_base_name="README"
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --base-name=* ]]; then
      LFBFL_base_name=${LFBFL_arg#--base-name=}
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "Searching md file: %s.\n" "${LFBFL_base_name}"
      fi
      break
    fi
  done
  readonly LFBFL_base_name

  # Remove line returns here to keep lines short.
  local LFBFL_sed_expression
  LFBFL_sed_expression='s/(\n[ ]*\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*<http[^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*- <http[^\n\\]*)\\\n/\1/Mg'
  readonly LFBFL_sed_expression
  # Since Markdown can contain code, it is limited to Markdown URLs.
  # Hence it is way more complicated than 's/\\\n//Mg' expression
  # in other files.
  # Another downside is that more than one pass is needed.

  if [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
    sed -Ez -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
      "${LFBFL_base_name}.md.tpl"\
      > "${LFBFL_base_name}.md.temp"
    overwrite_if_not_equal "${LFBFL_base_name}.md"\
      "${LFBFL_base_name}.md.temp"
  else
    printf "No file %s.md.tpl\n" "${LFBFL_base_name}"
    # Unfortunately, we cannot return early, since we want to propose
```

```
    # HTML file generation with pandoc from .md file,
    # if the file .md.tpl doesn't exist.
  fi

  if [[ ! -f "${LFBFL_base_name}.md" ]]; then
    printf "No file %s.md\n" "${LFBFL_base_name}"
    return 1
  fi

  pandoc --from markdown --to html --standalone\
    --shift-heading-level-by=-1 "${LFBFL_base_name}.md"\
    -o "${LFBFL_base_name}.html.temp"
  # Correcting pandoc HTML "by hand".
  # https://github.com/jgm/pandoc/issues/10957
  # With the most open-minded answer found on Internet ;) XD.
  # This second sed expression is what I need,
  # but it may be unsuitable for you.
  # Other problems:
  # https://github.com/jgm/pandoc/issues/11484
  local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
  LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
  readonly LFBFL_prewrap
  sed -i -E -e 's~(  <meta .*) />~\1>~g'\
            -e 's~<html .*>~<html lang="en">~'\
            -e '/    code\{white-space: pre-wrap;\}/d'\
    "${LFBFL_base_name}.html.temp"
  sed -i -Ez -e 's~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
             -e "${LFBFL_prewrap}"\
    "${LFBFL_base_name}.html.temp"
  overwrite_if_not_equal "${LFBFL_base_name}.html"\
    "${LFBFL_base_name}.html.temp"

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
  #   --verbose ${LFBFL_base_name}.html"
}

build_md_from_printable_md "$@"
```

3.4 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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#
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# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
  # $1=file_path
  #
  # declare -r LFBFL_file_path=$(basename "$1")
  # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
  #   return 0
  # fi
  # diff <(head -n 1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  # diff <(head -n 1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  declare -r LFBFL_head=$(head -n 1 "$1")
  [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
    || printf "%s:File has wrong shell script beginning.\n" "$1"
```

```
}

check_shell_scripts_beginnings(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  local LFBFL_head
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_beginnings: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script beginning.\n"\
          "${LFBFL_file_path}"
    LFBFL_head=$(head -n 1 "${LFBFL_file_path}")
    [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
      || printf "%s:File has wrong shell script beginning.\n"\
          "${LFBFL_file_path}"
  done
}
```

3.5 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
```

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_one_shell_script_indentation(){
  # $1=file_path
  declare -i LFBFL_i_file_with_error=0
  local LFBFL_some_line
  local LFBFL_previous_line="workeduntilsomeasshole"
  # created a command named "workeduntilsomeasshole"
  # and called it without neither indentation, nor parameters
  # in a .sh script XD.
  # Manual justification of the lines L-4 and L-3 was a serendipity.
  # I just tried a joke and was lucky ;) :).
  local LFBFL_line_end
  local LFBFL_previous_line_end
  declare -i LFBFL_i_line_length
  declare -i LFBFL_i_line_end_length
  declare -i LFBFL_i_line_start_length
  local LFBFL_spacing
  declare -i LFBFL_i_offset
  local LFBFL_substring1
```

```
  local LFBFL_substring2
  declare -r LFBFL_pattern='^ *-e '
  # This grep would have been enough without some sed commands.
  declare -r LFBFL_s_lines=$(
    grep -EHn -B 1 '^(  )* ([^ ]|$)' "$1"
  )
  if [[ -z "${LFBFL_s_lines}" ]]; then
    return
  fi

  declare -a LFBFL_arr_lines
  mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
  readonly LFBFL_arr_lines

  for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
    if [[ "${LFBFL_some_line}" == "--" ]]; then
      # When switching between code fragments, we reinit.
      LFBFL_previous_line="workeduntilsomeasshole"
      continue
    fi
    # Note how it's funny that this loop always work because we check
    # shell scripts beginning in another script and thus we always
    # have a line before.
    if [[ "${LFBFL_previous_line}" == "workeduntilsomeasshole" ]];
    then
      LFBFL_previous_line="${LFBFL_some_line}"
      continue
    fi
    # Remove the name of the file and then the line number
    LFBFL_line_end=${LFBFL_some_line#*:}
    LFBFL_line_end=${LFBFL_line_end#*:}
    # But for the previous line '-' or ':' is used instead of ':'
    # and filenames with '-' are much more frequent than with ':'...
    LFBFL_i_line_length=${#LFBFL_some_line}
    LFBFL_i_line_end_length=${#LFBFL_line_end}
    LFBFL_i_line_start_length=$((
      LFBFL_i_line_length - LFBFL_i_line_end_length
    ))
    LFBFL_i_offset=$((LFBFL_i_line_start_length - 2))
    # Checking if increasing number of digits in the line numbers
    # (99 -> 100).
    LFBFL_substring1=${LFBFL_previous_line:${LFBFL_i_offset}:1}
    # Nice puzzle below :) Look for comments above :)
    case ${LFBFL_substring1} in
      -) ((LFBFL_i_offset += 1));;
      :) ((LFBFL_i_offset += 1));;
```

```
      [0-9]) ((LFBFL_i_offset += 2));;
      *) printf ":-(\n";;
    esac
    LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_i_offset}}
    if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
      # Line starts with spacing and -e
      # Keep max spaces
      LFBFL_spacing=${LFBFL_line_end%%-*}
      LFBFL_i_offset=${#LFBFL_spacing}
      LFBFL_substring1=${LFBFL_line_end:${LFBFL_i_offset}:3}
      LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_i_offset}:3}
      if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
        printf "%s\n" "${LFBFL_some_line}"
        LFBFL_i_file_with_error=1
      fi
    else
      printf "%s\n" "${LFBFL_some_line}"
      LFBFL_i_file_with_error=1
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done

  if [[ LFBFL_i_file_with_error -eq 1 ]]; then
    printf "%s:File has some lines with odd number of spaces.\n" "$1"
  fi
}

check_shell_scripts_indentation(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
```

```
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_indentation: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script indentation.\n"\
          "${LFBFL_file_path}"
    check_one_shell_script_indentation "${LFBFL_file_path}"
  done
}
```

3.6 build_and_checks_dependencies/check_URLs.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
```

```
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS

  declare -Ar LFBFL_substitutions=(
    ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
  )

  local LFBFL_pattern
  local LFBFL_base_file_name
  local LFBFL_substitution
  local LFBFL_substitution2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "check_URLs: Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    fi
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | xargs grep -H 'http:'\
      | grep -v "| xargs grep -H 'htt"\
      | grep -vP "['\"]http(:[^'\"]*)['\"].*['\"]https\\1['\"]"\
      | grep -v 'http://www.w3.org/1999/xhtml'\
      | grep -v 'http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd'
```

```
    # Second grep after xargs grep just above will remove false positives
    # from substitutions that fit on one line.

    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_main_html_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]] && printf "check_URLs: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
      [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
        || continue
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "check_URLs: Handling the file: %s.\n" "${LFBFL_file_path}"
      fi
      for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
        LFBFL_substitution2=${LFBFL_substitutions[${LFBFL_substitution}]}
        if grep -q "${LFBFL_substitution}" "${LFBFL_file_path}"; then
          sed -i "s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
            "${LFBFL_file_path}"
        fi
      done
    done
  done
}
```

3.7 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
  declare -r LFBFL_dependencies_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_verbose
  readonly LFBFL_i_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download* ]]; then
    LFBFL_check_download="--check-download"
  fi
  readonly LFBFL_check_download
```

```
60
61    declare -ar LFBFL_some_common_options=(
62      "${LFBFL_verbose}"
63      # "$--work-directory={LFBFL_work_directory}" doesn't work
64      "${LFBFL_work_directory_option}"
65    )
66
67    source ./wget_sha512.libr.sh
68
69    local LFBFL_subdir="build_and_checks_dependencies"
70    declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
71    declare -r LFBFL_subdir3="build_and_checks_variables"
72    local LFBFL_file_name
73    local LFBFL_script_download_URL
74    local LFBFL_file_path
75    local LFBFL_correct_sha512
76
77    wrapped_wget_sha512(){
78      wget_sha512 "${LFBFL_file_path}"\
79        "${LFBFL_script_download_URL}"\
80        "${LFBFL_correct_sha512}"\
81        "${LFBFL_verbose}"\
82        "${LFBFL_check_download}"
83    }
84
85    LFBFL_file_name="build_md_from_printable_md.exec.sh"
86    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
87    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
88    LFBFL_correct_sha512='565323e37ba608d97d46b20ec976f087ff52315c3ca24'
89    LFBFL_correct_sha512+='5d1d15c587f349f70443a1ac33e6e72008f22cfca5fa'
90    LFBFL_correct_sha512+='ff2a87d5113c8e9ed2a576b095c35c8995b0447'
91    wrapped_wget_sha512
92    chmod +x "./${LFBFL_file_path}"
93
94    LFBFL_file_name="build_dependencies_notes.exec.php"
95    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
96    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
97    LFBFL_correct_sha512='2e8a54f8627a6f5a4e966043b91fd116209452ecf7aa6'
98    LFBFL_correct_sha512+='c09e144881fb2bf063f455c3563813aebd57b33a1ee7'
99    LFBFL_correct_sha512+='b85e9157a5bf6293831e391f607e8d887453896'
100   wrapped_wget_sha512
101   chmod +x "./${LFBFL_file_path}"
102
103   LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
104   LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
105   LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
LFBFL_correct_sha512='e5d12684ceb7e143e381743a9eb18e8ecb77bf0a7b8c4'
LFBFL_correct_sha512+='820a511495ef0b6e9fb16db4bec64d7548df2513f6f1'
LFBFL_correct_sha512+='3e13d1494a174a28a23914833e0cceaff6d9e10'
wrapped_wget_sha512

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='261b9cd75cde6f837e7b958d565db5ca7318852cb30e7'
LFBFL_correct_sha512+='47ed6a63bbf0bed9673c2cf0f6cae277b371562cf896'
LFBFL_correct_sha512+='baeb52cc1a509c1464e3901206d96b750bfaf94'
wrapped_wget_sha512

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='2027e3f9cdb847cc71b4b69149486779df666a1b18c30'
LFBFL_correct_sha512+='ab161b7c96ee2c48fed4ef78edb489d566d568fb7b1c'
LFBFL_correct_sha512+='8390a680a5212f5a04c9bd6dd05744374bbc5b3'
wrapped_wget_sha512

LFBFL_file_name="common_options.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='58f125f10f9c10cd38f04cdeec997bb7dc376ba6f0c09'
LFBFL_correct_sha512+='dfc1d2c0927855d709fbf002c027e67370cc0f8864b7'
LFBFL_correct_sha512+='8b23ab7685fc192413cc657df46475e28b989e2'
wrapped_wget_sha512

LFBFL_file_name="comparisons.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='eb2bfabf11d219a4f7a105c95ea5fc2b1f6ddd668fad7'
LFBFL_correct_sha512+='344963c62d574eb3ecb93b35aad8b5ee4c2558792478'
LFBFL_correct_sha512+='f7ef9da058fba1cfe2195b61c87e1ed8686f200'
wrapped_wget_sha512

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='1a9fcbf1e06d33bc92d065e0c0697de3ca332489b7b55'
LFBFL_correct_sha512+='b01b3758755298e1d718f7029b54c409ee1d1eba7be3'
LFBFL_correct_sha512+='b57d06fec08f25ce207fbb36793624f347d1cac'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"
```

```
LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='6369d2f6f28986d56398b2a9533fddbd14251fd8c8853'
LFBFL_correct_sha512+='203f3d8561065b2109abc8c328f0045b4dfb47895690'
LFBFL_correct_sha512+='a275d560ac684aed15e41307c1531c2e88a3643'
wrapped_wget_sha512

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='fc041abf937df4c7cb1241c73e776e8bd07a5525bc9a9'
LFBFL_correct_sha512+='29c83d597ddd6aa88a6e61a99d22ca88c11561f1cdbc'
LFBFL_correct_sha512+='0fbf5df4794a2384f33e7e7dd5b65a53ab46f4a'
wrapped_wget_sha512

LFBFL_file_name="grammar_and_spelling_check.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='48f0edc5841f59f92b8126240df13e651a887bb399640'
LFBFL_correct_sha512+='589fb4bdc56dfd9006e6908ad952f1ce037e6dd22293'
LFBFL_correct_sha512+='5667301e528a767d0ef00ad43d733990b1ae89f'
wrapped_wget_sha512

# /licenses_templates/ ---------------------------------------------
local LFBFL_dependencies_URL2
LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
readonly LFBFL_dependencies_URL2
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
LFBFL_correct_sha512='b521beeff9a3e45d7e188d0e1b7352689ac91a86b7355'
LFBFL_correct_sha512+='a802aabc56d88b57cd357738e8da89e7227aef52acf5'
LFBFL_correct_sha512+='50db4b2e67f189e8c64b9d1a2cc64eb1a7b8084'
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
LFBFL_correct_sha512='9f9aa56d9b90d5003a12c5ec3370ac31aab74fc4a2524'
LFBFL_correct_sha512+='2c5fbcb0197b0abbb8242f62dafc76eaf60faa5da173'
LFBFL_correct_sha512+='05924b31b64b037083532fced9f98445000e2d0'
wrapped_wget_sha512

LFBFL_file_name="license_file_header_GPLv3+.tpl"
```

```
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
LFBFL_correct_sha512='2bdef9a729a5a9ca5ed87ef8a518877c5ee6b31b45336'
LFBFL_correct_sha512+='08660316a4e595fb31465a4847f4d53a2f7264118bd4'
LFBFL_correct_sha512+='eab7c1d1892e00bc1861455ed02898f194db067'
wrapped_wget_sha512

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
LFBFL_correct_sha512='265fd0d086b48ab798f35a072335a4976e4eda8c256fe'
LFBFL_correct_sha512+='bba6c77c98ca86cdd844c6ac150c3445c80debbb3fc2'
LFBFL_correct_sha512+='574b5fceeeaf79ec734a759bdef5fe798067c62'
wrapped_wget_sha512
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='ec36e818b0aab4e386e4b01e643b07ad6bb1ef2ec5ccc'
LFBFL_correct_sha512+='4cfa697bac206f9a65d3cd710994a18db7397597a403'
LFBFL_correct_sha512+='463ec1353bc951ec4d79f54ccee61261dd279d8'
wrapped_wget_sha512

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='86a32b551709cf0743b69dabd1bbe091bda33a46b0207'
LFBFL_correct_sha512+='8aa2aa7c3539d86f8c70dd7fc526c06ed8fcc2511390'
LFBFL_correct_sha512+='dd9398bf27599405ba5112ffb679591eabb499c'
wrapped_wget_sha512

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='25c3e5082951260295b5f0e87ba1f2c978efdea1e8926'
LFBFL_correct_sha512+='072ce4d2062dc2a47bb92b9098f081e553eeae5d9154'
LFBFL_correct_sha512+='fdc50c9d88ae5365b776a23eb6e026b884a8374'
wrapped_wget_sha512

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='2e2e9c804ec8e1879fc96df595f30994a0685934483ce'
LFBFL_correct_sha512+='c9deed4c2fb98f79d4a393e13b6511b3b740ce9f073a'
LFBFL_correct_sha512+='e7ed124f08a60b35c896d60dd710223d2592d16'
```

```
wrapped_wget_sha512

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='abe0e5701ad6a56937b1a1e3687366e87e8cb2578a179'
LFBFL_correct_sha512+='8e15ce81e0aaafc208c1b325058287c331092765ef73'
LFBFL_correct_sha512+='2c584cadf33e3acbcc195f25b0b632a0ce229d3'
wrapped_wget_sha512

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='70542f00cb59ed52d6576e6c066c5dc701ac90274ab68'
LFBFL_correct_sha512+='475086182b13b8cdf75f6b0263ac813e0010dd6cdb86'
LFBFL_correct_sha512+='9239899c2a6161f912bf1f789ad30965b8c1fe5'
wrapped_wget_sha512

LFBFL_file_name="repository_data.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='e3e8b487c45bf3960c5eb739fd1d29229ca358bfecb23'
LFBFL_correct_sha512+='e585fa7abf65c9cdaa263e799e96a83a22c660323e31'
LFBFL_correct_sha512+='bf2cdbc2f857b43957ef9405d2b8bed95dac7c0'
wrapped_wget_sha512

LFBFL_file_name="shell_checks_complement.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='e70c9b931d1adb41be165c2d84e17dd9dbb23a3d6d48d'
LFBFL_correct_sha512+='1b609a9187e96a2faa831c78335e440edef26792b307'
LFBFL_correct_sha512+='346fb053c5c0dd1fac493e1db3657950fd6ae08'
wrapped_wget_sha512

LFBFL_file_name="split_score.exec.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='2f7ff107a43716109b1acbd7d89870d05577af00a981a'
LFBFL_correct_sha512+='05f162a054c96387fb99544f8225c333b1212f74326c'
LFBFL_correct_sha512+='f9ec6196e796f6257dfbba9347ac8c57b4f09db'
wrapped_wget_sha512

LFBFL_file_name="split_score.libr.php"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
LFBFL_correct_sha512='ca60f3621f0808f0ba28bdfb860f6f8d3d872fa26c93b'
```

```
290     LFBFL_correct_sha512+='94a9ce0b1b90f5ecb4fc4c4e1cea30368d99e7e7b3f7'
291     LFBFL_correct_sha512+='7ec5f80052754e987a51bceffc5bb8ee552c480'
292     wrapped_wget_sha512
293
294     LFBFL_file_name="strings_functions.libr.sh"
295     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
296     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
297     LFBFL_correct_sha512='1c5cb59a9ad306ca8cd49b398146dc6621de0e69cb510'
298     LFBFL_correct_sha512+='b9f41d4412d5b853f26eae7128410419be7dfb0d36cf'
299     LFBFL_correct_sha512+='1a6272f70f270e551406866b1e1fd209bdcce44'
300     wrapped_wget_sha512
301
302     LFBFL_file_name="too_long_code_lines.libr.sh"
303     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
304     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
305     LFBFL_correct_sha512='3835c4b66734f5e27313d9de7e7e94ca7d04e15d7e2e3'
306     LFBFL_correct_sha512+='981a83b9c96b8ad023cae15035ea268de4a3262451be'
307     LFBFL_correct_sha512+='90a660c425e1e0a5325f3e8a5fcfb341f172ba4'
308     wrapped_wget_sha512
309
310     LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
311     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
312     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
313     LFBFL_correct_sha512='9d9100a8527ea9717aff99f483c81b7921fd079ace4f9'
314     LFBFL_correct_sha512+='ead5cecd3c89da90f43959297ff78184eadf55bff79e'
315     LFBFL_correct_sha512+='b716060befe5eff9f00cc3b2f4dc69522e8bf8f'
316     wrapped_wget_sha512
317     chmod +x "./${LFBFL_file_path}"
318
319     # shellcheck source=check_shell_scripts_beginnings.libr.sh
320     source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
321     # shellcheck source=check_shell_scripts_indentation.libr.sh
322     source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
323     # shellcheck source=check_URLs.libr.sh
324     source "./${LFBFL_subdir}/check_URLs.libr.sh"
325     # shellcheck source=common_options.libr.sh
326     source "./${LFBFL_subdir}/common_options.libr.sh"
327     # shellcheck source=comparisons.libr.sh
328     source "./${LFBFL_subdir}/comparisons.libr.sh"
329     # shellcheck source=generate_from_template.libr.sh
330     source "./${LFBFL_subdir}/generate_from_template.libr.sh"
331     # shellcheck source=get_common_text_glob_patterns.libr.sh
332     source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
333     # shellcheck source=grammar_and_spelling_check.libr.sh
334     source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
335     # shellcheck source=lines_counts.libr.sh
```

```
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=python_black_complement.libr.sh
source "./${LFBFL_subdir}/python_black_complement.libr.sh"
# shellcheck source=python_isort_complement.libr.sh
source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
# shellcheck source=shell_checks_complement.libr.sh
source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

enhanced_set_bash_option extglob
# shellcheck source=repository_data.libr.sh
source "./${LFBFL_subdir}/repository_data.libr.sh"
# shellcheck disable=SC2154,SC2309
[[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
  && enhanced_unset_bash_option extglob

declare -a LFBFL_return_traps_stack
local LFBFL_previous_return_trap
init_return_trap

enhanced_set_shell_option pipefail --trap-unset

declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
keep_options LFBFL_some_common_options2 --verbose

printf "Building license headers\n"
"./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
  "${LFBFL_some_common_options[@]}"

printf "Building README.md\n"
"./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
  --work-directory="${LFBFL_work_directory}"\
  --base-name=README\
  "${LFBFL_verbose}"

printf "Building other MarkDown files\n"
local LFBFL_some_directory
```

```
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
    | relevant_find
  )
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
      printf "Found template %s.\n" "${LFBFL_file_path}"
      LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
      LFBFL_file_name=$(basename "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.md.tpl}
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        --work-directory="${LFBFL_some_directory}"\
        --base-name="${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  printf "Building dependencies_notes\n"
  php "./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
    "${LFBFL_work_directory}"

  declare -i LFBFL_i_directory_changed
  pushd_to_work_directory
  LFBFL_i_directory_changed=$?
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_name="${LFBFL_subdir3}/repository_data.txt"

  declare -i LFBFL_i_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix=LFBFL_i_

  declare -i LFBFL_i_upgrade_venvs=0
  local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
  readonly LFBFL_upgrade_venvs_ts_file
  declare -i LFBFL_i_upgrade_venvs_ts
  declare -i LFBFL_i_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
```

```
    "${LFBFL_some_common_options2[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_i_upgrade_venvs_ts=$(
      stat --format %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
    ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      LFBFL_i_upgrade_venvs=1
    fi
  else
    LFBFL_i_upgrade_venvs=1
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    local LFBFL_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix=LFBFL_
    if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
        "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options2[@]}"

  printf -- "---Python---\n"

```

```
maybe_deactivate(){
  if command -v deactivate
  then
    deactivate
  fi
}

source_without_return_trap(){
  declare -i LFBFL_i_no_return_traps=1
  # shellcheck disable=SC1090,SC1091
  source "$1"
}

printf "Running isort\n"
local LFBFL_isort_venv=""
grep_variable "${LFBFL_data_file_name}" isort_venv\
  --result-variable-prefix=LFBFL_\
  --replace-line-returns-by=""
if [[ -n "${LFBFL_isort_venv}" ]]; then
  maybe_deactivate
  source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
fi
if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
  pip install --upgrade isort
fi
isort --profile=black .
if [[ -n "${LFBFL_isort_venv}" ]]; then
  deactivate
fi
printf "Running python_isort_complement\n"
python_isort_complement "${LFBFL_some_common_options2[@]}"

printf "Running black\n"
# First, we update the configuration file with max_line_length.
local LFBFL_update_max_length="s/^line-length = [0-9]*$"
LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
sed -E -e "${LFBFL_update_max_length}"\
  "${LFBFL_subdir3}/black.toml"\
  > "${LFBFL_subdir3}/black.toml.temp"
overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
  "${LFBFL_subdir3}/black.toml.temp"

local LFBFL_black_venv=""
grep_variable "${LFBFL_data_file_name}" black_venv\
  --result-variable-prefix=LFBFL_\
  --replace-line-returns-by=""
```

```
  if [[ -n "${LFBFL_black_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config="${LFBFL_subdir3}/black.toml" .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
        mypy "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
```

```
fi
if [[ -n "${LFBFL_mypy_venv}" ]]; then
  deactivate
fi

printf "Running bandit\n"
LFBFL_bandit_venv=""
grep_variable "${LFBFL_data_file_name}" bandit_venv\
  --result-variable-prefix=LFBFL_\
  --replace-line-returns-by=""
if [[ -n "${LFBFL_bandit_venv}" ]]; then
  maybe_deactivate
  source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
fi
if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
  pip install --upgrade bandit
fi
bandit --ini="${LFBFL_subdir3}/bandit.ini"\
  --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
  --recursive .
# Saving new baseline in temp if necessary.
bandit --ini="${LFBFL_subdir3}/bandit.ini"\
  --format=json\
  --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
  --recursive .
if [[ -n "${LFBFL_bandit_venv}" ]]; then
  deactivate
fi

printf "Running pylint\n"
LFBFL_pylint_venv=""
grep_variable "${LFBFL_data_file_name}" pylint_venv\
  --result-variable-prefix=LFBFL_\
  --replace-line-returns-by=""
if [[ -n "${LFBFL_pylint_venv}" ]]; then
  maybe_deactivate
  source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
fi
if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
  pip install --upgrade pylint
fi
pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
if [[ -n "${LFBFL_pylint_venv}" ]]; then
  deactivate
fi

```

```
  printf "Running ruff\n"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_name}" ruff_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config="${LFBFL_subdir3}/ruff.toml"
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  printf -- "---Python end---\n"

  printf -- "---PHP---\n"
  printf "Running PHPMD\n"
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
  local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
  LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_temp_phpmd_baseline

  local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
  LFBFL_phpmd_rulesets+="design,naming,unusedcode"
  readonly LFBFL_phpmd_rulesets

  phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  # Saving new baseline in temp if necessary.
  rm "${LFBFL_temp_phpmd_baseline}"
  phpmd --color --generate-baseline\
    --baseline-file="${LFBFL_temp_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
  sed -i -E -e 's~(file=".*)/>~\1\n  />~g' "${LFBFL_temp_phpmd_baseline}"
  diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"
  printf -- "---PHP end---\n"

  printf -- "---JS---\n"
```

```
LFBFL_npm_lint_directories=""
grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
  --result-variable-prefix=LFBFL_
if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
  printf "Running ESLint\n"
  local LFBFL_JS_directory
  declare -a LFBFL_arr_paths
  mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
  for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
    (cd "${LFBFL_JS_directory}" && npm run lint)
  done
fi
printf -- "---JS end---\n"

printf "Analyzing too long lines\n"
# shellcheck disable=SC2248
too_long_code_lines "${LFBFL_some_common_options2[@]}" \
  --max-line-length=${LFBFL_i_max_line_length}

printf "Analyzing shell scripts beginnings\n"
check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
  | relevant_grep

printf "Analyzing shell scripts indentation\n"
check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
  | relevant_grep

printf "Analyzing URLs\n"
check_URLs "${LFBFL_some_common_options2[@]}" \
  | relevant_grep

printf "Analyzing strange characters: hover over in doubt\n"
# shellcheck disable=SC1111
LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»"
grep --binary-files=without-match --color=always --exclude-dir .git\
  --line-number --perl-regexp --recursive --\
  "[^${LFBFL_usual_characters}]" .

printf "Grammar and spelling check\n"
grammar_and_spelling_check "${LFBFL_data_file_name}"\
  "${LFBFL_some_common_options2[@]}"

[[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory

printf "Checking listed files\n"
"./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
```

```
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
    "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix=LFBFL_

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_i_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_i_vnu_jar_ts=$(stat --format %Y "${LFBFL_vnu_jar_path}")
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
    ((
      LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
    ))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    # vnu has false positives for xhtml
    local LFBFL_use_vnu_for_xhtml=""
    grep_variable "${LFBFL_data_file_name}" use_vnu_for_xhtml\
```

```
      --result-variable-prefix=LFBFL_
    local LFBFL_find_regexp
    if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
      LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
    else
      LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex "${LFBFL_find_regexp}"\
      | relevant_find
    )
    if [[ -n "${LFBFL_files_for_Nu}" ]]; then
      declare -a LFBFL_files_for_Nu2
      mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
      readonly LFBFL_files_for_Nu2
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose "${LFBFL_files_for_Nu2[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

3.8 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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#
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# that it will be useful,
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# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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```

```
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
  declare -r LFBFL_dependencies_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_verbose
  readonly LFBFL_i_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download* ]]; then
    LFBFL_check_download="--check-download"
  fi
  readonly LFBFL_check_download

  declare -ar LFBFL_some_common_options=(
    "${LFBFL_verbose}"
```

```
63      # "$--work-directory={LFBFL_work_directory}" doesn't work
64      "${LFBFL_work_directory_option}"
65    )
66
67    source ./wget_sha512.libr.sh
68
69    local LFBFL_subdir="build_and_checks_dependencies"
70    declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
71    declare -r LFBFL_subdir3="build_and_checks_variables"
72    local LFBFL_file_name
73    local LFBFL_script_download_URL
74    local LFBFL_file_path
75    local LFBFL_correct_sha512
76
77    wrapped_wget_sha512(){
78      wget_sha512 "${LFBFL_file_path}"\
79        "${LFBFL_script_download_URL}"\
80        "${LFBFL_correct_sha512}"\
81        "${LFBFL_verbose}"\
82        "${LFBFL_check_download}"
83    }
84
85    LFBFL_file_name="build_md_from_printable_md.exec.sh"
86    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
87    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
88    @sha512_build_md_from_printable_md.exec.sh@
89    wrapped_wget_sha512
90    chmod +x "./${LFBFL_file_path}"
91
92    LFBFL_file_name="build_dependencies_notes.exec.php"
93    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
94    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
95    @sha512_build_dependencies_notes.exec.php@
96    wrapped_wget_sha512
97    chmod +x "./${LFBFL_file_path}"
98
99    LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
100   LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
101   LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
102   @sha512_check_shell_scripts_beginnings.libr.sh@
103   wrapped_wget_sha512
104
105   LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
106   LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
107   LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
108   @sha512_check_shell_scripts_indentation.libr.sh@
```

```
wrapped_wget_sha512

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_URLs.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="common_options.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_common_options.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="comparisons.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_comparisons.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_create_PDF.exec.sh@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_generate_from_template.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_get_common_text_glob_patterns.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="grammar_and_spelling_check.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_grammar_and_spelling_check.libr.sh@
wrapped_wget_sha512

# /licenses_templates/ ----------------------------------------------
```

```
    local LFBFL_dependencies_URL2
    LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
    readonly LFBFL_dependencies_URL2
    LFBFL_file_name="build_licenses_templates.exec.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
    @sha512_build_licenses_templates.exec.sh@
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="license_file_header_AGPLv3+.tpl"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
    @sha512_license_file_header_AGPLv3+.tpl@
    wrapped_wget_sha512

    LFBFL_file_name="license_file_header_GPLv3+.tpl"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
    @sha512_license_file_header_GPLv3+.tpl@
    wrapped_wget_sha512

    LFBFL_file_name="license_file_header_LGPLv3+.tpl"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
    @sha512_license_file_header_LGPLv3+.tpl@
    wrapped_wget_sha512
    # ------------------------------------------------------------------

    LFBFL_file_name="lines_counts.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_lines_counts.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="lines_filters.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_lines_filters.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="lines_maps.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_lines_maps.libr.sh@
    wrapped_wget_sha512
```

```
201
202     LFBFL_file_name="overwrite_if_not_equal.libr.sh"
203     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
204     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
205     @sha512_overwrite_if_not_equal.libr.sh@
206     wrapped_wget_sha512
207
208     LFBFL_file_name="python_black_complement.libr.sh"
209     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
210     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
211     @sha512_python_black_complement.libr.sh@
212     wrapped_wget_sha512
213
214     LFBFL_file_name="python_isort_complement.libr.sh"
215     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
216     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
217     @sha512_python_isort_complement.libr.sh@
218     wrapped_wget_sha512
219
220     LFBFL_file_name="repository_data.libr.sh"
221     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
222     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
223     @sha512_repository_data.libr.sh@
224     wrapped_wget_sha512
225
226     LFBFL_file_name="shell_checks_complement.libr.sh"
227     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
228     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
229     @sha512_shell_checks_complement.libr.sh@
230     wrapped_wget_sha512
231
232     LFBFL_file_name="split_score.exec.php"
233     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
234     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
235     @sha512_split_score.exec.php@
236     wrapped_wget_sha512
237
238     LFBFL_file_name="split_score.libr.php"
239     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
240     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
241     @sha512_split_score.libr.php@
242     wrapped_wget_sha512
243
244     LFBFL_file_name="strings_functions.libr.sh"
245     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
246     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
@sha512_strings_functions.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_too_long_code_lines.libr.sh@
wrapped_wget_sha512

LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_update_or_check_files_names_listing.exec.sh@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=check_shell_scripts_indentation.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
# shellcheck source=check_URLs.libr.sh
source "./${LFBFL_subdir}/check_URLs.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=grammar_and_spelling_check.libr.sh
source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=python_black_complement.libr.sh
source "./${LFBFL_subdir}/python_black_complement.libr.sh"
# shellcheck source=python_isort_complement.libr.sh
source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
# shellcheck source=shell_checks_complement.libr.sh
source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
```

```
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

enhanced_set_bash_option extglob
# shellcheck source=repository_data.libr.sh
source "./${LFBFL_subdir}/repository_data.libr.sh"
# shellcheck disable=SC2154,SC2309
[[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
  && enhanced_unset_bash_option extglob

declare -a LFBFL_return_traps_stack
local LFBFL_previous_return_trap
init_return_trap

enhanced_set_shell_option pipefail --trap-unset

declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
keep_options LFBFL_some_common_options2 --verbose

printf "Building license headers\n"
"./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
  "${LFBFL_some_common_options[@]}"

printf "Building README.md\n"
"./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
  --work-directory="${LFBFL_work_directory}"\
  --base-name=README\
  "${LFBFL_verbose}"

printf "Building other MarkDown files\n"
local LFBFL_some_directory
declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
local LFBFL_s_files_paths
LFBFL_s_files_paths=$(
  find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
  | relevant_find
)
declare -a LFBFL_arr_files_paths
if [[ -n "${LFBFL_s_files_paths}" ]]; then
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
    printf "Found template %s.\n" "${LFBFL_file_path}"
    LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
```

```
      LFBFL_file_name=$(basename "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.md.tpl}
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        --work-directory="${LFBFL_some_directory}"\
        --base-name="${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  printf "Building dependencies_notes\n"
  php "./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
    "${LFBFL_work_directory}"

  declare -i LFBFL_i_directory_changed
  pushd_to_work_directory
  LFBFL_i_directory_changed=$?
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_name="${LFBFL_subdir3}/repository_data.txt"

  declare -i LFBFL_i_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix=LFBFL_i_

  declare -i LFBFL_i_upgrade_venvs=0
  local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
  readonly LFBFL_upgrade_venvs_ts_file
  declare -i LFBFL_i_upgrade_venvs_ts
  declare -i LFBFL_i_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options2[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_i_upgrade_venvs_ts=$(
      stat --format %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
    ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      LFBFL_i_upgrade_venvs=1
    fi
  else
```

```
    LFBFL_i_upgrade_venvs=1
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    local LFBFL_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix=LFBFL_
    if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
        "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options2[@]}"

  printf -- "---Python---\n"

  maybe_deactivate(){
    if command -v deactivate
    then
      deactivate
    fi
  }

  source_without_return_trap(){
    declare -i LFBFL_i_no_return_traps=1
    # shellcheck disable=SC1090,SC1091
    source "$1"
  }

```

```
  printf "Running isort\n"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort --profile=black .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options2[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    "${LFBFL_subdir3}/black.toml"\
    > "${LFBFL_subdir3}/black.toml.temp"
  overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
    "${LFBFL_subdir3}/black.toml.temp"

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config="${LFBFL_subdir3}/black.toml" .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"
```

```
477
478   printf "Probing if mypy should be runned\n"
479   local LFBFL_mypy_venv=""
480   grep_variable "${LFBFL_data_file_name}" mypy_venv\
481     --result-variable-prefix=LFBFL_\
482     --replace-line-returns-by=""
483   if [[ -n "${LFBFL_mypy_venv}" ]]; then
484     maybe_deactivate
485     source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
486   fi
487   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
488     pip install --upgrade mypy
489   fi
490   local LFBFL_directory_path
491   declare -i LFBFL_i_no_toml=1
492   LFBFL_s_files_paths=$(
493     find . -type f -name "pyproject.toml"\
494     | relevant_find
495   )
496   if [[ -n "${LFBFL_s_files_paths}" ]]; then
497     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
498     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
499       if grep --quiet "Typing :: Typed" "${LFBFL_file_path}"; then
500         printf "Running mypy\n"
501         LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
502         mypy "${LFBFL_directory_path}"
503       fi
504       LFBFL_i_no_toml=0
505     done
506   fi
507   if [[ LFBFL_i_no_toml -eq 1 ]]; then
508     printf "Running mypy\n"
509     mypy .
510   fi
511   if [[ -n "${LFBFL_mypy_venv}" ]]; then
512     deactivate
513   fi
514
515   printf "Running bandit\n"
516   LFBFL_bandit_venv=""
517   grep_variable "${LFBFL_data_file_name}" bandit_venv\
518     --result-variable-prefix=LFBFL_\
519     --replace-line-returns-by=""
520   if [[ -n "${LFBFL_bandit_venv}" ]]; then
521     maybe_deactivate
522     source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
```

```
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini="${LFBFL_subdir3}/bandit.ini"\
    --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
    --recursive .
  # Saving new baseline in temp if necessary.
  bandit --ini="${LFBFL_subdir3}/bandit.ini"\
    --format=json\
    --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
    --recursive .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  printf "Running pylint\n"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_name}" pylint_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  printf "Running ruff\n"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_name}" ruff_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config="${LFBFL_subdir3}/ruff.toml"
```

```
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  printf -- "---Python end---\n"

  printf -- "---PHP---\n"
  printf "Running PHPMD\n"
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
  local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
  LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_temp_phpmd_baseline

  local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
  LFBFL_phpmd_rulesets+="design,naming,unusedcode"
  readonly LFBFL_phpmd_rulesets

  phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  # Saving new baseline in temp if necessary.
  rm "${LFBFL_temp_phpmd_baseline}"
  phpmd --color --generate-baseline\
    --baseline-file="${LFBFL_temp_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
  sed -i -E -e 's~(file=".*)/>~\1\n  />~g' "${LFBFL_temp_phpmd_baseline}"
  diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"
  printf -- "---PHP end---\n"

  printf -- "---JS---\n"
  LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
    --result-variable-prefix=LFBFL_
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    printf "Running ESLint\n"
    local LFBFL_JS_directory
    declare -a LFBFL_arr_paths
    mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"
```

```
615
616   printf "Analyzing too long lines\n"
617   # shellcheck disable=SC2248
618   too_long_code_lines "${LFBFL_some_common_options2[@]}" \
619     --max-line-length=${LFBFL_i_max_line_length}
620
621   printf "Analyzing shell scripts beginnings\n"
622   check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
623     | relevant_grep
624
625   printf "Analyzing shell scripts indentation\n"
626   check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
627     | relevant_grep
628
629   printf "Analyzing URLs\n"
630   check_URLs "${LFBFL_some_common_options2[@]}" \
631     | relevant_grep
632
633   printf "Analyzing strange characters: hover over in doubt\n"
634   # shellcheck disable=SC1111
635   LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»"
636   grep --binary-files=without-match --color=always --exclude-dir .git\
637     --line-number --perl-regexp --recursive --\
638     "[^${LFBFL_usual_characters}]" .
639
640   printf "Grammar and spelling check\n"
641   grammar_and_spelling_check "${LFBFL_data_file_name}"\
642     "${LFBFL_some_common_options2[@]}"
643
644   [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory
645
646   printf "Checking listed files\n"
647   "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
648     "${LFBFL_some_common_options[@]}"
649
650   printf "Creating the PDF file of the listing of the source code\n"
651   "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"
652
653   pushd_to_work_directory --trap-popd
654   can_continue_after_enhanced_pushd || return 1
655
656   if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
657     "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
658   fi
659
660   # ------------------------------------------------------------------
```

```
661     # It is at the end that all the HTML files are generated and that
662     # we can check there are no mistakes.
663     printf "Running Nu W3C HTML CSS and SVG validator\n"
664     local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
665     grep_variable "${LFBFL_data_file_name}"\
666       upgrade_vnu_jar_time_interval_in_seconds\
667       --result-variable-prefix=LFBFL_
668
669     if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
670     then
671       LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
672     fi
673
674     declare -i LFBFL_i_vnu_jar_ts
675     local LFBFL_vnu_jar_path=""
676     grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
677       --result-variable-prefix=LFBFL_\
678       --replace-line-returns-by=""
679     if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
680       LFBFL_i_vnu_jar_ts=$(stat --format %Y "${LFBFL_vnu_jar_path}")
681       LFBFL_i_current_ts=$(date +%s)
682       ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
683       ((
684         LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
685       ))
686       if [[ LFBFL_i_current_ts -gt 0 ]]; then
687         printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
688         printf "Go get the latest version here:\n"
689         printf "https://github.com/validator/validator/releases\n"
690       fi
691       # vnu has false positives for xhtml
692       local LFBFL_use_vnu_for_xhtml=""
693       grep_variable "${LFBFL_data_file_name}" use_vnu_for_xhtml\
694         --result-variable-prefix=LFBFL_
695       local LFBFL_find_regexp
696       if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
697         LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
698       else
699         LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
700       fi
701       declare -r LFBFL_files_for_Nu=$(
702         find . -iregex "${LFBFL_find_regexp}"\
703         | relevant_find
704       )
705       if [[ -n "${LFBFL_files_for_Nu}" ]]; then
706         declare -a LFBFL_files_for_Nu2
```

```
      mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
      readonly LFBFL_files_for_Nu2
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose "${LFBFL_files_for_Nu2[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

3.9 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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get_where_was_i(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
```

```
  if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
    LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
  fi
  if [[ "${FUNCNAME[$1]}" != "" ]]; then
    LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
  fi
  readonly LFBFL_where_was_i
  # Comment out line above in case someone would use distinct
  # offsets in the same function.
}

# RETURN traps handling
# Add the following lines at the start of a function.
#   declare -a LFBFL_return_traps_stack
#   local LFBFL_previous_return_trap
#   init_return_trap

init_return_trap(){
  local LFBFL_previous_return_trap2
  LFBFL_previous_return_trap2=$(trap -p RETURN)
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s previous_return_trap is: %s.\n"\
      "${LFBFL_where_was_i}"\
      "${LFBFL_previous_return_trap2}"
  fi
  if [[ -z "${LFBFL_previous_return_trap2}" ]]; then
    LFBFL_previous_return_trap2="trap - RETURN"
  fi
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  LFBFL_previous_return_trap="${LFBFL_function_depth}:"
  LFBFL_previous_return_trap+="${LFBFL_previous_return_trap2}"
  trap 'execute_return_traps' RETURN
}

execute_return_traps(){
  if [[ "${FUNCNAME[1]}" == "init_return_trap" ]]; then
    return
  fi
  # ----------------------------------------------------------------------
  # See https://lists.gnu.org/archive/html/help-bash/2026-03/msg00052.html
  if [[ "${FUNCNAME[1]}" == "source" ]]; then
    # Not Working
    return
  fi
```

```
  if [[ ${LFBFL_i_no_return_traps:-0} -eq 1 ]]; then
    return
  fi
  # ----------------------------------------------------------------------
  local LFBFL_some_return_trap
  local LFBFL_where_was_i
  get_where_was_i 2
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  while [[ -n "${LFBFL_return_traps_stack}" ]]; do
    LFBFL_some_return_trap="${LFBFL_return_traps_stack[-1]}"
    if [[ "${LFBFL_some_return_trap}" != ${LFBFL_function_depth}:* ]]; then
      # We only treat the top of the stack that should correspond to the
      # current function.
      break
    fi
    LFBFL_some_return_trap=${LFBFL_some_return_trap#*:}
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "%s executing some return trap: %s.\n"\
        "${LFBFL_where_was_i}"\
        "${LFBFL_some_return_trap}"
    fi
    eval "${LFBFL_some_return_trap}"
    unset 'LFBFL_return_traps_stack[-1]'
  done
  if [[ "${LFBFL_previous_return_trap}" != ${LFBFL_function_depth}:* ]];
  then
    # We only set the previous return trap if such a trap was replaced
    # by an init_return_trap.
    return
  fi
  LFBFL_previous_return_trap=${LFBFL_previous_return_trap#*:}
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "%s setting back to previous_return_trap: %s.\n"\
      "${LFBFL_where_was_i}"\
      "${LFBFL_previous_return_trap}"
  fi
  eval "${LFBFL_previous_return_trap}"
}


is_top_dirstack_directory(){
  # This function should not be needed since when testing "${DIRSTACK[0]}"
  # is already a realpath. But I keep it, in case it is needed.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
  # $1=to_directory needs to be "realpathed" already
  if [[ -z "$1" ]]; then
```

```
    return 0
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  [[ "${LFBFL_real_path}" == "$1" ]]
}

enhanced_pushd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory must be sanitized with realpath before call
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_pushd requested: %s\n" "$1"
  declare -gi i_enhanced_pushd_result=0
  if [[ -z "$1" ]]; then
    i_enhanced_pushd_result=110
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  fi
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated1\n"
  is_top_dirstack_directory "$1" && {
    i_enhanced_pushd_result=111
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated2\n"

  pushd "$1" || {
    i_enhanced_pushd_result=$?
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s %s%s no such directory.\n" "${LFBFL_where_was_i}" "$3" "$1"
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd executed\n"
  # shellcheck disable=SC2248
  return ${i_enhanced_pushd_result}
}

enhanced_popd(){
  # Auxiliary function needed below.
```

```
  # See usages below if you want to reuse it.
  # $1=to_directory
  # $2=offset_for_where_was_i
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_popd requested: %s\n" "$1"
  if [[ -z "$1" ]]; then
    return 0
  fi

  declare -i LFBFL_i_popd_result
  popd || {
    LFBFL_i_popd_result=$?
    readonly LFBFL_i_popd_result
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s popd failed.\n" "${LFBFL_where_was_i}"
    # shellcheck disable=SC2248
    return ${LFBFL_i_popd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_popd executed\n"
  return 0
}

can_continue_after_enhanced_pushd(){
  [[ i_enhanced_pushd_result -eq 0
  || i_enhanced_pushd_result -eq 110
  || i_enhanced_pushd_result -eq 111
  ]]
}

# Main usage for the next functions relative to verbose mode:
# Add the two following lines at the start of a function.
#   declare -i LFBFL_i_verbose=0
#   get_verbose_option "$@"
# /!\ You really need all of the two lines. /!\
# Otherwise, the call to get_verbose_option may impact a function above;
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
```

```
    return
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

get_verbose_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --verbose
  # This test is faster than:
  #   for LFBFL_arg in "$@"; do
  #     if [[ "${LFBFL_arg}" == "--verbose" ]]; then
  # But it is not bug free.
  if [[ "$*" == *--verbose* ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s\n" "${LFBFL_where_was_i}" "$*"
    LFBFL_i_verbose=1
  fi
  # shellcheck disable=SC2034
  readonly LFBFL_i_verbose
}


get_some_option(){
  # $1=variable_name
  # $2=some_option_long_name
  # $3=option_default_value
  # $4=read_only 0 or 1 to make the variable readonly
  # This command is to be called in another one with same arguments on top
  # of the three previous.
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == "$2"=* ]]; then
      printf -v "$1" "%s" "${LFBFL_arg#"$2"=}"
      printf -v "$1" "%s" "${!1:-$3}"
      break
    fi
  done
  if [[ $4 -eq 1 ]]; then
    readonly "$1"
  fi
```

```
}

keep_options(){
  # $1=variable_name of array of options
  # Other arguments: the options to keep.
  declare -a LFBFL_array_of_filtered_options
  declare -n LFBFL_array_of_options=$1
  local LFBFL_some_argument
  local LFBFL_some_argument2
  for LFBFL_some_argument in "${LFBFL_array_of_options[@]}"; do
    for LFBFL_some_argument2 in "$@"; do
      if [[
        "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"
        || "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"=*
      ]]; then
        LFBFL_array_of_filtered_options+=("${LFBFL_some_argument}")
      fi
    done
  done
  LFBFL_array_of_options=("${LFBFL_array_of_filtered_options[@]}")
}


# Main usage for the next functions relative to work_directory:
# Add the five following lines at the start of a function.
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory\
#     && trap 'popd_from_work_directory' RETURN
#   can_continue_after_enhanced_pushd || return 1
# Or:
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory --trap-popd
#   can_continue_after_enhanced_pushd || return 1

get_work_directory_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --work-directory=""
  get_some_option LFBFL_work_directory --work-directory "." 0 "$@"
  # LFBFL_work_directory=${LFBFL_work_directory/#~/${HOME}} line below
  if [[ -n "${LFBFL_work_directory}" ]]; then
    LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  fi
  readonly LFBFL_work_directory
```

```
}

pushd_to_work_directory(){
  # Options:
  #   --trap-popd
  enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
  if [[ i_enhanced_pushd_result -eq 0 ]]; then
    local LFBFL_arg
    for LFBFL_arg in "$@"; do
      if [[ "${LFBFL_arg}" == --trap-popd ]]; then
        declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
        local LFBFL_some_return_trap
        LFBFL_some_return_trap="${LFBFL_function_depth}:"
        LFBFL_some_return_trap+="popd_from_work_directory 4"
        readonly LFBFL_some_return_trap
        LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
        break
      fi
    done
  fi
  # shellcheck disable=SC2248
  return ${i_enhanced_pushd_result}
}

popd_from_work_directory(){
  # $1=offset for where_was_i
  enhanced_popd "${LFBFL_work_directory}" "${1:-3}"
}

work_directory_is_top_dirstack_directory(){
  is_top_dirstack_directory "${LFBFL_work_directory}"
}


get_offset_option(){
  # This command is to be called in another one with same arguments on top
  # of the default value.
  # $1=option_default_value
  # Options:
  #   --offset=2
  get_some_option LFBFL_i_offset --offset "$1" 1 "$@"
}


# Functions relative to shell and bash options
# (See SHELLOPTS and BASHOPTS, set and shopt):
```

```
# - shell options usage, example with pipefail:
#     enhanced_set_shell_option pipefail\
#       && trap 'enhanced_unset_shell_option pipefail' RETURN
#   or maybe:
#     enhanced_unset_shell_option pipefail\
#       && trap 'enhanced_set_shell_option pipefail' RETURN
#   if you need to unset it in some function.
#   or better if you use my RETURN traps handler:
#     enhanced_set_shell_option pipefail --trap-unset
#     enhanced_unset_shell_option pipefail --trap-set
# - bash options usage, example with globstar:
#     enhanced_set_bash_option globstar\
#       && trap 'enhanced_unset_bash_option globstar' RETURN
#   or maybe:
#     enhanced_unset_bash_option globstar\
#       && trap 'enhanced_set_bash_option globstar' RETURN
#   or better if you use my RETURN traps handler:
#     enhanced_set_bash_option globstar --trap-unset
#     enhanced_unset_bash_option globstar --trap-set

enhanced_set_shell_option(){
  # $1=optname some shell option for set
  # Options:
  #   --offset=2
  #   --trap-unset
  declare -r LFBFL_result_name="i_enhanced_set_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set -o "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s shell option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-unset ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
```

```
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_unset_shell_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_shell_option(){
  # $1=optname some shell option for set
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ ! -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set +o "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s shell option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-set ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_set_shell_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
```

```
}

enhanced_set_bash_option(){
  # $1=optname some bash option for shopt
  # Options:
  #   --offset=2
  #   --trap-unset
  declare -r LFBFL_result_name="i_enhanced_set_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
  # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
  if shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -s "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s bash option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-unset ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_unset_bash_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_bash_option(){
  # $1=optname some bash option for shopt
  # Options:
  #   --offset=2
  #   --trap-set
```

```
  declare -r LFBFL_result_name="i_enhanced_unset_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if ! shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -u "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s bash option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --trap-set ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="enhanced_set_bash_option '$1' --offset=3"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
      break
    fi
  done
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

3.10 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
```

```
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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#
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# the GNU General Public License
# along with DevOrSysAdminScripts.
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#
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# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

equal(){
  # Returns 1 if all the arguments strings are equal.
  # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
  # J'ai rerepensé au 3-way-merge
  # et à mon article sur le principe de première différence
  # appliqué à la largeur modulaire, de clique ou de rang avec
  # l'idée d'un principe de première différence ternaire.
  declare -i LFBFL_i_first=1
  local LFBFL_current
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ LFBFL_i_first -eq 1 ]]; then
      LFBFL_current="${LFBFL_arg}"
      LFBFL_i_first=0
      continue
    fi
    if [[ "${LFBFL_current}" != "${LFBFL_arg}" ]]; then
      return 0
    fi
  done
  return 1
}

some_distinct(){
  ! equal "$@"
}

all_distinct(){
  # Returns 1 if all the arguments strings are distinct.
  declare -ir LFBFL_ij_max=$#
```

```
  declare -i LFBFL_i
  declare -i LFBFL_j
  for ((LFBFL_i=1; LFBFL_i<LFBFL_ij_max; ++LFBFL_i)); do
    for ((LFBFL_j = LFBFL_i + 1; LFBFL_j<=LFBFL_ij_max; ++LFBFL_j)); do
      [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
    done
  done
  return 1
}

all_distinct2(){
  # Returns 1 if all the arguments strings are distinct.
  local LFBFL_element
  local LFBFL_previous_element
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_sorted_arguments=$(
    printf "%s\n" "$@" \
    | sort
  )
  declare -a LFBFL_arr_sorted_arguments
  mapfile -t LFBFL_arr_sorted_arguments <<< "${LFBFL_s_sorted_arguments}"
  declare -i LFBFL_i
  declare -ir LFBFL_i_max=${#LFBFL_arr_sorted_arguments[@]}
  LFBFL_previous_element=${LFBFL_arr_sorted_arguments[0]}
  for ((LFBFL_i=1; LFBFL_i<LFBFL_i_max; ++LFBFL_i)); do
    LFBFL_element="${LFBFL_arr_sorted_arguments[${LFBFL_i}]}"
    if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]]; then
      return 0
    fi
    LFBFL_previous_element="${LFBFL_element}"
  done
  return 1
}

all_distinct3(){
  # Returns 1 if all the arguments strings are distinct.
  # all_distinct3 is slower than old all_distinct2 since a call to sort is
  # needed:
  # the manual of uniq says it doesn't detect non-adjacent repeated lines.
  # A few weeks ago, when testing without sort, I had no bug...
  # I may have tested all_distinct3 a a b and not all_distinct3 a b a
  # There is a sort also in all_distinct2, but uniq could have an option to
  # sort before "applying -d", or sort could have an option for printing
  # duplicates only. It would complement -u option of sort.
```

```
  # Both solutions would avoid one pipe and one more shell command call.
  # Note that old all_distinct2 was faster with pipes and read -r commands,
  # but read -r has small drawbacks. Using mapfile is a cleaner way to do
  # it, since it handles arguments with spaces from scratch.
  # Arguments with line-returns are still not handled.
  # Use C, PHP or Python for example for that.
  # Somehow Bash was made to push people to stop being lazy and do actual C
  # code, even if it needs compiling something, adding a makefile, etc. XD
  # Note that C code would be faster with all_distinct2 blueprint,
  # that's the reason I kept all 3 versions, see the (asymptotic)
  # complexity and simple algorithms ideas on this toy problem.
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_duplicated_arguments=$(
    printf "%s\n" "$@" \
    | sort\
    | uniq -d
  )
  [[ -n "${LFBFL_s_duplicated_arguments}" ]]
}

some_equal(){
  ! all_distinct "$@"
}

some_equal2(){
  ! all_distinct2 "$@"
}

some_equal3(){
  ! all_distinct3 "$@"
}

max(){
  # Returns the maximum string among the arguments.
  # Renvoie la chaîne la plus grande parmi celles proposées.
  # $1=sort_command with the flags you want to be applied.
  # $ max sort a b c
  # c
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
```

```
151     | eval "$1 -r"\
152     | head -1
153 }
154
155 min(){
156   # Returns the minimum string among the arguments.
157   # Renvoie la chaîne la plus petite parmi celles proposées.
158   # $1=sort_command with the flags you want to be applied.
159   # $ min sort a b c
160   # a
161   if [[ $# -eq 0 ]]; then
162     return 1
163   fi
164   enhanced_set_shell_option pipefail\
165     && trap 'enhanced_unset_shell_option pipefail' RETURN
166   printf "%s\n" "${@:2}" \
167     | eval "$1"\
168     | head -1
169 }
170
171 is_substring(){
172   # $1=string
173   # $2=substring
174   declare -r LFBFL_var_1=$(
175     printf "%s" "$1"\
176     | sed -e 's/\\/\\&/g' -e "s/'/\\\&/g"
177   )
178   declare -r LFBFL_var_2=$(
179     printf "%s" "$2"\
180     | sed -e 's/\\/\\&/g' -e "s/'/\\\&/g"
181   )
182   local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
183   LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
184   LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
185   LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
186   readonly LFBFL_var_3
187   php -r "${LFBFL_var_3}"
188   return $?
189 }
190
191 is_subfile(){
192   # $1=file_path
193   # $2=sub_file_path
194   local LFBFL_var_1="\$a = file_get_contents('$1');"
195   LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
196   LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
```

```
  LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_1
  php -r "${LFBFL_var_1}"
  return $?
}

alias min_from_int_sort="min 'sort --numeric-sort'"
alias max_from_int_sort="max 'sort --numeric-sort'"
```

3.11 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
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#
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# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
```

```
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

create_PDF(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  local LFBFL_subdir2="build_and_checks_variables"
  local LFBFL_data_file_name="${LFBFL_subdir2}/repository_data.txt"
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_repository_web_url=""
  grep_variable "${LFBFL_data_file_name}" repository_web_url\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_repository_git_url=""
  grep_variable "${LFBFL_data_file_name}" repository_git_url\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_abstract=""
  grep_variable "${LFBFL_data_file_name}" abstract\
    --result-variable-prefix="LFBFL_"

```

```
  local LFBFL_acknowledgments=""
  grep_variable "${LFBFL_data_file_name}" acknowledgments\
    --result-variable-prefix="LFBFL_"

  local LFBFL_author_email=""
  grep_variable "${LFBFL_data_file_name}" author_email\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  local LFBFL_author_full_name=""
  grep_variable "${LFBFL_data_file_name}" author_full_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=" "

  local LFBFL_author_website=""
  grep_variable "${LFBFL_data_file_name}" author_website\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  declare -i LFBFL_i_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_i_"

  declare -r LFBFL_current_date=$(date -I"date")

  declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)

  declare -ir LFBFL_i_number_of_commits=$(
    git shortlog\
    | space_starting_lines\
    | wc -l
  )

  local LFBFL_redirect="/dev/stdout"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_redirect="/dev/null"
  fi
  {
    code_lines_count_all "${LFBFL_common_options[@]}"
    code_lines_count_not_empty "${LFBFL_common_options[@]}"
    code_lines_count_empty "${LFBFL_common_options[@]}"
  } &> "${LFBFL_redirect}"

  local LFBFL_number_of_lines="${i_code_lines_count_all_result}"
  LFBFL_number_of_lines+=" total lines,"
  LFBFL_number_of_lines+=" ${i_code_lines_count_not_empty_result}"
```

```
LFBFL_number_of_lines+=" not empty lines,"
LFBFL_number_of_lines+=" ${i_code_lines_count_empty_result}"
LFBFL_number_of_lines+=" empty lines."
readonly LFBFL_number_of_lines

declare -r LFBFL_temp_path="${LFBFL_subdir2}/temp"

# see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
# ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
# ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
tree -a --gitignore\
  -I "node_modules/"\
  -I "__pycache__/"\
  -I ".mypy_cache/"\
  -I ".ruff_cache/"\
  -I ".git/"\
  | replace_non_ascii_spaces\
  > "${LFBFL_temp_path}/current_tree_light.txt"

tree -a -DFh --gitignore\
  -I "node_modules/"\
  -I "__pycache__/"\
  -I ".mypy_cache/"\
  -I ".ruff_cache/"\
  -I ".git/"\
  | replace_non_ascii_spaces\
  > "${LFBFL_temp_path}/current_tree.txt.temp"

local LFBFL_s_files_paths
# Remove line returns that are here to keep lines short.
LFBFL_s_files_paths=$(
  sed -Ez 's/(.)\\\n/\1/Mg' "${LFBFL_subdir2}/files_names_listing.txt"\
  | grep -v '^//'
)
readonly LFBFL_s_files_paths
declare -a LFBFL_arr_files_paths
if [[ -n "${LFBFL_s_files_paths}" ]]; then
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
fi
readonly LFBFL_arr_files_paths

local LFBFL_license=""
grep_variable "${LFBFL_data_file_name}" license\
  --result-variable-prefix="LFBFL_"

local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
```

```
  LFBFL_temp_files_listing+="files_listing.tex.sub.temp"
  readonly LFBFL_temp_files_listing
  cp "${LFBFL_temp_path}/license_file_header_${LFBFL_license}.tex"\
    "${LFBFL_temp_files_listing}"
  local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
  LFBFL_temp_files_listing2+="files_listing.html.sub.temp"
  readonly LFBFL_temp_files_listing2
  : > "${LFBFL_temp_files_listing2}"
  # HTML <li> elements, hence "lis".
  local LFBFL_temp_files_lis
  LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.sub.temp"
  readonly LFBFL_temp_files_lis
  : > "${LFBFL_temp_files_lis}"
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_line_length} /
  declare -r LFBFL_score_command="${get_split_score_result}"
  declare -i LFBFL_i_score_command_properties
  LFBFL_i_score_command_properties=${i_get_split_score_result2}
  readonly LFBFL_i_score_command_properties
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_line_length} ':'
  declare -r LFBFL_score_command2="${get_split_score_result}"
  declare -i LFBFL_i_score_command_properties2
  LFBFL_i_score_command_properties2=${i_get_split_score_result2}
  readonly LFBFL_i_score_command_properties2
  declare -r LFBFL_suffix='%'
  declare -i LFBFL_i=0
  local LFBFL_cleaned_path1
  local LFBFL_cleaned_path2
  local LFBFL_new_lines
  local LFBFL_new_lines2
  local LFBFL_new_lines3
  local LFBFL_file_path
  local LFBFL_s_format
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    printf "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
    # printf "Listing file for tex : %s\n" "${LFBFL_file_path}"
    # LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_cleaned_path1="${LFBFL_file_path//_/\\_}"
    LFBFL_cleaned_path2=$(
      printf "%s" "${LFBFL_file_path}"\
      | sed -e 's/\//:/g' -e 's/\.//g'
    )
    LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
    if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
```

```
      repeated_split_last_line "${LFBFL_new_lines}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}\
        "\\"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
    if [[ ${#LFBFL_new_lines2} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines2}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command2}"\
        ${LFBFL_i_score_command_properties2}
      LFBFL_new_lines2=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines3="${LFBFL_file_path}"
    if [[ ${#LFBFL_new_lines3} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines3}"\
        ""\
        ${LFBFL_i_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}
      LFBFL_new_lines3=${repeated_split_last_line_result}
    fi
    LFBFL_s_format="\\subsection{\n%s\n}\n\\label{\n%s\n}\n\n"
    LFBFL_s_format+="\\VerbatimInput[numbers=left,xleftmargin=-5mm]"
    LFBFL_s_format+="{\n%s\n}\n\n\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}"\
      "${LFBFL_new_lines}"\
      "${LFBFL_new_lines2}"\
      "${LFBFL_new_lines3}"\
      >> "${LFBFL_temp_files_listing}"

    # printf "Listing file for HTML : %s\n" "${LFBFL_file_path}"
    ((++LFBFL_i))
    LFBFL_new_lines="${LFBFL_file_path}"
    if [[ ${#LFBFL_file_path} -gt LFBFL_i_max_line_length ]]; then
      # shellcheck disable=SC2248
```

```
      repeated_split_last_line "${LFBFL_new_lines}"\
        "-->"\
        ${LFBFL_i_max_line_length}\
        "<!--"\
        "${LFBFL_score_command}"\
        ${LFBFL_i_score_command_properties}
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    {
      LFBFL_s_format="<h3 id=\"subsection3.%s\">3.%s\n%s\n</h3>\n"
      LFBFL_s_format+="<pre class=\"numbered_lines\">\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_i}"\
        "${LFBFL_i}"\
        "${LFBFL_new_lines}"
      sed -e 's/&/\&amp;/g' -e 's/</\&lt;/g'\
        < "${LFBFL_file_path}"
      printf "</pre>\n\n\n"
    } >> "${LFBFL_temp_files_listing2}"
    LFBFL_s_format="      <li><a href=\"#subsection3.%s\">\n%s\n"
    LFBFL_s_format+="      </a></li>\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}" "${LFBFL_i}" "${LFBFL_new_lines}"\
      >> "${LFBFL_temp_files_lis}"
  done
  overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.tex.sub"\
    "${LFBFL_temp_files_listing}"
  # Example of generated content of files_listing.tex.sub:
  # ----------------------------------------------------------------------
  # \subsection{
  #   build\_and\_checks\_dependencies/licenses\_templates/%
  # build\_licenses\_templates.exec.sh
  # }
  # \label{
  #   build_and_checks_dependencies:licenses_templates:%
  # build_licenses_templatesexecsh
  # }
  #
  # \VerbatimInput[numbers=left,xleftmargin=-5mm]{
  # build_and_checks_dependencies/licenses_templates/%
  # build_licenses_templates.exec.sh
  # }
  # ----------------------------------------------------------------------
  sed -i -e "s|\\n</pre>|</pre>|g" "${LFBFL_temp_files_listing2}"
  overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.html.sub"\
```

```
312       "${LFBFL_temp_files_listing2}"
313     # Example of generated content of files_listing.html.sub:
314     # ---------------------------------------------------------------------
315     # <h3 id="subsection3.15">3.15
316     # build_and_checks_dependencies/licenses_templates/<!--
317     # -->build_licenses_templates.exec.sh
318     # </h3>
319     # ...
320     # <h3 id="subsection3.33">3.33
321     # build_and_checks_variables/.gitignore
322     # </h3>
323     # <pre class="numbered_lines">
324     # temp/
325     # upgrade_venvs_ts
326     # </pre>
327     # ---------------------------------------------------------------------
328     overwrite_if_not_equal "${LFBFL_temp_path}/files_lis.html.sub"\
329       "${LFBFL_temp_files_lis}"
330     # Example of generated content of files_lis.html.sub:
331     # ---------------------------------------------------------------------
332     #       <li><a href="#subsection3.15">
333     # build_and_checks_dependencies/licenses_templates/<!--
334     # -->build_licenses_templates.exec.sh
335     #       </a></li>
336     # ---------------------------------------------------------------------
337
338     # We verify if some lines are beyond max_line_length characters
339     # in current_tree_light.txt and current_tree.txt.
340     declare -ar LFBFL_trees=(
341       "current_tree_light.txt"
342       "current_tree.txt.temp"
343     )
344     local LFBFL_tree_path
345     local LFBFL_line
346     local LFBFL_prefix
347     declare -i LFBFL_i_prefix_last_position
348     local LFBFL_char
349     local LFBFL_line_start
350     declare -ir LFBFL_i_overlength=$((LFBFL_i_max_line_length+1))
351     local LFBFL_file_name
352     local LFBFL_s_lines
353     declare -a LFBFL_arr_lines
354     declare -i LFBFL_i_keep_length
355     for LFBFL_tree in "${LFBFL_trees[@]}"; do
356       LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
357       # shellcheck disable=SC2248
```

```
    LFBFL_s_lines=$(
      grep '.\{'${LFBFL_i_overlength}'\}' "${LFBFL_tree_path}"
    )
    if [[ -z "${LFBFL_s_lines}" ]]; then
      continue
    fi
    mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
    for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
      # printf "LFBFL_line: %s\n" "${LFBFL_line}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E -e 's/(.*)──[^─]+$/\1/g'
      )
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_i_prefix_last_position=$((${#LFBFL_prefix}-1))
      LFBFL_char="${LFBFL_prefix:${LFBFL_i_prefix_last_position}:1}"
      # printf "LFBFL_char: %s\n" "${LFBFL_char}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_prefix}"\
        | sed -E -e 's/[^ ]+$//g'
      )
      if [[ "${LFBFL_char}" == "└" ]]; then
        LFBFL_prefix+="  "
      else
        LFBFL_prefix+="│ "
      fi
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_file_name=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E 's|.* ([^ ]+)$|\1|g'
      )
      # printf "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
      LFBFL_i_keep_length=$((${#LFBFL_line} - ${#LFBFL_file_name}))
      LFBFL_line_start=${LFBFL_line:0:${LFBFL_i_keep_length}}
      LFBFL_line_start=$(
        printf "%s" "${LFBFL_line_start}"\
        | sed -e 's/ *$//g'
      )
      # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"
      LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
      if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
        # shellcheck disable=SC2248
        repeated_split_last_line "${LFBFL_new_lines}"\
          "${LFBFL_prefix}"\
          ${LFBFL_i_max_line_length}\
          ""\
```

```
          ""\
          ""
        LFBFL_new_lines=${repeated_split_last_line_result}
      fi
      printf "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
        > "${LFBFL_tree_path}.some_line.temp"
      insert_file_at_token "${LFBFL_tree_path}"\
        "${LFBFL_line}"\
        "${LFBFL_tree_path}.some_line.temp"\
        ""\
        --quiet
    done
    rm -f "${LFBFL_tree_path}.some_line.temp"
  done

  overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
    "${LFBFL_temp_path}/current_tree.txt.temp" 1 1

  declare -r LFBFL_tex_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
  declare -r LFBFL_html_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.html"

  if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]]; then
    # If there is a template, we init the process from it.
    cp "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
      "${LFBFL_tex_path_start}.1"
  elif [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]]; then
    # Otherwise if there is a tex file, we init the process from it.
    cp "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.1"
  else
    printf "Neither .tex.tpl, nor .tex in ./%s/\n" "${LFBFL_subdir2}"
  fi

  # Same logic with repository HTML file.
  if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]]; then
    cp "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
      "${LFBFL_html_path_start}.1"
  elif [[ -f "${LFBFL_repository_name}.html" ]]; then
    cp "${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
  else
    printf "Neither .html.tpl in ./%s/, nor .html in ./\n"\
      "${LFBFL_subdir2}"
  fi

```

```
450     # shellcheck disable=SC2001
451     printf "%s\n" "${LFBFL_abstract}"\
452       | sed -e 's/\\n/\n/g'\
453       > "${LFBFL_temp_path}/abstract_temp"
454
455     # shellcheck disable=SC2001
456     printf "%s\n" "${LFBFL_acknowledgments}"\
457       | sed -e 's/\\n/\n/g'\
458       > "${LFBFL_temp_path}/acknowledgments_temp"
459
460     # But the filling still occurs, in case the dev want to refill
461     # part of the tex/html file that he modified with @token@
462     # using some computed results.
463     # Tex filling:
464     if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
465       sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
466              -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
467              -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
468              -e "s|@author_email@|${LFBFL_author_email}|g"\
469              -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
470              -e "s|@author_website@|${LFBFL_author_website}|g"\
471              -e "s|@current_date@|${LFBFL_current_date}|g"\
472              -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
473              -e "s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
474              -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
475         "${LFBFL_tex_path_start}.1"
476
477       insert_file_at_token "${LFBFL_tex_path_start}.1"\
478         @abstract@\
479         "${LFBFL_temp_path}/abstract_temp"\
480         "${LFBFL_tex_path_start}.2"
481
482       insert_file_at_token "${LFBFL_tex_path_start}.2"\
483         @acknowledgments@\
484         "${LFBFL_temp_path}/acknowledgments_temp"\
485         "${LFBFL_tex_path_start}.3"
486
487       insert_file_at_token "${LFBFL_tex_path_start}.3"\
488         @current_tree_light@\
489         "${LFBFL_temp_path}/current_tree_light.txt"\
490         "${LFBFL_tex_path_start}.4"
491
492       insert_file_at_token "${LFBFL_tex_path_start}.4"\
493         @current_tree@\
494         "${LFBFL_temp_path}/current_tree.txt"\
495         "${LFBFL_tex_path_start}.5"
```

```
496
497     insert_file_at_token "${LFBFL_tex_path_start}.5"\
498       @dependencies_notes@\
499       "${LFBFL_temp_path}/dependencies_notes.tex.sub"\
500       "${LFBFL_tex_path_start}.6"
501
502     insert_file_at_token "${LFBFL_tex_path_start}.6"\
503       @files_listing_VerbatimInput@\
504       "${LFBFL_temp_path}/files_listing.tex.sub"\
505       "${LFBFL_tex_path_start}.7"
506
507     overwrite_if_not_equal\
508       "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
509       "${LFBFL_tex_path_start}.7" 1
510   fi
511
512   # HTML filling:
513   declare -i LFBFL_i_HTML_updated=0
514   if [[ -f "${LFBFL_html_path_start}.1" ]]; then
515     sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
516            -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
517            -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
518            -e "s|@author_email@|${LFBFL_author_email}|g"\
519            -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
520            -e "s|@author_website@|${LFBFL_author_website}|g"\
521            -e "s|@current_date@|${LFBFL_current_date}|g"\
522            -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
523            -e "s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
524            -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
525       "${LFBFL_html_path_start}.1"
526
527     insert_file_at_token "${LFBFL_html_path_start}.1"\
528       @abstract@\
529       "${LFBFL_temp_path}/abstract_temp"\
530       "${LFBFL_html_path_start}.2"
531
532     insert_file_at_token "${LFBFL_html_path_start}.2"\
533       @acknowledgments@\
534       "${LFBFL_temp_path}/acknowledgments_temp"\
535       "${LFBFL_html_path_start}.3"
536
537     insert_file_at_token "${LFBFL_html_path_start}.3"\
538       @files_lis@\
539       "${LFBFL_temp_path}/files_lis.html.sub"\
540       "${LFBFL_html_path_start}.4"
541
```

```
    insert_file_at_token "${LFBFL_html_path_start}.4"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_html_path_start}.5"

    insert_file_at_token "${LFBFL_html_path_start}.5"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_html_path_start}.6"

    insert_file_at_token "${LFBFL_html_path_start}.6"\
      @dependencies_notes@\
      "${LFBFL_temp_path}/dependencies_notes.html.sub"\
      "${LFBFL_html_path_start}.7"

    insert_file_at_token "${LFBFL_html_path_start}.7"\
      @files_listing_HTMLPreInput@\
      "${LFBFL_temp_path}/files_listing.html.sub"\
      "${LFBFL_html_path_start}.8"

    overwrite_if_not_equal "${LFBFL_repository_name}.html"\
      "${LFBFL_html_path_start}.8" 1
    LFBFL_i_HTML_updated=$?
  fi

  # The HTML contains everything including listings with source code.
  # If it doesn't get updated,
  # then it is useless to recompute the PDF.
  if [[ LFBFL_i_HTML_updated -eq 0 ]]; then
    return
  fi

  local LFBFL_latex_command=""
  grep_variable "${LFBFL_data_file_name}" latex_command\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""

  case ${LFBFL_latex_command} in
    pdflatex) : ;;
    lualatex) : ;;
    *) printf "Unsupported latex command" ; return 1 ;;
  esac

  local LFBFL_command
  LFBFL_command="${LFBFL_latex_command}"
  LFBFL_command+=" '${LFBFL_subdir2}/${LFBFL_repository_name}.tex'"
```

```
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_command+=" > /dev/null"
  fi
  readonly LFBFL_command

  for ((i=0; i<3; i++)); do
    eval "${LFBFL_command}"
  done

  LFBFL_files_to_temp=(\
    "${LFBFL_repository_name}.aux"\
    "${LFBFL_repository_name}.log"\
    "${LFBFL_repository_name}.out"\
    "${LFBFL_repository_name}.toc"
  )
  for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
    mv "${LFBFL_file_name}" "${LFBFL_temp_path}/"
  done
}

create_PDF "$@"
```

3.12 build_and_checks_dependencies/generate_from_template.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
```

```
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
  {
    printf "%s\n" "$3"
    cat "$1"
    printf "%s\n" "$4"
  } > "${LFBFL_temp}"
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

generate_from_template_with_line_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=line_comment_prefix
  # $4=optional_post_processing
  # $5=file_prefix
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
  readonly LFBFL_temp
  if [[ -n "$5" ]]; then
    printf "%s\n" "$5"\
      > "${LFBFL_temp}"
  fi
  sed -e "s/^/$3/g" "$1"\
    >> "${LFBFL_temp}"
  if [[ -n "$4" ]]; then
    eval "$4"
  fi
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}
```

```
69
70  split_file_in_two(){
71    # $1=file_path
72    # $2=token assume that token is the only thing on his line,
73    #    except for a possible indentation.
74    # $3=path_of_part1_of_file
75    # $4=path_of_part2_of_file
76    # We cannot add --line-regexp with --fixed-strings because of the
77    # possible indentation in file.
78    declare -ir LFBFL_i_line_number=$(
79      grep_fixed_string_with_anchor "$1" "$2"\
80        --enforce-line-ends-with-fixed-string --fixed-strings --line-number\
81      | head --lines=1\
82      | cut -f 1 -d ':'
83    )
84    declare -ir LFBFL_i_line_count=$(ll_wc -l -n "$1")
85    declare -ir LFBFL_i_lines_after=$((
86      LFBFL_i_line_count - LFBFL_i_line_number
87    ))
88    head --lines="$((LFBFL_i_line_number - 1))" "$1"\
89      > "$3"
90    tail --lines="${LFBFL_i_lines_after}" "$1"\
91      > "$4"
92  }
93
94  insert_file_at_token(){
95    # $1=file_path
96    # $2=token assume that token is the only thing on his line,
97    #    except for a possible indentation.
98    # $3=path_of_file_to_insert
99    # $4=path_of_result_file
100   # Options:
101   #   $5=--quiet to suppress diff output in overwrite_if_not_equal
102   declare -r LFBFL_start_file_name="$1.insert_file_at_token1.temp"
103   declare -r LFBFL_end_file_name="$1.insert_file_at_token2.temp"
104   local LFBFL_result_file_name
105   if [[ -n "$4" ]]; then
106     LFBFL_result_file_name="$4"
107   else
108     LFBFL_result_file_name="$1.insert_file_at_token3.temp"
109   fi
110   readonly LFBFL_result_file_name
111   split_file_in_two "$1" "$2" "${LFBFL_start_file_name}"\
112     "${LFBFL_end_file_name}"
113   cat "${LFBFL_start_file_name}" "$3" "${LFBFL_end_file_name}"\
114     > "${LFBFL_result_file_name}"
```

```
  rm "${LFBFL_start_file_name}" "${LFBFL_end_file_name}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_name}" "" "" "$5"
  fi
}
```

3.13 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# This file was renamed from "get_common_text_glob_patterns.sh"
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
    '**/*.htm'
    '**/*.html'
    '**/*.ini'
```

```
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.php'
    '**/*.py'
    '**/*.sh'
    '**/*.sql'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
    '**/*.xml'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
  )
}


get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
    '*.c'
    '*.css'
    '*.h'
    '*.htm'
    '*.html'
    '*.ini'
    '*.js'
    '*.json'
    '*.md'
    '*.php'
    '*.py'
    '*.sh'
    '*.sql'
    '*.tex'
    '*.toml'
    '*.tpl'
```

```
    '*.ts'
    '*.txt'
    '*.xml'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

3.14 build_and_checks_dependencies/grammar_and_spelling_check.libr.sh

```
#!/usr/bin/env bash
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2026-03-22T15:30+01:00: This file was renamed from
# "grammar_and_spell_check.libr.sh"
# to
# "grammar_and_spelling_check.libr.sh".
# This file was renamed from "grammar_and_spell_check.sh"
# to "grammar_and_spell_check.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
```

```
32 # shellcheck source=common_options.libr.sh
33 source "./${LFBFL_subdir}/common_options.libr.sh"
34 # shellcheck source=get_common_text_glob_patterns.libr.sh
35 source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
36 # shellcheck source=lines_filters.libr.sh
37 source "./${LFBFL_subdir}/lines_filters.libr.sh"
38
39 grammar_and_spelling_check(){
40   # $1=configuration_path build_and_checks_variables/repository_data.txt
41   # Options:
42   #   --verbose
43   #   --work-directory=""
44   declare -i LFBFL_i_verbose=0
45   get_verbose_option "$@"
46   local LFBFL_work_directory=""
47   get_work_directory_option "$@"
48   declare -a LFBFL_return_traps_stack
49   local LFBFL_previous_return_trap
50   init_return_trap
51   pushd_to_work_directory --trap-popd
52   can_continue_after_enhanced_pushd || return 1
53
54   enhanced_set_shell_option pipefail --trap-unset
55
56   get_COMMON_TEXT_FILES_GLOB_PATTERNS
57   local LFBFL_grammar_and_spelling_checker_command=""
58   grep_variable "$1" grammar_and_spelling_checker_command\
59     --result-variable-prefix=LFBFL_\
60     --replace-line-returns-by=""
61   declare -r LFBFL_command="${LFBFL_grammar_and_spelling_checker_command}"
62   if [[ -z "${LFBFL_command}" ]]; then
63     if [[ LFBFL_i_verbose -eq 1 ]]; then
64       printf "grammar_and_spelling_check: No command in configuration.\n"
65     fi
66     return
67   fi
68   if [[ LFBFL_i_verbose -eq 1 ]]; then
69     printf "grammar_and_spelling_check: Command found: %s.\n"\
70       "${LFBFL_command}"
71   fi
72   local LFBFL_file_path
73   local LFBFL_s_files_paths
74   declare -a LFBFL_arr_files_paths
75   local LFBFL_pattern
76   local LFBFL_eval_string
77   for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
```

```
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "grammar_and_spelling_check: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "grammar_and_spelling_check: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval "${LFBFL_eval_string}"
    done
  done
}
```

3.15 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_licenses_templates(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap

  enhanced_set_shell_option pipefail --trap-unset

  local LFBFL_s_format

  # Source
  local LFBFL_license_source_subdir
  LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
  readonly LFBFL_license_source_subdir
  local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
  LFBFL_source_license_prefix+="license_file_header_"
  readonly LFBFL_source_license_prefix
  # Target
  local LFBFL_license_target_subdir
  LFBFL_license_target_subdir="${LFBFL_work_directory}/"
  LFBFL_license_target_subdir="build_and_checks_variables/temp/"
  readonly LFBFL_license_target_subdir
  local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
```

```
70    LFBFL_target_license_prefix+="license_file_header_"
71    readonly LFBFL_target_license_prefix
72
73    declare -ar LFBFL_licenses=(
74      "AGPLv3+"
75      "GPLv3+"
76      "LGPLv3+"
77    )
78
79    # ------------------------------------------------------------------
80    # Various language definitions for text file formats
81    declare -ar LFBFL_block_comment_languages=(
82      "c"
83      "py"
84    )
85    declare -ar LFBFL_block_comment_enters=(
86      '/*'
87      '"""'
88    )
89    declare -ar LFBFL_block_comment_exits=(
90      '*/'
91      '"""'
92    )
93    equal "${#LFBFL_block_comment_languages[@]}"\
94          "${#LFBFL_block_comment_enters[@]}"\
95          "${#LFBFL_block_comment_exits[@]}"
96    declare -i LFBFL_i_equal_result=$?
97    if [[ LFBFL_i_equal_result -eq 0 ]]; then
98      printf $'/!\\Problem with the definitions of languages arrays 1/!\\\n'
99      return 1
100   fi
101
102   declare -ar LFBFL_line_comment_languages=(
103     "sh"
104     "sql"
105     "tex"
106   )
107   declare -ar LFBFL_line_comment_prefixes=(
108     '# '
109     '-- '
110     '% '
111   )
112   equal "${#LFBFL_line_comment_languages[@]}"\
113         "${#LFBFL_line_comment_prefixes[@]}"
114   LFBFL_i_equal_result=$?
115   if [[ LFBFL_i_equal_result -eq 0 ]]; then
```

```
    printf $'/!\\Problem with the definitions of languages arrays 2/!\\\n'
    return 1
  fi
  # ------------------------------------------------------------------

  # ------------------------------------------------------------------
  # We generate the license header templates adapted to various text
  # files formats.
  # Kamoulox do endfor, Jacques Beauregard XD
  # (C'est toujours mieux que l'inverse :) !)
  # for do endfor -> for do done (Fort Doux Donne).
  local LFBFL_some_license
  local LFBFL_license_prefix
  local LFBFL_license_prefix2
  local LFBFL_extension
  local LFBFL_license_file_name
  local LFBFL_enter_string
  local LFBFL_exit_string
  local LFBFL_temp2
  local LFBFL_prefix_string
  local LFBFL_intermediate_file_name
  local LFBFL_file_prefix
  declare -i LFBFL_i_generate_from_template_result
  declare -i LFBFL_i
  for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
    LFBFL_license_prefix="${LFBFL_source_license_prefix}"
    LFBFL_license_prefix+="${LFBFL_some_license}"
    LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
    LFBFL_license_prefix2+="${LFBFL_some_license}"
    for ((
      LFBFL_i = 0;
      LFBFL_i < ${#LFBFL_block_comment_languages[@]};
      ++LFBFL_i
    )); do
      LFBFL_extension=${LFBFL_block_comment_languages[LFBFL_i]}
      LFBFL_license_file_name="${LFBFL_license_prefix2}"
      LFBFL_license_file_name+=".${LFBFL_extension}"
      LFBFL_enter_string=${LFBFL_block_comment_enters[LFBFL_i]}
      LFBFL_exit_string=${LFBFL_block_comment_exits[LFBFL_i]}
      generate_from_template_with_block_comments\
        "${LFBFL_license_prefix}.tpl"\
        "${LFBFL_license_file_name}.tpl"\
        "${LFBFL_enter_string}" "${LFBFL_exit_string}"
      LFBFL_i_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
```

```
162           printf "License template %s.tpl updated.\n"\
163             "${LFBFL_license_file_name}"
164         elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
165           printf "License template %s.tpl created.\n"\
166             "${LFBFL_license_file_name}"
167         fi
168       fi
169     done
170     LFBFL_temp2=".generate_from_template_with_line_comments.temp"
171     for ((
172       LFBFL_i = 0;
173       LFBFL_i < ${#LFBFL_line_comment_languages[@]};
174       ++LFBFL_i
175     )); do
176       LFBFL_extension=${LFBFL_line_comment_languages[LFBFL_i]}
177       LFBFL_license_file_name="${LFBFL_license_prefix2}"
178       LFBFL_license_file_name+=".${LFBFL_extension}"
179       LFBFL_prefix_string=${LFBFL_line_comment_prefixes[LFBFL_i]}
180       LFBFL_intermediate_file_name="${LFBFL_license_file_name}.tpl"
181       LFBFL_intermediate_file_name+="${LFBFL_temp2}"
182       LFBFL_file_prefix=""
183       if [[ "${LFBFL_extension}" == "sh" ]]; then
184         LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
185       fi
186       # The sed command is here to remove trailing spaces of previously
187       # empty lines after comment prefix is added to the line.
188       generate_from_template_with_line_comments\
189         "${LFBFL_license_prefix}.tpl"\
190         "${LFBFL_license_file_name}.tpl"\
191         "${LFBFL_prefix_string}"\
192         "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_name}'"\
193         "${LFBFL_file_prefix}"
194       LFBFL_i_generate_from_template_result=$?
195       if [[ LFBFL_i_verbose -eq 1 ]]; then
196         if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
197           printf "License template %s.tpl updated.\n"\
198             "${LFBFL_license_file_name}"
199         elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
200           printf "License template %s.tpl created.\n"\
201             "${LFBFL_license_file_name}"
202         fi
203       fi
204     done
205   done
206   # ------------------------------------------------------------------
207
```

```
208   # Now that base license templates generated languages license
209   # templates in working directory, we can go there.
210   pushd_to_work_directory --trap-popd
211   can_continue_after_enhanced_pushd || return 1
212
213   local LFBFL_data_file_name="build_and_checks_variables/"
214   LFBFL_data_file_name+="repository_data.txt"
215   readonly LFBFL_data_file_name
216   local LFBFL_repository_name=""
217   grep_variable "${LFBFL_data_file_name}" repository_name\
218     --result-variable-prefix="LFBFL_"\
219     --replace-line-returns-by=""
220   local LFBFL_license=""
221   grep_variable "${LFBFL_data_file_name}" license\
222     --result-variable-prefix="LFBFL_"
223   local LFBFL_license2=""
224   grep_variable "${LFBFL_data_file_name}" license2\
225     --result-variable-prefix="LFBFL_"
226   local LFBFL_author_full_name=""
227   grep_variable "${LFBFL_data_file_name}" author_full_name\
228     --result-variable-prefix="LFBFL_"\
229     --replace-line-returns-by=" "
230
231   LFBFL_license_prefix="build_and_checks_variables/temp/"
232   LFBFL_license_prefix+="license_file_header_"
233   readonly LFBFL_license_prefix
234   LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
235   readonly LFBFL_license_prefix2
236   declare -r\
237     LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"
238
239   # First year according to current state of git repository.
240   declare -ir LFBFL_i_first_year=$(
241     git log\
242     | grep 'Date:'\
243     | tail -1\
244     | cut -f 8 -d ' '
245   )
246   # Last year according to current state of git repository.
247   declare -ir LFBFL_i_last_year=$(
248     git log\
249     | grep 'Date:'\
250     | head -1\
251     | cut -f 8 -d ' '
252   )
253   local LFBFL_copyright_string
```

```
LFBFL_copyright_string="${LFBFL_i_first_year}-${LFBFL_i_last_year}"
LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
readonly LFBFL_copyright_string

prepare_filled_license_file_block(){
  # $1=license_file_name
  sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
      -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
      "$1.tpl"\
    > "$1.temp"
  overwrite_if_not_equal "$1" "$1.temp"
  declare -ir LFBFL_i_result=$?
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    if [[ LFBFL_i_result -eq 1 ]]; then
      printf "License template %s updated.\n" "$1"
    elif [[ LFBFL_i_result -eq 2 ]]; then
      printf "License template %s created.\n" "$1"
    fi
  fi
  head --lines=-1 "$1"\
    | tail --lines=+2\
    > "$1.temp"
  # Strange world: tail --lines=-1 was working
  # to exclude first line.
  # I tested it months ago, and saw with my eyes the start and end
  # block comments removed with:
  # head --lines=-1 "$1" | tail --lines=-1.
  # Now, man page says to use --lines=+2 instead.
  # Welcome in the world where AI change your environment and your
  # documentation on the fly...
}

declare -Ar LFBFL_all_block_comment_languages=(
  ["c"]="c"
  ["php"]="c"
  ["py"]="py"
  ["js"]="c"
)

local LFBFL_key
local LFBFL_dest
local LFBFL_license_file_name2
declare -i LFBFL_i_not_subfile
declare -i LFBFL_i_not_subfile2
local LFBFL_file_path
local LFBFL_s_files_paths
```

```
300   declare -a LFBFL_arr_files_paths
301   for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
302     LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
303     LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
304     prepare_filled_license_file_block "${LFBFL_license_file_name}"
305     if [[ -n "${LFBFL_license2}" ]]; then
306       LFBFL_license_file_name2="${LFBFL_license_prefix3}"
307       LFBFL_license_file_name2+=".${LFBFL_dest}"
308       prepare_filled_license_file_block "${LFBFL_license_file_name2}"
309     fi
310
311     LFBFL_s_files_paths=$(
312       find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
313         -printf '%P\n'\
314       | relevant_find
315     )
316     if [[ -z "${LFBFL_s_files_paths}" ]]; then
317       if [[ LFBFL_i_verbose -eq 1 ]]; then
318         LFBFL_s_format="build_licenses_templates: "
319         LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
320         # shellcheck disable=SC2059
321         printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
322       fi
323       continue
324     fi
325     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
326     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
327       is_subfile "${LFBFL_file_path}"\
328         "${LFBFL_license_file_name}.temp"
329       LFBFL_i_not_subfile=$?
330       LFBFL_i_not_subfile2=1
331       if [[ -n "${LFBFL_license2}" ]]; then
332         is_subfile "${LFBFL_file_path}"\
333           "${LFBFL_license_file_name2}.temp"
334         LFBFL_i_not_subfile2=$?
335       fi
336       if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
337       then
338         printf "File %s has no/wrong license header?\n"\
339           "${LFBFL_file_path}"
340       fi
341     done
342     rm "${LFBFL_license_file_name}.temp"
343     if [[ -n "${LFBFL_license2}" ]]; then
344       rm "${LFBFL_license_file_name2}.temp"
345     fi
```

```
346   done
347
348   prepare_filled_license_file_line(){
349     # $1=LFBFL_license_file_name
350     sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
351         -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
352         "$1.tpl"\
353       > "$1.temp"
354     overwrite_if_not_equal "$1" "$1.temp"
355     declare -ir LFBFL_i_result=$?
356     if [[ LFBFL_i_verbose -eq 1 ]]; then
357       if [[ LFBFL_i_result -eq 1 ]]; then
358         printf "License template %s updated.\n" "$1"
359       elif [[ LFBFL_i_result -eq 2 ]]; then
360         printf "License template %s created.\n" "$1"
361       fi
362     fi
363   }
364
365   declare -Ar LFBFL_all_line_comment_languages=(
366     ["sh"]="sh"
367     ["sql"]="sql"
368     ["tex"]="tex"
369   )
370
371   for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
372     LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
373     LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
374     prepare_filled_license_file_line "${LFBFL_license_file_name}"
375     if [[ -n "${LFBFL_license2}" ]]; then
376       LFBFL_license_file_name2="${LFBFL_license_prefix3}"
377       LFBFL_license_file_name2+=".${LFBFL_dest}"
378       prepare_filled_license_file_line "${LFBFL_license_file_name2}"
379     fi
380
381     LFBFL_s_files_paths=$(
382       find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
383         -printf '%P\n'\
384       | relevant_find
385     )
386     if [[ -z "${LFBFL_s_files_paths}" ]]; then
387       if [[ LFBFL_i_verbose -eq 1 ]]; then
388         LFBFL_s_format="build_licenses_templates: "
389         LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
390         # shellcheck disable=SC2059
391         printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
```

```
      fi
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name}"
      LFBFL_i_not_subfile=$?
      LFBFL_i_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name2}"
        LFBFL_i_not_subfile2=$?
      fi
      if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
      then
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
  done
}

build_licenses_templates "$@"
```

3.16 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

```
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```

```
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```

3.17 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

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```

3.18 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

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3.19 build_and_checks_dependencies/lines_counts.libr.sh

```
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# This file was renamed from "lines_counts.sh" to
# "lines_counts.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

all_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
```

```
39    declare -i LFBFL_i_verbose=0
40    get_verbose_option "$@"
41    local LFBFL_work_directory=""
42    get_work_directory_option "$@"
43    declare -a LFBFL_return_traps_stack
44    local LFBFL_previous_return_trap
45    init_return_trap
46    pushd_to_work_directory --trap-popd
47    can_continue_after_enhanced_pushd || return 1
48
49    enhanced_set_shell_option pipefail --trap-unset
50
51    get_COMMON_TEXT_FILES_GLOB_PATTERNS
52    local LFBFL_pattern
53    for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
54      if [[ LFBFL_i_verbose -eq 1 ]]; then
55        printf "all_code_lines: Iterating on pattern: %s.\n"\
56          "${LFBFL_pattern}"
57      fi
58      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
59        | xargs grep --invert-match --perl-regexp --with-filename 'a(?!a)a'
60    done
61  }
62
63  all_self_code_lines(){
64    # Options:
65    #   --verbose
66    #   --work-directory=""
67    declare -i LFBFL_i_verbose=0
68    get_verbose_option "$@"
69    local LFBFL_work_directory=""
70    get_work_directory_option "$@"
71    declare -a LFBFL_return_traps_stack
72    local LFBFL_previous_return_trap
73    init_return_trap
74    pushd_to_work_directory --trap-popd
75    can_continue_after_enhanced_pushd || return 1
76
77    enhanced_set_shell_option pipefail --trap-unset
78
79    declare -a LFBFL_common_options=("$@")
80    keep_options LFBFL_common_options --verbose
81
82    # below and another lines, see SC2145 false positive
83    all_code_lines "${LFBFL_common_options[@]}" \
84      | relevant_grep\
```

```
    | not_license_grep\
    | not_main_tex_grep\
    | not_main_html_grep
}

# shellcheck disable=SC2120
all_self_empty_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  all_self_code_lines "${LFBFL_common_options[@]}" \
    | empty_lines_after_file_name
}

# shellcheck disable=SC2120
all_self_not_empty_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

```

```
131   declare -a LFBFL_common_options=("$@")
132   keep_options LFBFL_common_options --verbose
133
134   all_self_code_lines "${LFBFL_common_options[@]}" \
135     | not_empty_lines_after_file_name
136 }
137
138 code_lines_count_all(){
139   # Options:
140   #   --verbose
141   #   --work-directory=""
142   declare -i LFBFL_i_verbose=0
143   get_verbose_option "$@"
144   local LFBFL_work_directory=""
145   get_work_directory_option "$@"
146   declare -a LFBFL_return_traps_stack
147   local LFBFL_previous_return_trap
148   init_return_trap
149   pushd_to_work_directory --trap-popd
150   can_continue_after_enhanced_pushd || return 1
151
152   enhanced_set_shell_option pipefail --trap-unset
153
154   declare -gi i_code_lines_count_all_result
155   i_code_lines_count_all_result=$(
156     all_self_code_lines\
157     | wc --lines
158   )
159   # The printf is for command line use.
160   # A script should use the result value instead.
161   # shellcheck disable=SC2248
162   printf "%s\n" ${i_code_lines_count_all_result}
163 }
164
165 code_lines_count_empty(){
166   # Options:
167   #   --verbose
168   #   --work-directory=""
169   declare -i LFBFL_i_verbose=0
170   get_verbose_option "$@"
171   local LFBFL_work_directory=""
172   get_work_directory_option "$@"
173   declare -a LFBFL_return_traps_stack
174   local LFBFL_previous_return_trap
175   init_return_trap
176   pushd_to_work_directory --trap-popd
```

```
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -gi i_code_lines_count_empty_result
  # shellcheck disable=SC2119
  i_code_lines_count_empty_result=$(
    all_self_empty_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_empty_result}
}

code_lines_count_not_empty(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -gi i_code_lines_count_not_empty_result
  # shellcheck disable=SC2119
  i_code_lines_count_not_empty_result=$(
    all_self_not_empty_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_not_empty_result}
}
```

3.20 build_and_checks_dependencies/lines_filters.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_filters.sh" to
# "lines_filters.libr.sh".

# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
```

```
45    # -w --words
46    # --help
47    # --version
48    # New option by Laurent Lyaudet (and probably other before):
49    # -n --no-filenames display only number(s)
50    # I think I did that the first time many years ago (Teliae ?).
51    # Since then, they added it as ignored... (-n)
52    enhanced_set_shell_option pipefail\
53      && trap 'enhanced_unset_shell_option pipefail' RETURN
54    declare -a ll_wc_var_args=()
55    declare -i ll_wc_var_i_number_only=0
56    declare -i ll_wc_var_i=0
57    for ll_wc_var_arg in "$@"; do
58      # printf "%s\n" "${ll_wc_var_arg}"
59      if [[ "${ll_wc_var_arg}" == "-n" ]]; then
60        ll_wc_var_i_number_only=1
61      elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
62        ll_wc_var_i_number_only=1
63      elif [[ ll_wc_var_i_number_only -eq 0 ]]; then
64        ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
65        ll_wc_var_i=$((ll_wc_var_i + 1))
66      elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
67        ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
68        ll_wc_var_i=$((ll_wc_var_i + 1))
69      fi
70    done
71    # typeset -p ll_wc_var_args
72    # printf "%s\n" "${ll_wc_var_i_number_only}"
73    if [[ ll_wc_var_i_number_only -gt 0 ]]; then
74      # Too simple code, only the base use case is handled now.
75      # Convenient but incomplete.
76      # With more time,
77      # the simpler would be to submit a patch to GNU coreutils,
78      # unless blockade from assholes.
79      # Otherwise, it may be more simply handled in PHP or Python
80      # (argparse, etc.).
81      # You'll need to start by improving cat first,
82      # with --files0-from handling.
83      # wc does not display file_name when stream...
84      # hence cat of last arg "${!#}"
85      cat "${!#}"\
86        | wc "${ll_wc_var_args[@]}"
87    else
88      # printf "normal\n"
89      wc "${ll_wc_var_args[@]}"
90    fi
```

```
}

# I submitted a patch for wc :).
# printf "https://domain/list-path/2024-05/msg00013.html" | \
#   sed --expression='s|domain|lists.gnu.org|' \
#       --expression='s|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
  alias ll_wc='wc'
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41

in_place_grep(){
  # Same arguments and options as grep.
  declare -r LFBFL_temp="${!#}.in_place_grep.temp"
  grep "$@" \
    > "${LFBFL_temp}"
  # declare -ir LFBFL_i_lines_before=$(ll_wc -l -n "${!#}")
  # declare -ir LFBFL_i_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
  # printf "%s\n" "${LFBFL_i_lines_before}"
  # printf "%s\n" "${LFBFL_i_lines_after}"
  # if [[ LFBFL_i_lines_before -eq LFBFL_i_lines_after ]]; then
  # Almost as fast (on large files) and correct in all cases :).
  # Thanks Paul Eggert for the suggestion :).
  # https://debbugs.gnu.org/cgi/bugreport.cgi?bug=80547
  if cmp --silent "${LFBFL_temp}" "${!#}"; then
    rm "${LFBFL_temp}"
    return
  fi
  mv "${LFBFL_temp}" "${!#}"
}

```

```
grep_fixed_string_with_anchor(){
  # $1=file_path
  # $2=some_fixed_string
  # Options:
  #   --enforce-line-starts-with-fixed-string
  #   --enforce-line-ends-with-fixed-string
  #   Apart from $1 and $2 and the options above,
  #   all other arguments are considered grep_options like --quiet
  #   for example.
  #   But note that --fixed-strings will be added below, you don't need to
  #   supply it.
  declare -a LFBFL_grep_options=()
  declare -a LFBFL_grep_options2=()
  declare -i LFBFL_i=0
  declare -i LFBFL_i_enforce_line_starts=0
  declare -i LFBFL_i_enforce_line_ends=0
  declare -i LFBFL_i_quiet=0
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    ((++LFBFL_i))
    if [[ LFBFL_i -lt 3 ]]; then
      continue
    fi
    if [[ "${LFBFL_arg}" == "--enforce-line-starts-with-fixed-string" ]];
    then
      LFBFL_i_enforce_line_starts=1
      continue
    fi
    if [[ "${LFBFL_arg}" == "--enforce-line-ends-with-fixed-string" ]];
    then
      LFBFL_i_enforce_line_ends=1
      continue
    fi
    if [[ "${LFBFL_arg}" == "-q" || "${LFBFL_arg}" == "--quiet" ]];
    then
      LFBFL_i_quiet=1
      LFBFL_grep_options+=("${LFBFL_arg}")
      continue
    fi
    LFBFL_grep_options2+=("${LFBFL_arg}")
  done

  if [[
    LFBFL_i_enforce_line_starts -eq 0
    && LFBFL_i_enforce_line_ends -eq 0
  ]]; then
```

```
183       grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
184       return
185     fi
186     declare -r LFBFL_s_lines=$(
187       grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
188         "$2" "$1"
189     )
190     declare -i LFBFL_i_result=1
191     if [[ -z "${LFBFL_s_lines}" ]]; then
192       # shellcheck disable=SC2248
193       return ${LFBFL_i_result}
194     fi
195     declare -a LFBFL_arr_lines
196     mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
197     readonly LFBFL_arr_lines
198
199     declare -i LFBFL_i_fixed_string_length=${#2}
200     local LFBFL_some_line
201     local LFBFL_line_part
202     for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
203       if [[ "${LFBFL_some_line}" == "$2" ]]; then
204         LFBFL_i_result=0
205         if [[ LFBFL_i_quiet -eq 1 ]]; then
206           break
207         else
208           printf "%s\n" "${LFBFL_some_line}"
209         fi
210         continue
211       fi
212       if [[
213         LFBFL_i_enforce_line_starts -eq 1
214         && LFBFL_i_enforce_line_ends -eq 1
215       ]]; then
216         continue
217       fi
218       if [[ LFBFL_i_enforce_line_starts -eq 1 ]]; then
219         LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_i_fixed_string_length}}
220       fi
221       if [[ LFBFL_i_enforce_line_ends -eq 1 ]]; then
222         LFBFL_line_part=${LFBFL_some_line: -${LFBFL_i_fixed_string_length}}
223       fi
224       if [[ "${LFBFL_line_part}" == "$2" ]]; then
225         LFBFL_i_result=0
226         if [[ LFBFL_i_quiet -eq 1 ]]; then
227           break
228         else
```

```
229         printf "%s\n" "${LFBFL_some_line}"
230       fi
231     fi
232   done
233   LFBFL_i_result=$((LFBFL_i_result | $?))
234   # shellcheck disable=SC2248
235   return ${LFBFL_i_result}
236 }
237
238 grep_variable(){
239   # $1=file_path
240   # $2=variable_name
241   # Options :
242   #   --replace-line-returns-by=""
243   #   --result-variable-prefix="LFBFL_" for example
244   declare -r LFBFL_regexp="(?<=^$2=).*$"
245   local LFBFL_arg
246   # printf "%s\n" "${LFBFL_regexp}"
247   local LFBFL_variable_value
248   if [[ "$*" == *--replace-line-returns-by=* ]]; then
249     local LFBFL_s_replace=""
250     for LFBFL_arg in "$@"; do
251       if [[ "${LFBFL_arg}" == --replace-line-returns-by=* ]]; then
252         LFBFL_s_replace=${LFBFL_arg#--replace-line-returns-by=}
253         break
254       fi
255     done
256     # sed -Ez -e...
257     LFBFL_variable_value=$(
258       grep --only-matching --perl-regexp "${LFBFL_regexp}" "$1"\
259       | sed --null-data --regexp-extended\
260         --expression='s/\n/'"${LFBFL_s_replace}"'/Mg'
261     )
262     # We remove the trailing replace string,
263     # since the intent is only to replace the intermediate \n.
264     if [[ -n "${LFBFL_s_replace}" ]]; then
265     LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
266     fi
267   else
268     LFBFL_variable_value=$(
269       grep --only-matching --perl-regexp "${LFBFL_regexp}" "$1"
270     )
271   fi
272   local LFBFL_prefix=""
273   for LFBFL_arg in "$@"; do
274     if [[ "${LFBFL_arg}" == --result-variable-prefix=* ]]; then
```

```
      LFBFL_prefix=${LFBFL_arg#--result-variable-prefix=}
      break
    fi
  done
  declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
  # printf "%s : %s\n " "${LFBFL_variable_name}" "${LFBFL_variable_value}"
  # The declare line may be dropped by a local or declare
  # in outer scope.
  # declare -g "${LFBFL_variable_name}"
  # So the assignment must be done on another line.
  printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
  # printf "%s\n" "${!LFBFL_variable_name}"
}

empty_lines(){
  grep '^$'
}

not_empty_lines(){
  grep --invert-match '^$'
}

space_starting_lines(){
  grep '^ '
}

not_space_starting_lines(){
  grep --invert-match '^ '
}

# Space starting or empty lines = spampty
spampty_lines(){
  grep --extended-regexp '^( |$)'
}

not_spampty_lines(){
  grep '^[^ ]'
  # grep --extended-regexp --invert-match '^( |$)'
}

all_space_lines(){
  grep '^[ ]\+$'
}

not_all_space_lines(){
  grep --invert-match '^[ ]\+$'
```

```
}

# All space or empty lines = aspampty
aspampty_lines(){
  grep '^[ ]*$'
}

not_aspampty_lines(){
  grep '[^ ]'
  # grep --invert-match '^[ ]*$'
}

empty_lines_after_file_name(){
  grep '^[^:]\+:$'
}

not_empty_lines_after_file_name(){
  grep --invert-match '^[^:]\+:$'
}

space_starting_lines_after_file_name(){
  grep '^[^:]\+: '
}

not_space_starting_lines_after_file_name(){
  grep --invert-match '^[^:]\+: '
}

spampty_lines_after_file_name(){
  grep --extended-regexp '^[^:]+:( |$)'
}

not_spampty_lines_after_file_name(){
  # The two variants results differ on lines without filename before.
  # Since the intent of the function is only on lines "blabla.sh:whatever",
  # use the fastest one. I didn't do extensive testing and just assumed
  # that the "positive" one is faster as usual when querying database.
  # But there is no index here ;) XD.
  grep '^[^:]\+:[^ ]'
  # grep --extended-regexp --invert-match '^[^:]+:( |$)'
}

all_space_lines_after_file_name(){
  grep '^[^:]\+:[ ]\+$'
}

```

```
not_all_space_lines_after_file_name(){
  grep --invert-match '^[^:]\+:[ ]\+$'
}

aspampty_lines_after_file_name(){
  grep '^[^:]\+:[ ]*$'
}

not_aspampty_lines_after_file_name(){
  grep --invert-match '^[^:]\+:[ ]*$'
}

# If you need to test the previous 20 functions,
# look at file spampty_data.txt in miscellaneous.
# And give a try at something like:
# cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
# -E = --show-ends :). Nice command :).

not_JS_dependencies_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)node_modules/"\
    --regexp="(^|/)package-lock\.json$"
}

not_JS_dependencies_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?node_modules/"\
    --regexp="^([^:]+/)?package-lock\.json:"
}

not_dependencies_find(){
  not_JS_dependencies_find
}

not_dependencies_grep(){
  not_JS_dependencies_grep
}

not_python_cache_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)__pycache__/"\
    --regexp="(^|/)\.mypy_cache/"\
    --regexp="(^|/)\.ruff_cache/"
}

not_python_cache_grep(){
```

```
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?__pycache__/"\
    --regexp="^([^:]+/)?\.mypy_cache/"\
    --regexp="^([^:]+/)?\.ruff_cache/"
}

not_cache_find(){
  not_python_cache_find
}

not_cache_grep(){
  not_python_cache_grep
}

not_git_find(){
  grep --extended-regexp --invert-match "(^|/)\.git/"
}

not_git_grep(){
  grep --extended-regexp --invert-match "^([^:]+/)?\.git/"
}

not_archive_find(){
  grep --extended-regexp --invert-match "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
}

not_archive_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
}

not_license_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)COPYING$"\
    --regexp="(^|/)COPYING\.LESSER$"
}

not_license_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?COPYING:"\
    --regexp="^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
```

```
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.tex$"
}

not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

not_main_html_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.html$"
}

not_main_html_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?${LFBFL_repository_name}\.html:"
}

not_temp_file_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)build_and_checks_variables/temp/"
}

not_temp_file_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
}
```

```

relevant_find(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_find\
    | not_cache_find\
    | not_git_find\
    | not_archive_find\
    | not_temp_file_find
}

relevant_grep(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_grep\
    | not_cache_grep\
    | not_git_grep\
    | not_archive_grep\
    | not_temp_file_grep
}
```

3.21 build_and_checks_dependencies/lines_maps.libr.sh

```
#!/usr/bin/env bash
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```

```
# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
# ]
replace_non_ascii_spaces(){
  sed -e 's/\xc2\xa0/ /g'\
      -e 's/\xe2\x80\x80/ /g'\
      -e 's/\xe2\x80\x81/ /g'\
      -e 's/\xe2\x80\x82/ /g'\
      -e 's/\xe2\x80\x83/ /g'\
      -e 's/\xe2\x80\x84/ /g'\
      -e 's/\xe2\x80\x85/ /g'\
      -e 's/\xe2\x80\x86/ /g'\
      -e 's/\xe2\x80\x87/ /g'\
      -e 's/\xe2\x80\x88/ /g'\
      -e 's/\xe2\x80\x89/ /g'\
      -e 's/\xe2\x80\x8a/ /g'\
      -e 's/\xe2\x80\x8b/ /g'\
      -e 's/\xe2\x80\x8c/ /g'\
      -e 's/\xe2\x80\x8d/ /g'\
      -e 's/\xe2\x80\x8e/ /g'\
      -e 's/\xe2\x80\x8f/ /g'\
      -e 's/\xe2\x80\xa8/ /g'\
      -e 's/\xe2\x80\xa9/ /g'\
      -e 's/\xe2\x80\xaa/ /g'\
      -e 's/\xe2\x80\xab/ /g'\
      -e 's/\xe2\x80\xac/ /g'\
      -e 's/\xe2\x80\xad/ /g'\
      -e 's/\xe2\x80\xae/ /g'\
      -e 's/\xe2\x80\xaf/ /g'\
      -e 's/\xe2\x81\x9f/ /g'\
      -e 's/\xe2\x81\xa0/ /g'\
      -e 's/\xe2\x81\xa1/ /g'\
      -e 's/\xe2\x81\xa2/ /g'\
      -e 's/\xe2\x81\xa3/ /g'\
      -e 's/\xe2\x81\xa4/ /g'\
      -e 's/\xe2\x81\xa5/ /g'\
```

```
      -e 's/\xe2\x81\xa6/ /g'\
      -e 's/\xe2\x81\xa7/ /g'\
      -e 's/\xe2\x81\xa8/ /g'\
      -e 's/\xe2\x81\xa9/ /g'\
      -e 's/\xe2\x81\xaa/ /g'\
      -e 's/\xe2\x81\xab/ /g'\
      -e 's/\xe2\x81\xac/ /g'\
      -e 's/\xe2\x81\xad/ /g'\
      -e 's/\xe2\x81\xae/ /g'\
      -e 's/\xe2\x81\xaf/ /g'
}
```

3.22 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

overwrite_if_not_equal(){
  # $1=target_file_path
  # $2=temp_file_path
  # $3=keep_temp
  # $4=tree_mode : when the two files are the output of tree command,
  # we ignore if all the differences are only on directories timestamps.
```

```
33    # $5=--quiet : we even don't want the message of diff saying that
34    #              files are distinct
35    # Return 0 target was the same
36    #        1 target was different
37    #        2 target did not already exist
38    if [[ ! -f "$1" ]]; then
39      if [[ -n "$3" ]]; then
40        cp -p "$2" "$1"
41      else
42        mv "$2" "$1"
43      fi
44      return 2
45    fi
46    declare -i LFBFL_i_is_equal
47    if [[ -n "$4" ]]; then
48      # Not setting pipefail since the result would still be incorrect.
49      # shellcheck disable=SC2312
50      diff "$1" "$2"\
51        | grep --extended-regexp "^(>|<)"\
52        | grep --invert-match "/$"
53      LFBFL_i_is_equal=${PIPESTATUS[2]}
54      if [[ LFBFL_i_is_equal -eq 1 ]]; then
55        grep --only-matching "[^ ]*/$" "$1"\
56          > "overwrite_if_not_equal.file1.temp"
57        grep --only-matching "[^ ]*/$" "$2"\
58          > "overwrite_if_not_equal.file2.temp"
59        if [[ -n "$5" ]]; then
60          diff "overwrite_if_not_equal.file1.temp"\
61            "overwrite_if_not_equal.file2.temp"\
62            > /dev/null
63        else
64          diff "overwrite_if_not_equal.file1.temp"\
65            "overwrite_if_not_equal.file2.temp"
66        fi
67        LFBFL_i_is_equal=1-$?
68        rm "overwrite_if_not_equal.file1.temp"\
69          "overwrite_if_not_equal.file2.temp"
70      fi
71    else
72      if [[ -n "$5" ]]; then
73        diff --brief "$1" "$2" > /dev/null
74      else
75        diff --brief "$1" "$2"
76      fi
77      LFBFL_i_is_equal=1-$?
78    fi
```

```
  if [[ LFBFL_i_is_equal -eq 1 ]]; then
    if [[ -z "$3" ]]; then
      rm "$2"
    fi
    return 0
  fi
  if [[ -n "$3" ]]; then
    cp -p "$2" "$1"
  else
    mv "$2" "$1"
  fi
  return 1
}
```

3.23 build_and_checks_dependencies/python_black_complement.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
```

```
# https://github.com/psf/black/issues/2370
# And moreover, at some point, I saw the reverse rule
# in black or pylint to add an empty line...
# This regexp may give false positives,
# but that's not the end of the world.
check_no_empty_line_after_python_function_docstring(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking empty lines after Python function docstrings\n"
  pcre2grep -M --\
    $'def ([^"]|"(?!"))*"""([^"]|"(?!"))*"""\n\n(?!\s*def)'\
    **/*.py
}

python_black_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_no_empty_line_after_python_function_docstring\
```

```
    "${LFBFL_common_options[@]}"
}
```

3.24 build_and_checks_dependencies/python_isort_complement.libr.sh

```
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# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

check_collections_abc_place(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
```

```
42    get_work_directory_option "$@"
43    declare -a LFBFL_return_traps_stack
44    local LFBFL_previous_return_trap
45    init_return_trap
46    pushd_to_work_directory --trap-popd
47    can_continue_after_enhanced_pushd || return 1
48
49    [[ LFBFL_i_verbose -eq 1 ]]\
50      && printf "Checking import of _collections_abc is at the right place\n"
51
52    enhanced_set_shell_option pipefail --trap-unset
53
54    declare -r LFBFL_temp=".check_collections_abc_place.temp"
55    local LFBFL_file_path
56    declare -r LFBFL_s_files_paths=$(
57      find . -type f -name "*.py" -printf '%P\n'\
58      | relevant_find
59    )
60    if [[ -z "${LFBFL_s_files_paths}" ]]; then
61      [[ LFBFL_i_verbose -eq 1 ]]\
62        && printf "check_collections_abc_place: No file found.\n"
63      return
64    fi
65    declare -a LFBFL_arr_files_paths
66    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
67    readonly LFBFL_arr_files_paths
68    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
69      sed -Ez 's/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
70        "${LFBFL_file_path}"\
71        > "${LFBFL_file_path}${LFBFL_temp}"
72      overwrite_if_not_equal "${LFBFL_file_path}"\
73        "${LFBFL_file_path}${LFBFL_temp}"
74    done
75  }
76
77  python_isort_complement(){
78    # Options:
79    #   --verbose
80    #   --work-directory=""
81    declare -i LFBFL_i_verbose=0
82    get_verbose_option "$@"
83    local LFBFL_work_directory=""
84    get_work_directory_option "$@"
85    declare -a LFBFL_return_traps_stack
86    local LFBFL_previous_return_trap
87    init_return_trap
```

```
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_collections_abc_place "${LFBFL_common_options[@]}"
}
```

3.25 build_and_checks_dependencies/repository_data.libr.sh

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

# /!\ You need to activate extglob before parsing this file. /!\
# And I'm kind enough to activate it inside functions that need it below.

get_upgrade_venvs_time_interval_in_seconds(){
  # $1=data_file_path
```

```
  # Options:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap

  enhanced_set_bash_option extglob --trap-unset

  grep_variable "$1"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "You say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"

  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    +([0-9])) : ;;
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) printf "No wat for you?\n";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "I say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
}
```

3.26 build_and_checks_dependencies/shell_checks_complement.libr.sh

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

check_no_size_of_array_first_element(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  local LFBFL_s="Checking that shell scripts do not"
  LFBFL_s+=" contain wrong array size expansion."
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "%s\n" "${LFBFL_s}"

  pcre2grep -M -- $'\$\{#.*(\b|_)arr_(?!.*\[@\])' **/*.sh
}

check_no_misplaced_then(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
```

```
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_arg
  declare -i LFBFL_i_max_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-line-length=}
      LFBFL_i_max_line_length=$((LFBFL_arg))
      break
    fi
  done
  declare -i LFBFL_i_max_previous_line_length
  LFBFL_i_max_previous_line_length=$((LFBFL_i_max_line_length - 5))

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf\
        "Checking that shell scripts do not contain misplaced 'then'.\n"

  local LFBFL_regexp
  LFBFL_regexp="^[^\n]{1,${LFBFL_i_max_previous_line_length}}"
  LFBFL_regexp+="(?<=\]\];)"
  LFBFL_regexp+="(?<!^\s{0,${LFBFL_i_max_previous_line_length}}\]\];)"
  LFBFL_regexp+="\n\s*then"

  pcre2grep -M -- "${LFBFL_regexp}" **/*.sh
}

shell_checks_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
```

```
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_no_size_of_array_first_element "${LFBFL_common_options[@]}"
  check_no_misplaced_then "${LFBFL_common_options[@]}"
}
```

3.27 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php
/*
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.exec.php"
to
"split_score.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.exec.php" to
"split_line_at_most.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.php" to
```

```
"call_split_score_after_before_or_before_after.exec.php".
*/

require_once('./build_and_checks_dependencies/split_score.libr.php');

/*
php -r\
 $'require_once(\'./build_and_checks_dependencies/'\
$'split_score.libr.php\');'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'
*/

// The first cast "(int) generate_split_score" is to avoid false
// positive errors on possible XSS if output is unescaped.
echo(
  (int) generate_split_score(
    (bool) $argv[1], (int) $argv[2], explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
);
```

3.28 build_and_checks_dependencies/split_score.libr.php

```
<?php
/*
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```

```
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".
*/

function generate_split_score(
  bool $b_after_before,
  int $i_max_length,
  array $a_delimiter_strings_domain,
) : Closure {
  return function(
    string $s_delimiter_string,
    int $i_cut_position,
    bool $b_is_cut_after,
  ) use ($b_after_before, $i_max_length, $a_delimiter_strings_domain)
  : int {
    if(
      !in_array(
        $s_delimiter_string, $a_delimiter_strings_domain, true
      )
    ){
      // I do not handle regexps for the moment.
      // I'm too brainfucked. I just want to finish this feature fast
      // for HTML and Tex/PDF listings generation with HTML and Tex on
      // at most 70 characters.
      return 0;
    }
    if($b_is_cut_after === $b_after_before){
      // When $b_after_before,
      // always way better to cut after '/' than before it.
      // When !$b_after_before (aka before_after),
      // always way better to cut before '/' than after it.
      return 1 + $i_max_length + $i_cut_position;
    }
    return 1 + $i_cut_position;
  };
}
```

3.29 build_and_checks_dependencies/strings_functions.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
```

```
#
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# I think I created a file strings_functions.sh or renamed to
# strings_functions.sh at some point, before committing.
# Hence, someone must have changed or reversed the name of this file
# to string_functions.sh at some point.
# This file was renamed from "string_functions.sh" to
# "strings_functions.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

split_line_at(){
  # $1=line
  # $2=position : number of characters in split_line_at_result_beginning
  declare -g split_line_at_result_beginning="${1:0:$2}"
  declare -g split_line_at_result_end="${1:$2}"
  # printf "%s\n" "$split_line_at_result_beginning"
  # printf "%s\n" "$split_line_at_result_end"
}

#-----------------------------------------------------------------------------
```

```
#-----------------------------------------------------------------------
# Properties for the score for a possible split/cut in the string between
# two characters.
#-----------------------------------------------------------------------
# Some preliminary remarks:
# a) If the character before the split position is a forbidden previous
#   character, the score is -1.
#   (Usually, either we have no or one forbidden previous character that is
#   a backslash.)
# b) If the previous character is not forbidden,
#   and either the character before the position is among the delimiters,
#   or the character after the position is among the delimiters,
#   then the score is computed according to the delimiter found,
#   the position, and the fact that the position is after or before the
#   delimiter.
# c) If two consecutive delimiters are found in the string,
#   the maximum score between the one after the first delimiter
#   and before the second delimiter is taken for the score between the
#   two delimiters.
# d) Otherwise, by default,
#   the score for cutting/splitting at some position is 0.
# e) A substring is given by the goal length given for the resulting
#   strings/lines after split.
# f) When some scores are equal,
#   we split at the rightmost position with maximum score in the substring.
# g) Remark f) implies that below we do not need to consider strict
#    inequalities for the score.
# h) The score function SF(position) which will dictate our choice for
#   split/cut position is in reality
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   )
#   but at the cost of expliciting its internal and adding verbosity,
#   we can see that the score really depends on the following arguments
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   )
#   If it is needed,
#   then matched_delimiter_before or matched_delimiter_after is NULL.
#   Somehow previous_character can be seen as position.previous_character,
#   and next_character can be seen as position.next_character.
#   We can see that the previous or next character without
#   knowing that it matches a delimiter is irrelevant.
#   The fact that there is 1 or more delimiters may be interesting.
#   The fact that there is 0 or more forbidden previous characters
```

```
#   may be interesting, but we haven't integrated it in the split_score
#   functions.
#   SF1a when the previous character is not forbidden
#   needs a score function SF2a with the following parameters
#   SF2a(position, current_character, delimiter, is_after):
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   ) = max(
#     For all delimiter in delimiters,
#     SF2a(position, previous_character, delimiter, TRUE),
#     SF2a(position, next_character, delimiter, FALSE),
#   )
#   =  max(
#     For all delimiter in delimiters,
#     For all couple C in {
#       (position.previous_character, TRUE),
#       (position.next_character, FALSE)
#     }
#     SF2a(position, C.1, delimiter, C.2),
#   )
#   Similarly,
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2b(position, matched_delimiter_before, TRUE),
#     SF2b(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2b has the following parameters
#   SF2b(cut_position, matched_delimiter, is_after).
#   We have also:
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2c(position-1, matched_delimiter_before, TRUE),
#     SF2c(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2c has the following parameters
#   SF2c(character_position, matched_delimiter, is_after).
#   current_character is only used to check if it is equal to the
#   delimiter, otherwise the score is zero;
#   but when expressing properties it makes sense to compare after
#   and before the same character position.
```

```
#   position, delimiter, and is_after may all three impact the score
#   once the current character matches the delimiter.
#   Position is an integer from 1 to goal length.
#   Cutting at position 0 would be stupid,
#   and moreover there would be no previous character.
#   Since we ask for a split, the current length is above goal length,
#   and there is a next character at the last considered position.
#   So, in search of properties that may have an impact on the algorithm,
#   we discard the characters to only keep matched delimiters.
#   Let us use p1 and p2 for position1 and position2,
#   mdb1 and mdb2 for matched_delimiter_before1, etc.,
#   mda1 and mda2 for matched_delimiter_after1, etc.,
#   pcif for previous_character_is_forbidden, etc.,
#   md1 and md2 for matched_delimiter1, etc.,
#   ia1 and ia2 for is_after1, etc.
#
#   What we consider as properties are of 3 forms.
#   The first form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all mdb1 in delimiters union {NULL},
#     for all mdb2 in delimiters union {NULL},
#     for all mda1 in delimiters union {NULL},
#     for all mda2 in delimiters union {NULL},
#     RelationP(p1, p2) AND RelationM(mdb1, mdb2, mda1, mda2)
#     implies RelationS(
#       SF1b(p1, mdb1, mda1, false),
#       SF1b(p2, mdb2, mda2, false),
#     )
#   There are many choices for RelationP, RelationM and RelationS.
#   We can apply the without loss of generality argument only once when
#   breaking symmetry between equivalent properties.
#   Breaking symmetry on another relation that RelationS first
#   is a challenge that we do not tackle.
#   Indeed RelationS is the only relation where all variables appear.
#   And breaking symmetry on RelationP would not allow to break symmetry
#   on RelationM freely since some variables are correlated by the
#   suffixes 1 or 2.
#   But some choices for some relation may allow to break symmetry on
#   another relation.
#   Breaking symmetry on the result, there are only 2 choices for
#   RelationS:
#   RS= : SF1b(p1, mdb1, mda1, false) = SF1b(p2, mdb2, mda2, false),
#   RS<= : SF1b(p1, mdb1, mda1, false) <= SF1b(p2, mdb2, mda2, false).
#   But then there are 5+1 choices for RelationP:
#   RP1 : p1 = p2,
#   RP2 : p1 < p2,
```

```
#   RP3 : p1 <= p2,
#   RP4 : p1 > p2,
#   RP5 : p1 >= p2,
#   plus the choice that p1 and p2 are not necessarily related
#   (empty RP or RPe).
#   We can add RP6 : p1 = p2 + 1 and RP7 : p2 = p1 + 1, since it implies
#   some equalities between matched delimiters.
#   Note that  p1 = p2 + i or p2 = p1 + i with i >= 2 brings no additional
#   logical relations compared to p1 < p2 or p2 < p1;
#   it only yields subslices of other properties and open a slightly silly
#   door on infinity.
#   If we have RS=, by symmetry we can consider only RP1, RP2, RP3 and RPe,
#   and maybe RP6.
#   Note that RSe or empty RS would have no interest at all.
#   There are many choices for RelationM:
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is NULL is not
#   interesting at all since both scores are 0 or -1.
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is not NULL
#   implies that second score is always larger, that's not an additional
#   property, that is a theorem.
#   Hence, at least one of mdb1 or mda1, resp. mdb2 or mda2,
#   must (have the possibility to) be non-NULL.
#   If we start listing the remaining relations, we have:
#   1) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#   RMA1A2
#   2) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 = mda2
#   RMA1A2=
#   3) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 != mda2 -> This case has no meaning:
#     If we only know they are different, we cannot say anything on
#     properties related to delimiters that are different unless the score
#     doesn't depend on the delimiter, and that case is already handled by
#     1). The same remark applies to some other choices.
#     The only thing that could be said would be if we had an order on
#     delimiters scores that are uniform for the position, or the score
#     before some delimiter would always be =, >=, <=, etc. the score
#     before or after another one. So it would only make sense with RPe,
#     and mda1 <= mda2 for example, but that would trivially imply RS<=.
#     It would end up with max(mda1, mdb1) <= max(mda2, mdb2) where max
#     ignores NULL value. You do not deduce or specify anything new here.
#   4) mdb1 is not NULL AND mdb2 is not NULL AND mda1 is NULL
#     AND mda2 is NULL
```

```
#   RMB1B2
#   and so on. So if we use the short notation RM followed by A1, resp. A2,
#   resp. B1, resp. B2, when mda1, resp. mda2, resp. mdb1, resp. mdb2,
#   is not null, we have the following possibilities:
#   RMA1A2, RMB1B2, RMA1B2, RMA2B1, all 4 with variant with = like RMA1A2=,
#   RMA1A2B1, RMA1A2B2, RMA1B1B2, RMA2B1B2, all 4 (RMXXYYZZ) with variants
#   with suffixes _1 (XX = YY), _2 (XX = ZZ), _3 (YY = ZZ),
#   and _4 (XX = YY = ZZ)
#   for equalities on the 1st, 2nd or 3rd couple or between the three
#   variables by transitivity,
#   RMALL where none is null, with suffixes A1A2 (A1 = A2), etc.
#   RMALL_A1A2, RMALL_A1B1, RMALL_A1B2, RMALL_A2B1, RMALL_A2B2, RMALL_B1B2,
#   for one edge;
#   RMALL_A1A2B1, RMALL_A1A2B2, RMALL_A1B1B2, RMALL_A2B1B2, one triangle;
#   RMALL_A1A2_B1B2, RMALL_A1B1_A2B2, RMALL_A1B2_A2B1, 2 disjoint edges;
#   RMALL_ALL square with diagonals.
#   So we see that we listed at least 39 possibilities only for RM.
#   We did not list yet the possibilities where the variables not present
#   in some equality can be NULL or not NULL, etc.
#   We see 9 of them : RMe (40), and the following 8:
#   RMA1A2+ = RMA1A2= OR RMA1A2B1_1 OR RMA1A2B2_1 OR RMALL_A1A2
#   RMB1B2+ = RMB1B2= OR RMA1B1B2_3 OR RMA2B1B2_3 OR RMALL_B1B2
#   RMA1B2+ = RMA1B2= OR RMA1A2B2_2 OR RMA1B1B2_2 OR RMALL_A1B2
#   RMA2B1+ = RMA2B1= OR RMA1A2B1_3 OR RMA2B1B2_1 OR RMALL_A2B1
#   RMA1A2B1+ = RMA1A2B1_4 OR RMALL_A1A2B1
#   RMA1A2B2+ = RMA1A2B2_4 OR RMALL_A1A2B2
#   RMA1B1B2+ = RMA1B1B2_4 OR RMALL_A1B1B2
#   RMA2B1B2+ = RMA2B1B2_4 OR RMALL_A2B1B2
#   We don't do complete analysis at the moment.
#   If we wanted to filter out some possibilities,
#   we would keep only the RM possibilities, among the 38,
#   that are symmetric between 1 and 2:
#   RMA1A2, RMB1B2, RMA1A2=, RMB1B2=,
#   RMALL, RMALL_A1A2, RMALL_A2B2, RMALL_A1A2_B1B2, RMALL_A1B1_A2B2,
#   RMALL_ALL. Still 10 possibilities.
#   We have 1 (RS=) times 5 times 10 + 1 (RS<=) times 8 times 10 = 130
#   combinations (life is too short ;) :) );
#   if we keep 48 RM possibilities, there is 624 combinations,
#   but we did not analyze totally which RM relations are useless.
#
#   The second form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
```

```
#     implies RelationS(
#       SF2b(p1, md1, ia1)
#       SF2b(p2, md2, ia2)
#     )
#   The third form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationPP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2c(p1, md1, ia1)
#       SF2c(p2, md2, ia2)
#     )
#   This time the combinatorics is much more pleasant :).
#   As above, we have RS= and RS<=, RPe, RP1 to RP7;
#   For RelationMM, assume that the possible delimiters correspond to
#   integer numbers 1 to i; hence md1, resp. md2, can be seen as an integer
#   in [0,i], either 0 for NULL or the delimiter index.
#   Clearly, if md1 = k in [1,i] and md2 = l in [1,i], then the only
#   information we can use is k = l or k != l.
#   If we allow that md1 != 0 and md2 = 0,
#   then both RS= and RS<= are false.
#   Thus RMMe is not possible.
#   If we allow that md1 = 0 and md2 != 0, then RS<= is always true,
#   whilst RS= is false.
#   Thus properties with RS<= can trivially be extended with
#   RMM-"allow-md1-0", not useful to explicit this for the moment.
#   If both md1 = 0 and md2 = 0, then both scores are 0 and we also learn
#   nothing more.
#   Thus we can focus on md1 and md2 are not NULL,
#   and are equal RMM= or not RMM!=. It seems that a property with RMM!=
#   could be extended to a property RMM=or!=/RMMNotNULL.
#   and finally RIA0 to RIA8 where TRUE = 1, FALSE = 0 and the digit is
#   3 * ia1 + ia2, and both variables can be 2 if they can be either TRUE
#   or FALSE. Wrong ! Not pleasant ! XD
#   So for RIA, we keep either: empty RIAe (RIA8),
#   ia1 = ia2 RIA= (RIA0 or RIA4),
#   ia1 = ia2 = TRUE RIA=T (RIA4), ia1 = ia2 = FALSE RIA=F (RIA0),
#   ia1 != ia2 RIA!= (RIA1 or RIA3), ia1 != ia2 = FALSE RIA!=T1 (RIA3),
#   ia1 != ia2 = TRUE RIA!=T2 (RIA1),
#   ia1 = TRUE (RIA5), ia1=FALSE (RIA2),
#   ia2 = TRUE (RIA7), ia2=FALSE (RIA6),
#   even less pleasant XD : 11 possibilities.
#   In total, we have now 1 (RS=) times 5 times 2 times 11
#   + 1 (RS<=) times 8 times 2 times 11 = 286
```

```
#   combinations (life is even shorter ;) :) ).
#   We only listed below properties of the third form.
# i) By the previous remark, there is a good chance that we missed some
#   possible property.
#   We hope we did found the properties that had a simple meaning,
#   in particular related to the algorithm where either you
#   loop on the string forward and keep last position with maximum score,
#   or you loop backward and abort as soon as possible as soon you know
#   that you found some position at the end that must have maximum score.
# j) These properties are "boolean" flags (integers that are powers of 2).
#   We first put the exact integers,
#   then wondered about switching everything to exponentiation 2**i,
#   then we took the solution that minimizes the character length between
#   [1-9][0-9]* and 2**i, then reverted to initial before committing.
#   Most efficient interpreted code is what we did first,
#   if there is no opcode cache for constants.
#   Most safe code is when the language hides the details and handles
#   families of flags,
#   but beware if some flag goes into the database.
#   Most readable code is a mix: 1 is easier to read than 2**0 ;) :P,
#   or $((2<<0))...
#   What I would love to see in many languages is a flag type, as a subtype
#   of integers, flag32, flag64 for C for example.
#   I think there is no such thing because sooner or later you play with
#   pseudo-flags that are combination of flags, hence integers.
#   But just for the declaration to let the type checker check that the
#   integer is a power of 2: flag32 MY_FLAG = 4;
#   Or better flag32 MY_FLAG = 2; as an equivalent of int32 MY_FLAG = 1<<2;
#   Another fact is that practice makes perfect, so working with the powers
#   of 2 written out in base 10 gets you accustomed to them.
#   Safe code should be favored since (pre-)compilation optimizes it,
#   but for this project, I prefer to keep the integers in decimal and
#   check that it doubles when reading my listing, because I'm familiar
#   with that and the powers of 2 doesn't go beyond what I can check fast
#   enough in my brain.
# k) Some of the properties below are union of other properties,
#   and are here to simplify not only code but understanding.
#   split_score_properties_logical_closure() function below checks that
#   all flags that are consequences of already activated flags are set.
# l) We use SSP_ prefix for SPLIT_SCORE_PROPERTY_.
# Some of these remarks are repeated below in the corresponding functions,
# and some remarks are added just after the constants.
#-----------------------------------------------------------------------
# Any score after a character that matches a delimiter
# is larger than or equal to
# any score before a character that matches a delimiter.
```

```
# RPPe RMMNotNull RIA!=T2 RS<=
declare -gir SSP_ALWAYS_LARGER_AFTER=1

# Any score before a character that matches a delimiter
# is larger than or equal to
# any score after a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T1 RS<=
declare -gir SSP_ALWAYS_LARGER_BEFORE=2

# The score doesn't depend on the delimiter that is found.
# RPP1 RMMNotNull RIA= RS=
declare -gir SSP_DELIMITER_UNIFORM=4

# Whenever a character that matches a delimiter is found,
# the score after is equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is equal to
# the score before the character.
# RPP1 RMM= RIA!= RS=
declare -gir SSP_EQUAL_AFTER_OR_BEFORE=8
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_EQUAL_AFTER_OR_BEFORE=

# Whenever a character that matches a delimiter is found,
# the score after is larger than or equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is larger than or equal to
# the score before the character.
# RPP1 RMM= RIA!=T2 RS<=
declare -gir SSP_LARGER_AFTER=16
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_AFTER=

# Whenever a character that matches a delimiter is found,
# the score before is larger than or equal to the score after.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score before the character
# is larger than or equal to
# the score after the character.
# RPP1 RMM= RIA!=T1 RS<=
```

```
declare -gir SSP_LARGER_BEFORE=32
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_BEFORE=

# The score after, resp. before, a character that matches a delimiter
# doesn't decrease when the position in the string increase.
# (RPP2 or RPP3) RMM= RIA= RS<=
declare -gir SSP_POSITION_NOT_DECREASING=64

# The score after a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=T RS<=
declare -gir SSP_POSITION_NOT_DECREASING_AFTER=128

# The score before a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=F RS<=
declare -gir SSP_POSITION_NOT_DECREASING_BEFORE=256

# The score after, resp. before, a character that matches a delimiter
# doesn't depend on the position in the string.
# (RPP2 or RPP3) RMM= RIA= RS=
declare -gir SSP_POSITION_UNIFORM=512

# The score after a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=T RS=
declare -gir SSP_POSITION_UNIFORM_AFTER=1024

# The score before a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=F RS=
declare -gir SSP_POSITION_UNIFORM_BEFORE=2048

# For any delimiter,
# any score after a character that matches this delimiter
# is equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!= RS=
# <=> RPPe RMM= RIAe RS=
declare -gir SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE=4096

# For any delimiter,
# any score after a character that matches this delimiter
# is larger than or equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!=T2 RS<=
```

```
declare -gir SSP_SAME_DELIMITER_LARGER_AFTER=8192

# For any delimiter,
# any score before a character that matches this delimiter
# is larger than or equal to
# any score after a character that matches this delimiter.
# RPPe RMM= RIA!=T1 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_BEFORE=16384

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!= RS=
declare -gir SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE=32768

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is larger than or equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T2 RS<=
declare -gir SSP_SAME_POSITION_LARGER_AFTER=65536

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score before a character at this position that matches a delimiter
# is larger than or equal to
# the score after a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T1 RS<=
declare -gir SSP_SAME_POSITION_LARGER_BEFORE=131072

# For all possible split/cut positions, the score is equal as long as
# at least one of the previous and next characters is a delimiter.
# Or equivalently:
# Any score after a character that matches a delimiter
# is equal to
# any score before a character that matches a delimiter.
# is_after doesn't change anything.
# So we cut on the last position adjacent to a delimiter without
# forbidden previous character (edge case, on last position).
# RPPe RMMNotNull RIAe RS=
declare -gir SSP_UNIFORM=262144
# declare -gir SSP_ALWAYS_EQUAL_AFTER_OR_BEFORE=1

```

```
#-----------------------------------------------------------------------
# m) Note that a property telling that the score is always the same
#   even if the current character isn't among delimiters is not needed,
#   since, in that case, a null split score command will split at the goal
#   length.
#-----------------------------------------------------------------------
#-----------------------------------------------------------------------

split_score_properties_logical_closure(){
  # $1=split_score_properties
  # Since this code is not performance critical,
  # we followed lexicographic order on constants,
  # instead of an order to minimize the operations.
  # It is also safer to avoid forgetting some relation.
  declare -i LFBFL_i_result=$(($1))
  declare -i LFBFL_i_result_old=0
  declare -i LFBFL_i_all_flags=$((
    SSP_ALWAYS_LARGER_AFTER
    | SSP_ALWAYS_LARGER_BEFORE
    | SSP_DELIMITER_UNIFORM
    | SSP_EQUAL_AFTER_OR_BEFORE
    | SSP_LARGER_AFTER
    | SSP_LARGER_BEFORE
    | SSP_POSITION_NOT_DECREASING
    | SSP_POSITION_NOT_DECREASING_AFTER
    | SSP_POSITION_NOT_DECREASING_BEFORE
    | SSP_POSITION_UNIFORM
    | SSP_POSITION_UNIFORM_AFTER
    | SSP_POSITION_UNIFORM_BEFORE
    | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
    | SSP_SAME_DELIMITER_LARGER_AFTER
    | SSP_SAME_DELIMITER_LARGER_BEFORE
    | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
    | SSP_SAME_POSITION_LARGER_AFTER
    | SSP_SAME_POSITION_LARGER_BEFORE
    | SSP_UNIFORM
  ))
  until [[ LFBFL_i_result_old -eq LFBFL_i_result ]]; do
    LFBFL_i_result_old=$((LFBFL_i_result))
    if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_LARGER_AFTER
        | SSP_SAME_DELIMITER_LARGER_AFTER
        | SSP_SAME_POSITION_LARGER_AFTER
      ))
```

```
555       fi
556       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE)); then
557         LFBFL_i_result=$((
558           LFBFL_i_result
559           | SSP_LARGER_BEFORE
560           | SSP_SAME_DELIMITER_LARGER_BEFORE
561           | SSP_SAME_POSITION_LARGER_BEFORE
562         ))
563       fi
564       if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING)); then
565         LFBFL_i_result=$((
566           LFBFL_i_result
567           | SSP_POSITION_NOT_DECREASING_AFTER
568           | SSP_POSITION_NOT_DECREASING_BEFORE
569         ))
570       fi
571       if ((LFBFL_i_result & SSP_POSITION_UNIFORM)); then
572         LFBFL_i_result=$((
573           LFBFL_i_result
574           | SSP_POSITION_UNIFORM_AFTER
575           | SSP_POSITION_UNIFORM_BEFORE
576           | SSP_POSITION_NOT_DECREASING
577           | SSP_POSITION_NOT_DECREASING_AFTER
578           | SSP_POSITION_NOT_DECREASING_BEFORE
579         ))
580       fi
581       if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER)); then
582         LFBFL_i_result=$((
583           LFBFL_i_result
584           | SSP_POSITION_NOT_DECREASING_AFTER
585         ))
586       fi
587       if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
588         LFBFL_i_result=$((
589           LFBFL_i_result
590           | SSP_POSITION_NOT_DECREASING_BEFORE
591         ))
592       fi
593       if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
594         LFBFL_i_result=$((
595           LFBFL_i_result
596           | SSP_EQUAL_AFTER_OR_BEFORE
597           | SSP_LARGER_AFTER
598           | SSP_LARGER_BEFORE
599           | SSP_SAME_DELIMITER_LARGER_AFTER
600           | SSP_SAME_DELIMITER_LARGER_BEFORE
```

```
601         ))
602       fi
603       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
604         LFBFL_i_result=$((
605           LFBFL_i_result
606           | SSP_LARGER_AFTER
607         ))
608       fi
609       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
610         LFBFL_i_result=$((
611           LFBFL_i_result
612           | SSP_LARGER_BEFORE
613         ))
614       fi
615       if ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
616         LFBFL_i_result=$((
617           LFBFL_i_result
618           | SSP_DELIMITER_UNIFORM
619           | SSP_EQUAL_AFTER_OR_BEFORE
620           | SSP_LARGER_AFTER
621           | SSP_LARGER_BEFORE
622           | SSP_SAME_POSITION_LARGER_AFTER
623           | SSP_SAME_POSITION_LARGER_BEFORE
624         ))
625       fi
626       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
627         LFBFL_i_result=$((
628           LFBFL_i_result
629           | SSP_LARGER_AFTER
630         ))
631       fi
632       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
633         LFBFL_i_result=$((
634           LFBFL_i_result
635           | SSP_LARGER_BEFORE
636         ))
637       fi
638       if ((LFBFL_i_result & SSP_UNIFORM)); then
639         LFBFL_i_result=$((LFBFL_i_all_flags))
640         break
641       fi
642       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER))\
643       && ((
644         LFBFL_i_result & (
645           SSP_ALWAYS_LARGER_BEFORE
646           | SSP_EQUAL_AFTER_OR_BEFORE
```

```
        | SSP_LARGER_BEFORE
        | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
        | SSP_SAME_DELIMITER_LARGER_BEFORE
        | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
        | SSP_SAME_POSITION_LARGER_BEFORE
      )
    )); then
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE))\
    && ((
      LFBFL_i_result & (
        SSP_EQUAL_AFTER_OR_BEFORE
        | SSP_LARGER_AFTER
        | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
        | SSP_SAME_DELIMITER_LARGER_AFTER
        | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
        | SSP_SAME_POSITION_LARGER_AFTER
      )
    )); then
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_LARGER_AFTER)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_LARGER_AFTER
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
    && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_LARGER_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
```

```
693     && ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
694       LFBFL_i_result=$((LFBFL_i_all_flags))
695       break
696     fi
697     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
698     && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
699       LFBFL_i_result=$((
700         LFBFL_i_result
701         | SSP_ALWAYS_LARGER_AFTER
702       ))
703     fi
704     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
705     && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
706       LFBFL_i_result=$((
707         LFBFL_i_result
708         | SSP_ALWAYS_LARGER_BEFORE
709       ))
710     fi
711     if ((LFBFL_i_result & SSP_LARGER_AFTER))\
712     && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
713       LFBFL_i_result=$((
714         LFBFL_i_result
715         | SSP_EQUAL_AFTER_OR_BEFORE
716       ))
717     fi
718     if ((LFBFL_i_result & SSP_LARGER_AFTER))\
719     && ((
720       LFBFL_i_result & (
721         SSP_POSITION_UNIFORM
722         | SSP_POSITION_UNIFORM_AFTER
723       )
724     )); then
725       LFBFL_i_result=$((
726         LFBFL_i_result
727         | SSP_SAME_DELIMITER_LARGER_AFTER
728       ))
729     fi
730     if ((LFBFL_i_result & SSP_LARGER_BEFORE))\
731     && ((
732       LFBFL_i_result & (
733         SSP_POSITION_UNIFORM
734         | SSP_POSITION_UNIFORM_BEFORE
735       )
736     )); then
737       LFBFL_i_result=$((
738         LFBFL_i_result
```

```
        | SSP_SAME_DELIMITER_LARGER_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_AFTER))\
    && ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_POSITION_NOT_DECREASING
      ))
    fi
    if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
    && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_AFTER
      ))
    fi
    if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
    && ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_POSITION_UNIFORM
      ))
    fi
    if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
      LFBFL_i_result=$((
```

```
785         LFBFL_i_result
786         | SSP_ALWAYS_LARGER_AFTER
787       ))
788     fi
789     if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE))\
790     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
791       LFBFL_i_result=$((
792         LFBFL_i_result
793         | SSP_ALWAYS_LARGER_BEFORE
794       ))
795     fi
796     if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE))\
797     && ((
798       LFBFL_i_result & (
799         SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
800         | SSP_SAME_POSITION_LARGER_AFTER
801         | SSP_SAME_POSITION_LARGER_BEFORE
802       )
803     )); then
804       LFBFL_i_result=$((LFBFL_i_all_flags))
805       break
806     fi
807     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
808     && ((
809       LFBFL_i_result & (
810         SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
811         | SSP_SAME_POSITION_LARGER_BEFORE
812       )
813     )); then
814       LFBFL_i_result=$((LFBFL_i_all_flags))
815       break
816     fi
817     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
818     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
819       LFBFL_i_result=$((
820         LFBFL_i_result
821         | SSP_ALWAYS_LARGER_AFTER
822       ))
823     fi
824     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
825     && ((
826       LFBFL_i_result & (
827         SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
828         |  SSP_SAME_POSITION_LARGER_AFTER
829       )
830     )); then
```

```
      LFBFL_i_result=$((LFBFL_i_all_flags))
      break
    fi
    if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_ALWAYS_LARGER_BEFORE
      ))
    fi
    if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER))\
    && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
      LFBFL_i_result=$((
        LFBFL_i_result
        | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
      ))
    fi
  done
  declare -gi i_split_score_properties_logical_closure_result=$((
    LFBFL_i_result
  ))
}

split_score_simple(){
  # $1=after_before
  # $2=delimiters_strings_domain concatenated characters/delimiters
  # $3=delimiter_string a single character
  # $4=is_cut_after
  declare -gi i_split_score_result
  if [[ "$2" != *$3* ]]; then
    i_split_score_result=0
    return
  fi
  if [[ "$4" == "$1" ]]; then
    i_split_score_result=2
    return
  fi
  i_split_score_result=1
}

get_split_score_simple(){
  # $1=after_before
  # $2=max_length
  # $3=delimiters_strings_domain
  declare -g get_split_score_result="split_score_simple $1 '$3'"
  declare -gi i_get_split_score_result2
```

```
  # i_get_split_score_result2=$((11-$1*4))
  # if [[ $1 -eq 1 ]]; then
  #   i_get_split_score_result2=7
  # else
  #   i_get_split_score_result2=11
  # fi
  i_get_split_score_result2=$((
    SSP_DELIMITER_UNIFORM
    + SSP_POSITION_UNIFORM
    + SSP_LARGER_BEFORE
    + (
      $1 * (
        SSP_LARGER_AFTER
        - SSP_LARGER_BEFORE
      )
    )
  ))
  # shellcheck disable=SC2248
  split_score_properties_logical_closure ${i_get_split_score_result2}
  i_get_split_score_result2=$((
    i_split_score_properties_logical_closure_result
  ))
}

get_split_score_exec(){
  declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
  SPLIT_SCORE_EXEC+="/split_score.exec.php"
}

split_score(){
  # $1=after_before
  # $2=max_length
  # $3=delimiters_strings_domain concatenated characters/delimiters
  # $4=delimiter_string a single character
  # $5=cut_position
  # $6=is_cut_after
  get_split_score_exec
  declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
  declare -gi i_split_score_result
  i_split_score_result=$(eval "${LFBFL_command}")
}

# This function is currently not called,
# hence we need this shellcheck disable.
# shellcheck disable=SC2034
get_split_score(){
```

```
  # $1=after_before
  # $1=max_length
  # $2=delimiters_strings_domain
  declare -g get_split_score_result="split_score $1 $2 '$3'"
  declare -gi i_get_split_score_result2
  # i_get_split_score_result2=$((9-$1*4))
  i_get_split_score_result2=$((
    SSP_DELIMITER_UNIFORM
    + SSP_LARGER_BEFORE
    + (
      $1 * (
        SSP_LARGER_AFTER
        - SSP_LARGER_BEFORE
      )
    )
  ))
  # shellcheck disable=SC2248
  split_score_properties_logical_closure ${i_get_split_score_result2}
  i_get_split_score_result2=$((
    i_split_score_properties_logical_closure_result
  ))
}

split_line_at_most(){
  # $1=line
  # $2=max_length of beginning result string
  # $3=split_score_command
  #   negative score means do not consider as a possible split
  # $4=split_score_command_properties :
  #   These properties are flags.
  #   If someday some property is not a flag,
  #   then create split_score_command_properties2.
  #   Old values to get a simple idea without reading everything at the
  #   beginning of file:
  #   - null or 0 no property;
  #   - 1 split score doesn't depend on which delimiter matches;
  #   - 2 split score doesn't depend on the current position;
  #   - 4 any split score after is always larger
  #     than any split score before;
  #   - 8 any split score before is always larger
  #     than any split score after;
  # $5=forbidden_previous_character
  # $6=split_score_command_delimiter_strings_domain
  #   optimisation if small domain
  #   - this argument can be "null", then we will use $3 only instead,
  #     in that case, only mono-character strings are considered,
```

```
#   - this argument can be an array of characters,
#     optimize accordingly,
#   - this argument can be an array of strings,
#     optimize accordingly,
#   - this argument can be an array of regexps,
#     optimize accordingly.
#     Not sure an array of regexps makes more sense than a single
#     one. Maybe they could be both cases with corresponding
#     optimizations. Same for a single character or substring.
#
# Note that we're always looking for the last position to split
# among the positions with maximum score;
# if some position gets a score from 2 delimiters,
# its score is always the maximum of the two corresponding scores.
declare -g split_line_at_most_result_start
declare -g split_line_at_most_result_end
declare -A LFBFL_positions
# printf "%s %s %s %s\n" "$1" "$2" "$3" "$4"
# For my use case in Bash scripts, I will need only an array of
# characters. See get_split_score().
if [[ -n "$6" ]]; then
  printf $'split_line_at_most() $6 NOT IMPLEMENTED YET\n'
fi
if [[ $2 -ge ${#1} ]]; then
  split_line_at_most_result_start=$1
  split_line_at_most_result_end=""
  return
fi
# We know that ${#1} > $2, hence ${#1} - 1 >= $2
# Edge cases:
# There is always at least one character that is overlength,
# but we need to look only at the score before that character that is
# overlength, the score after that character is invalid,
# whatever ${#1} - 1 = $2, or ${#1} - 1 > $2.
# That first character that is overlength has index $2.
# The cut position after that character has index $2 + 1.
# Looping from character position going from 0 to $2, we need to drop
# score before character 0 and drop score after character $2.
# We do this with forbidden score -1.
declare -ir LFBFL_i_max=$(($2))
declare -i LFBFL_i
declare -i LFBFL_j
local LFBFL_current_char

# At the beginning, each position has score 0.
LFBFL_positions["0"]="-1"
```

```
1015    for ((LFBFL_i = 1; LFBFL_i <= LFBFL_i_max; ++LFBFL_i)) do
1016      LFBFL_positions["${LFBFL_i}"]="0"
1017    done
1018    LFBFL_positions["${LFBFL_i}"]="-1"
1019
1020    # If we have a forbidden previous character,
1021    # we mark the forbidden positions with -1.
1022    if [[ -n "$5" ]]; then
1023      for ((LFBFL_i = 0; LFBFL_i < LFBFL_i_max;)) do
1024        LFBFL_current_char="${1:${LFBFL_i}:1}"
1025        ((++LFBFL_i))
1026        if [[ "${LFBFL_current_char}" == "$5" ]]; then
1027          LFBFL_positions["${LFBFL_i}"]="-1"
1028        fi
1029      done
1030    fi
1031
1032    declare -r LFBFL_sort_command="sort --numeric-sort"
1033
1034    if (($4 & SSP_DELIMITER_UNIFORM))\
1035    && (($4 & SSP_LARGER_AFTER))\
1036    && (($4 & SSP_POSITION_NOT_DECREASING_AFTER)); then
1037      for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1038        LFBFL_current_char="${1:${LFBFL_i}:1}"
1039        eval "$3 '${LFBFL_current_char}' 1"
1040        # needs j=i+1 printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1041        if [[ i_split_score_result -ge 1 ]]; then
1042          LFBFL_j=$((LFBFL_i + 1))
1043          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1044          if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
1045            LFBFL_positions["${LFBFL_j}"]=$(
1046              max "${LFBFL_sort_command}"\
1047                "${LFBFL_positions["${LFBFL_j}"]}"\
1048                "${i_split_score_result}"
1049            )
1050            break
1051          fi
1052        fi
1053        eval "$3 '${LFBFL_current_char}' 0"
1054        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1055        if [[ i_split_score_result -ge 1 ]]; then
1056          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1057          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1058            LFBFL_positions["${LFBFL_i}"]=$(
1059              max "${LFBFL_sort_command}"\
1060                "${LFBFL_positions["${LFBFL_i}"]}"\
```

```
1061                "${i_split_score_result}"
1062            )
1063          fi
1064        fi
1065      done
1066    elif (($4 & SSP_DELIMITER_UNIFORM))\
1067    && (($4 & SSP_LARGER_BEFORE))\
1068    && (($4 & SSP_POSITION_NOT_DECREASING_BEFORE)); then
1069      for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1070        LFBFL_current_char="${1:${LFBFL_i}:1}"
1071        # We still need to break in second if, after computing score after,
1072        # because SSP_LARGER_BEFORE implies large inequality instead of
1073        # strict inequality.
1074        # And since the algorithm is to take the largest position with
1075        # maximum score, there is a possibility that we'll break on a found
1076        # score before but for splitting on a score after.
1077        eval "$3 '${LFBFL_current_char}' 1"
1078        # needs j=i+1 printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1079        if [[ i_split_score_result -ge 1 ]]; then
1080          LFBFL_j=$((LFBFL_i + 1))
1081          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_j}"
1082          if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
1083            LFBFL_positions["${LFBFL_j}"]=$(
1084              max "${LFBFL_sort_command}"\
1085                "${LFBFL_positions["${LFBFL_j}"]}"\
1086                "${i_split_score_result}"
1087            )
1088          fi
1089        fi
1090        eval "$3 '${LFBFL_current_char}' 0"
1091        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1092        if [[ i_split_score_result -ge 1 ]]; then
1093          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1094          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1095            LFBFL_positions["${LFBFL_i}"]=$(
1096              max "${LFBFL_sort_command}"\
1097                "${LFBFL_positions["${LFBFL_i}"]}"\
1098                "${i_split_score_result}"
1099            )
1100            break
1101          fi
1102        fi
1103      done
1104    else
1105      for ((LFBFL_i = 0; LFBFL_i <= LFBFL_i_max;)) do
1106        LFBFL_current_char="${1:${LFBFL_i}:1}"
```

```
1107        eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
1108        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1109        if [[ i_split_score_result -ge 1 ]]; then
1110          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1111          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1112            LFBFL_positions["${LFBFL_i}"]=$(
1113              max "${LFBFL_sort_command}"\
1114                "${LFBFL_positions["${LFBFL_i}"]}"\
1115                "${i_split_score_result}"
1116            )
1117          fi
1118        fi
1119        ((++LFBFL_i))
1120        eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 1"
1121        # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1122        if [[ i_split_score_result -ge 1 ]]; then
1123          # printf "%s|%s\n" "${i_split_score_result}" "${LFBFL_i}"
1124          if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
1125            LFBFL_positions["${LFBFL_i}"]=$(
1126              max "${LFBFL_sort_command}"\
1127                "${LFBFL_positions["${LFBFL_i}"]}"\
1128                "${i_split_score_result}"
1129            )
1130          fi
1131        fi
1132      done
1133    fi
1134    declare -ir LFBFL_i_max_score=$(
1135      max 'sort --numeric-sort' "${LFBFL_positions[@]}"
1136    )
1137    local LFBFL_value
1138    declare -i LFBFL_i_best_position
1139    for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
1140      if [[ LFBFL_i -eq 0 ]]; then
1141        printf $'/!\\split_line_at_most bug: no split position found./!\\\n'
1142      fi
1143      local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
1144      # printf "%s %s\n" "${LFBFL_i} ${LFBFL_value}"
1145      if [[ "${LFBFL_value}" == "${LFBFL_i_max_score}" ]]; then
1146        LFBFL_i_best_position=$((LFBFL_i))
1147        break
1148      fi
1149    done
1150    # shellcheck disable=SC2248
1151    split_line_at "$1" ${LFBFL_i_best_position}
1152    split_line_at_most_result_start="${split_line_at_result_beginning}"
```

```
  split_line_at_most_result_end="${split_line_at_result_end}"
  # printf "%s\n" "$split_line_at_most_result_start"
  # printf "%s\n" "$split_line_at_most_result_end"
}

split_last_line(){
  # $1=new_lines
  # $2=prefix
  # $3=max_length
  # $4=suffix
  # $5=split_score_command
  # $6=split_score_command_properties
  # $7=forbidden_previous_character ('\' usually)
  # Options:
  #   --verbose
  # In general, a prefix/suffix can be a required line prefix/suffix
  # for all final lines like a comment prefix,
  # or it can be a continuation line prefix/suffix applied only
  # when splitting a line in 2 lines,
  # or it can be a prefix/suffix for the string made of all the lines.
  # Clearly, there is no gain to couple handling a prefix/suffix for
  # the string made of all the lines in this function.
  # Since we add a prefix/suffix when splitting,
  # it is easy to see that we do not need to distinguish between
  # required line prefix/suffix and continuation line prefix/suffix.
  # The $2 and $4 arguments are given as concatenation of the
  # required and continuation line prefixes/suffixes as needed by the
  # user.
  # Thus lines arguments should be given already prefixed
  # and suffixed.
  # This is logic, since it would be cumbersome to handle prefixing
  # or not already splitted lines in this function
  # (see repeated_split_last_line).
  # The given max_length is the goal.
  # This function adapts the effective max position for the split by
  # taking into account the length of the suffix.
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap

  enhanced_set_shell_option pipefail --trap-unset

  declare -g split_last_line_result="$1"
  declare -ir LFBFL_i_overlength=$(($3 + 1))
```

```
1199    declare -ir LFBFL_i_length2=$(($3 - ${#4}))
1200    # shellcheck disable=SC2248
1201    get_overlength_regexp ${LFBFL_i_overlength}
1202
1203    # Testing if some line is too long.
1204    printf "%s" "$1"\
1205      | grep -q "${get_overlength_regexp_result}";
1206    if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
1207      return
1208    fi
1209
1210    declare -r LFBFL_start=$(
1211      printf "%s" "$1"\
1212      | head --lines=-1
1213    )
1214    # printf "start: %s\n" "${LFBFL_start}"
1215    declare -r LFBFL_last_line=$(
1216      printf "%s" "$1"\
1217      | tail --lines=1
1218    )
1219    # printf "last_line: %s\n" "${LFBFL_last_line}"
1220    split_last_line_result=""
1221    if [[ -n "${LFBFL_start}" ]]; then
1222      split_last_line_result="${LFBFL_start}"
1223      # Last line-return of LFBFL_start seems dropped by $() and not ="${}".
1224      # printf "%s %s\n" "${#split_last_line_result}"
1225      #   "${split_last_line_result}"
1226      split_last_line_result+=$'\n'
1227      # something="${split_last_line_result}"
1228      # printf "%s %s\n" "${#something}" "${something}"
1229    fi
1230    if [[ -n "$5" ]]; then
1231      # shellcheck disable=SC2248
1232      split_line_at_most "${LFBFL_last_line}" ${LFBFL_i_length2}\
1233        "$5" "$6" "$7"
1234      split_last_line_result+="${split_line_at_most_result_start}"
1235      split_last_line_result+="$4"
1236      split_last_line_result+=$'\n'
1237      split_last_line_result+="$2${split_line_at_most_result_end}"
1238    else
1239      split_last_line_result+="${LFBFL_last_line:0:${LFBFL_i_length2}}"
1240      split_last_line_result+="$4"
1241      split_last_line_result+=$'\n'
1242      split_last_line_result+="$2${LFBFL_last_line:${LFBFL_i_length2}}"
1243    fi
1244    # printf "result: %s\n" "${split_last_line_result}"
```

```
}

repeated_split_last_line(){
  # See split_last_line for the first arguments.
  # $8=final_suffix to handle when the end of the original string
  #    is the suffix.
  declare -g repeated_split_last_line_result="$1"
  local LFBFL_result="$1x"
  until
    [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
  do
    split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
      "$4" "$5" "$6" "$7"
    LFBFL_result="${repeated_split_last_line_result}"
    repeated_split_last_line_result="${split_last_line_result}"
    # printf "%s\n" "${repeated_split_last_line_result}"
  done
  if [[ -n "$4" && -n "$8" ]]; then
    declare -ir LFBFL_i_offset=$((-${#4}))
    local LFBFL_lines_end
    LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_i_offset}}"
    if [[ "${LFBFL_lines_end}" == "$4" ]]; then
      repeated_split_last_line_result+="$8"
    fi
  fi
}
```

3.30 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
```

```
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#
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# This file was renamed from "too_long_code_lines.sh"
# to "too_long_code_lines.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

get_overlength_regexp(){
  declare -g get_overlength_regexp_result=".\{$1\}"
  # declare -g get_overlength_regexp_result="^.\{$1\}"
  # declare -g get_overlength_regexp_result=".\{$1\}\$"
  # First one seems to be faster by a few percents when testing like this:
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}' COPYING; done
  # time for ((i = 0; i < 1000; ++i)); do grep '^.{70}' COPYING; done
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}$' COPYING; done
}

too_long_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  local LFBFL_previous_return_trap
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_arg
  declare -i LFBFL_i_max_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
```

```
64        LFBFL_arg=${LFBFL_arg#--max-line-length=}
65        LFBFL_i_max_line_length=$((LFBFL_arg))
66        break
67      fi
68    done
69
70    declare -ir LFBFL_i_overlength=$((LFBFL_i_max_line_length+1))
71    # shellcheck disable=SC2248
72    get_overlength_regexp ${LFBFL_i_overlength}
73
74    enhanced_set_shell_option pipefail --trap-unset
75
76    get_COMMON_TEXT_FILES_GLOB_PATTERNS
77    local LFBFL_pattern
78    local LFBFL_long_line
79    local LFBFL_file_path
80    local LFBFL_line
81    local LFBFL_extension
82    local LFBFL_base_name
83    local LFBFL_s_long_lines
84    declare -a LFBFL_arr_long_lines
85    for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
86      if [[ LFBFL_i_verbose -eq 1 ]]; then
87        printf "too_long_code_lines: Iterating on pattern: %s.\n"\
88          "${LFBFL_pattern}"
89      fi
90      # shellcheck disable=SC2248
91      LFBFL_s_long_lines=$(
92        find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
93        | relevant_find\
94        | not_license_find\
95        | xargs grep --line-number --with-filename\
96          "${get_overlength_regexp_result}"
97      )
98      if [[ -z "${LFBFL_s_long_lines}" ]]; then
99        [[ LFBFL_i_verbose -eq 1 ]]\
100         && printf "too_long_code_lines: No file found with long lines.\n"
101       continue
102     fi
103     mapfile -t LFBFL_arr_long_lines <<< "${LFBFL_s_long_lines}"
104     for LFBFL_long_line in "${LFBFL_arr_long_lines[@]}"; do
105       LFBFL_file_path=${LFBFL_long_line%%:*} # Drop long ':*' suffix
106       # Remove file_path and line_number
107       LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
108       LFBFL_line=${LFBFL_line#*:} # Drop short '*:' prefix
109       # LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
```

```
      LFBFL_extension=${LFBFL_file_path##*.} # Drop long '*.' prefix
      LFBFL_base_name=${LFBFL_file_path%.*} # Drop short '.*' suffix
      if [[ "${LFBFL_extension}" == "html" ]]; then
        if [[ -f "${LFBFL_base_name}.md" ]]; then
          continue
        fi
        if grep --quiet --fixed-strings -- "${LFBFL_line}"\
          "./build_and_checks_variables/temp/files_listing.html.sub"
        then
          continue
        fi
        printf "%s\n" "${LFBFL_long_line}"
      elif [[ "${LFBFL_extension}" == "md" ]]; then
        if ! [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
          printf "%s\n" "${LFBFL_long_line}"
        fi
      else
        printf "%s\n" "${LFBFL_long_line}"
      fi
    done
  done
}
```

3.31 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
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```

```
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".

source "overwrite_if_not_equal.libr.sh"

update_common_build_and_checks(){
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_i_verbose

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && printf "pipefail option activated\n"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  cp "./${LFBFL_common_file_name}.tpl"\
    "./${LFBFL_common_file_name}.temp"

  declare -ar LFBFL_files_names=(
    "build_md_from_printable_md.exec.sh"
    "build_dependencies_notes.exec.php"
    "check_shell_scripts_beginnings.libr.sh"
    "check_shell_scripts_indentation.libr.sh"
    "check_URLs.libr.sh"
    "common_options.libr.sh"
    "comparisons.libr.sh"
    "create_PDF.exec.sh"
    "generate_from_template.libr.sh"
    "get_common_text_glob_patterns.libr.sh"
    "grammar_and_spelling_check.libr.sh"
    "licenses_templates/build_licenses_templates.exec.sh"
    "licenses_templates/license_file_header_AGPLv3+.tpl"
    "licenses_templates/license_file_header_GPLv3+.tpl"
    "licenses_templates/license_file_header_LGPLv3+.tpl"
    "lines_counts.libr.sh"
    "lines_filters.libr.sh"
    "lines_maps.libr.sh"
    "update_or_check_files_names_listing.exec.sh"
    "overwrite_if_not_equal.libr.sh"
```

```
68      "python_black_complement.libr.sh"
69      "python_isort_complement.libr.sh"
70      "repository_data.libr.sh"
71      "shell_checks_complement.libr.sh"
72      "split_score.exec.php"
73      "split_score.libr.php"
74      "strings_functions.libr.sh"
75      "too_long_code_lines.libr.sh"
76    )
77
78    local LFBFL_file_name
79    local LFBFL_file_sha512
80    local LFBFL_file_sha512_all
81    local LFBFL_base_file_name
82    for LFBFL_file_name in "${LFBFL_files_names[@]}"; do
83      LFBFL_file_sha512=$(
84        sha512sum "./${LFBFL_file_name}"\
85        | cut -f1 -d' '
86      )
87      LFBFL_file_sha512_all="LFBFL_correct_sha512="
88      LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
89      LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
90      LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
91      LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
92      LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
93      LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
94      sed -i\
95        "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
96        "./${LFBFL_common_file_name}.temp"
97    done
98
99    overwrite_if_not_equal "./${LFBFL_common_file_name}"\
100     "./${LFBFL_common_file_name}.temp"
101
102   declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
103   cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"
104
105   LFBFL_file_sha512=$(
106     sha512sum "./${LFBFL_common_file_name}"\
107     | cut -f1 -d' '
108   )
109   LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
110   LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
111   LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
112   LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
113   LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
```

```
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed -i\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}

update_common_build_and_checks "$@"
```

3.32 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
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# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
# to "update_or_check_files_names_listing.sh"?
```

```
33
34 LFBFL_subdir="build_and_checks_dependencies"
35 # shellcheck source=common_options.libr.sh
36 source "./${LFBFL_subdir}/common_options.libr.sh"
37 # shellcheck source=lines_filters.libr.sh
38 source "./${LFBFL_subdir}/lines_filters.libr.sh"
39 # shellcheck source=strings_functions.libr.sh
40 source "./${LFBFL_subdir}/strings_functions.libr.sh"
41
42 update_or_check_files_names_listing(){
43   # Options:
44   #   --verbose
45   #   --work-directory=""
46   declare -i LFBFL_i_verbose=0
47   get_verbose_option "$@"
48   local LFBFL_work_directory=""
49   get_work_directory_option "$@"
50   declare -a LFBFL_return_traps_stack
51   local LFBFL_previous_return_trap
52   init_return_trap
53   pushd_to_work_directory --trap-popd
54   can_continue_after_enhanced_pushd || return 1
55
56   declare -i LFBFL_i_write=0
57   if [[ "$*" == *--write* ]]; then
58     LFBFL_i_write=1
59   fi
60   readonly LFBFL_i_write
61
62   declare -i LFBFL_i_append=0
63   if [[ "$*" == *--append* ]]; then
64     LFBFL_i_append=1
65   fi
66   readonly LFBFL_i_append
67
68   enhanced_set_shell_option pipefail --trap-unset
69
70   LFBFL_subdir2="build_and_checks_variables"
71   local LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
72   readonly LFBFL_listing
73   local LFBFL_data_file_name="./${LFBFL_subdir2}/repository_data.txt"
74   readonly LFBFL_data_file_name
75
76   declare -i LFBFL_i_max_line_length
77   grep_variable "${LFBFL_data_file_name}" max_line_length\
78     --result-variable-prefix="LFBFL_i_"
```

```
79
80    if [[ LFBFL_i_write -eq 1 ]]; then
81      : > "${LFBFL_listing}"
82    fi
83    # Remove line returns that are here to keep lines short.
84    sed -Ez 's/(.)\\\n/\1/Mg' "${LFBFL_listing}"\
85      > "${LFBFL_listing}.temp"
86    # shopt -s dotglob was needed at some point but I don't see why now.
87    # shellcheck disable=SC2248
88    get_split_score_simple 1 ${LFBFL_i_max_line_length} /
89    declare -r LFBFL_split_score_command="${get_split_score_result}"
90    declare -i LFBFL_i_split_score_command_properties
91    LFBFL_i_split_score_command_properties=${i_get_split_score_result2}
92    readonly LFBFL_i_split_score_command_properties
93    local LFBFL_suffix=\\ # instead of '\' to avoid shellcheck SC1003
94    readonly LFBFL_suffix
95    # There is always a $'\n' final suffix from >>.
96    # But when origin string ends with the suffix we must add something.
97    declare -r LFBFL_final_suffix=$'\n\\'
98
99    local LFBFL_s_files_paths
100   LFBFL_s_files_paths=$(
101     find . -type f -printf '%P\n'\
102     | relevant_find\
103     | sort
104   )
105   # printf "%s\n" "${LFBFL_s_files_paths}"
106   declare -a LFBFL_arr_files_paths
107   if [[ -n "${LFBFL_s_files_paths}" ]]; then
108     if [[ LFBFL_i_verbose -eq 1 ]]; then
109       printf\
110         "update_or_check_files_names_listing: Some files to process.\n"
111     fi
112     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
113   fi
114   # printf "%s\n" "${LFBFL_arr_files_paths[@]}"
115
116   local LFBFL_file_path
117   local LFBFL_base_file_name
118   for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
119     # if [[ LFBFL_i_verbose -eq 1 ]]; then
120     #   printf\
121     #     "update_or_check_files_names_listing: Maybe ignored file: %s\n"\
122     #     "${LFBFL_file_path}"
123     # fi
124     git check-ignore -q "${LFBFL_file_path}" && continue
```

```
    LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
      || continue
    if [[ "${LFBFL_base_file_name}" == *.md ]]; then
      if [[ -f "${LFBFL_file_path}.tpl" ]]; then
        continue
      fi
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "update_or_check_files_names_listing: Non-ignored file: %s\n"\
        "${LFBFL_file_path}"
    fi
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_file_path}"\
      ""\
      ${LFBFL_i_max_line_length}\
      "${LFBFL_suffix}"\
      "${LFBFL_split_score_command}"\
      ${LFBFL_i_split_score_command_properties}\
      ""\
      "${LFBFL_final_suffix}"
    if [[ LFBFL_i_write -eq 1 ]]; then
      # shellcheck disable=SC2001
      printf "%s\n" "${repeated_split_last_line_result}"\
        >> "${LFBFL_listing}"
      continue
    fi
    if grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
        "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;
    then
      :
    else
      printf "The file %s is not listed in %s.\n"\
        "${LFBFL_file_path}"\
        "${LFBFL_listing}"
      if [[ LFBFL_i_append -eq 1 ]]; then
        # shellcheck disable=SC2001
        printf "%s\n" "${repeated_split_last_line_result}"\
          >> "${LFBFL_listing}"
      fi
    fi
  done

  LFBFL_s_files_paths=$(cat "${LFBFL_listing}.temp")
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
```

```
  fi

  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ "${LFBFL_file_path}" != '//'* ]] || continue
    if [[ ! -f "${LFBFL_file_path}" ]]; then
      printf "The non-file %s is listed in %s.\n"\
        "${LFBFL_file_path}"\
        "${LFBFL_listing}"
    fi
  done

  rm "${LFBFL_listing}.temp"

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    cat "${LFBFL_listing}"
  fi
}

update_or_check_files_names_listing "$@"
```

3.33 build_and_checks_variables/.gitignore

```
temp/
upgrade_venvs_ts
```

3.34 build_and_checks_variables/bandit.ini

```
# FILE: .bandit
[bandit]
```

3.35 build_and_checks_variables/bandit_baseline.json

```
{"results":[]}
```

3.36 build_and_checks_variables/black.toml

```
[tool.black]
line-length = 75
```

3.37 build_and_checks_variables/dependencies_data.json

```
[
  { "dependency_name": "Bash", "URLs": [
      "https://www.gnu.org/software/bash/",
      "https://tiswww.case.edu/php/chet/bash/bashtop.html",
      "https://savannah.gnu.org/projects/bash/"
```

```
6     ] },
7     { "dependency_name": "GNU coreutils", "URLs": [
8         "https://www.gnu.org/software/coreutils/",
9         "https://savannah.gnu.org/projects/coreutils/"
10    ] },
11    { "dependency_name": "uutils coreutils", "URLs": [
12        "https://uutils.github.io/coreutils/",
13        "https://github.com/uutils/coreutils"
14    ] },
15    { "dependency_name": "diff (GNU diffutils)", "URLs": [
16        "https://www.gnu.org/software/diffutils/",
17        "https://savannah.gnu.org/projects/diffutils/"
18    ] },
19    { "dependency_name": "find, xargs (GNU findutils)", "URLs": [
20        "https://www.gnu.org/software/findutils/",
21        "https://savannah.gnu.org/projects/findutils/"
22    ] },
23    { "dependency_name": "grep (GNU grep)", "URLs": [
24        "https://www.gnu.org/software/grep/",
25        "https://savannah.gnu.org/projects/grep/"
26    ] },
27    { "dependency_name": "sed (GNU sed)", "URLs": [
28        "https://www.gnu.org/software/sed/",
29        "https://savannah.gnu.org/projects/sed/"
30    ] },
31    { "dependency_name": "ShellCheck", "URLs": [
32        "https://www.shellcheck.net/",
33        "https://github.com/koalaman/shellcheck"
34    ] },
35    { "dependency_name": "GNU Wget", "URLs": [
36        "https://www.gnu.org/software/wget/",
37        "https://savannah.gnu.org/projects/wget/"
38    ] },
39    { "dependency_name": "more of util-linux", "URLs": [
40        "https://www.kernel.org/pub/linux/utils/util-linux/",
41        "https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/",
42        {
43          "URL": "https://github.com/util-linux/util-linux",
44          "comment": "official?"
45        }
46    ] },
47    { "dependency_name": "pandoc", "URLs": [
48        "https://pandoc.org/",
49        "https://github.com/jgm/pandoc"
50    ] },
51    { "dependency_name": "pdftex/pdflatex", "URLs": [
```

```
52          "https://www.tug.org/applications/pdftex/",
53          "https://tug.org/texlive/svn/"
54      ] },
55      { "dependency_name": "tree", "URLs": [
56          "https://oldmanprogrammer.net/source.php?dir=projects/tree",
57          "https://gitlab.com/OldManProgrammer/unix-tree",
58          "https://github.com/Old-Man-Programmer/tree"
59      ] },
60      { "dependency_name": "git", "URLs": [
61          "https://git-scm.com/about",
62          "https://github.com/git/git"
63      ] },
64      { "dependency_name": "pylint", "URLs": [
65          "https://www.pylint.org/",
66          "https://pylint.readthedocs.io/en/stable/",
67          "https://github.com/pylint-dev/pylint",
68          "https://pypi.org/project/pylint/"
69      ] },
70      { "dependency_name": "notify-send", "URLs": [
71          "https://gitlab.gnome.org/GNOME/libnotify/",
72          "https://gnome.pages.gitlab.gnome.org/libnotify/"
73      ] },
74      { "dependency_name": "black", "URLs": [
75          "https://github.com/psf/black",
76          "https://black.readthedocs.io/en/stable/",
77          "https://pypi.org/project/black/"
78      ] },
79      { "dependency_name": "isort", "URLs": [
80          "https://pycqa.github.io/isort/",
81          "https://github.com/pycqa/isort/",
82          "https://pypi.org/project/isort/"
83      ] },
84      { "dependency_name": "mypy", "URLs": [
85          "https://mypy-lang.org/",
86          "https://mypy.readthedocs.io/en/stable/",
87          "https://github.com/python/mypy",
88          "https://pypi.org/project/mypy/"
89      ] },
90      { "dependency_name": "pip", "URLs": [
91          "https://pip.pypa.io/en/stable/",
92          "https://github.com/pypa/pip",
93          {
94            "URL": "https://pypi.org/project/pip/",
95            "comment": "(egg and chicken ;) )"
96          }
97      ] },
```

```
  { "dependency_name": "ruff", "URLs": [
      "https://docs.astral.sh/ruff/",
      "https://github.com/astral-sh/ruff",
      "https://pypi.org/project/ruff/"
  ] },
  { "dependency_name": "bandit", "URLs": [
      "https://bandit.readthedocs.io/en/latest/",
      "https://github.com/PyCQA/bandit",
      "https://pypi.org/project/bandit/"
  ] },
  { "dependency_name": "phpmd", "URLs": [
      "https://phpmd.org/",
      "https://github.com/phpmd/phpmd",
      "https://packagist.org/packages/phpmd/phpmd"
  ] },
  { "dependency_name": "php", "URLs": [
      "https://www.php.net/",
      "https://github.com/php/php-src"
  ] },
  { "dependency_name": "tex", "URLs": [
      "https://ctan.org/tex-archive/systems/knuth/dist/tex"
  ] },
  { "dependency_name": "pcregrep, pcre2grep", "URLs": [
      "https://www.pcre.org/",
      "https://pcre2project.github.io/pcre2/",
      "https://github.com/PCRE2Project/pcre1",
      "https://github.com/PCRE2Project/pcre2",
      "https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
  ] },
  { "dependency_name": "python3", "URLs": [
      "https://www.python.org/",
      "https://docs.python.org/3/"
  ] }
]
```

3.38 build_and_checks_variables/dependencies_notes.txt

```
Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

```

```
uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
https://savannah.gnu.org/projects/findutils/

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
https://pandoc.org/
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

```

```
git can be found at the following URLs:
https://git-scm.com/about
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
https://github.com/psf/black
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/
https://github.com/pypa/pip
https://pypi.org/project/pip/ (egg and chicken ;) )

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
https://pypi.org/project/ruff/

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpmd can be found at the following URLs:
```

```
https://phpmd.org/
https://github.com/phpmd/phpmd
https://packagist.org/packages/phpmd/phpmd

php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
https://github.com/PCRE2Project/pcre2
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

python3 can be found at the following URLs:
https://www.python.org/
https://docs.python.org/3/

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

3.39 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
This file is part of DevOrSysAdminScripts library.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed in the hope
that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

```

```
You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<meta name="generator" content="DevOrSysAdminScripts">
<style>
.title{
  margin: 2ex auto;
  text-align: center;
}
.titlemain{
  margin: 1ex 2ex 2ex 1ex;
}
.titlerest{
  font-size: 1.17em;
  margin: 0ex 2ex;
}
table{
  border-collapse: collapse;
}
td{
  padding: 0;
}
pre{
  text-align: left;
  margin-left: 0ex;
  margin-right: auto;
}
pre.numbered_lines:before{
  background: snow;
  border-right: 1px black solid;
  color: #010101;
  content: attr(line-numbers);
  float: left;
  padding: 0 1ch;
  text-align: right;
}
blockquote{
  margin-left: 4ex;
  margin-right: 4ex;
  text-align: left;
```

```
}
ol.level1, ol.level2{
  list-style-type: none;
  /*
  list-style-type: decimal;
  Previous rule is shorter and more efficient if you need only level1.
  But you don't have an accessible counter if you use decimal.
  */
}
ol.level1{
  counter-reset: level1;
}
ol.level2{
  counter-reset: level2;
}
ol.level1 li:before{
  content: counter(level1) ". ";
  counter-increment: level1;
}
ol.level2 li:before{
  content: counter(level1) "." counter(level2) " ";
  counter-increment: level2;
}
</style>
<title>@repository_name@</title>
</head>
<body >
<table class="title">
  <tr>
    <td style="padding:1ex">
      <h1 class="titlemain">@repository_name@</h1>
      <h2 class="titlerest">
        @author_full_name@
        <br>
        <a href="@author_website@">
          <span style="font-family:monospace">
          @author_website@
          </span>
        </a>
        <br>
        <a href="mailto:@author_email@">
          <span style="font-family:monospace">
          @author_email@
          </span>
        </a>
      </h2>
```

```
    </td>
  </tr>
</table>

<blockquote class="abstract">
  <span style="font-weight:bold">Abstract:</span>
  @abstract@
</blockquote>
<p>Current version: @current_date@</p>
<p>Current number of commits: @number_of_commits@</p>
<p>Current git SHA1: @current_git_SHA1@</p>
<p>
Code lines: @number_of_lines@
</p>


<h2>Contents</h2>

<ol class="level1">
  <li><a href="#section1">Files Tree</a></li>
  <li><a href="#section2">Notes on licenses and dependencies</a></li>
  <li><a href="#section3">Listing of files</a><br>
    <ol class="level2">
      @files_lis@
    </ol>
  </li>
</ol>


<h2 id="section1">1 Files tree</h2>
<pre class="verbatim">
@current_tree_light@
</pre>
<pre class="verbatim">
@current_tree@
</pre>


<h2 id="section2">2 Notes on licenses and dependencies</h2>

<p>
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.
</p>
```

```

<p>
The text of the licenses is available at:
<a href="https://www.gnu.org/licenses/"><span
  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
>@repository_git_url@</span>.
</p>

<div>
<p>
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, pip, ruff;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
<li> python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
</ul>
</div>

<p>
```

```
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

@dependencies_notes@


<h2 id="section3">3 Listing of files</h2>

@files_listing_HTMLPreInput@

<p>
@acknowledgments@
</p>

<script>
"use strict";
const pres = Array.from(
  document.getElementsByClassName('numbered_lines')
);
pres.forEach(
  (pre) => {
    const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
    let numbers = "";
    for(let i = 1; i <= nb; ++i){
      numbers += i.toString();
      if(i < nb){ numbers += "\n "; }
    }
    pre.setAttribute('line-numbers', numbers);
  }
);
</script>
</html>
```

3.40 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts library.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it under the terms
% of the GNU Lesser General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
```

```
%
% DevOrSysAdminScripts is distributed in the hope
% that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
% "./latex"
% to
% "./build_and_checks_variables".

\documentclass{article}

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{subfigure}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage[pdftex]{hyperref}
\usepackage{tikz}
\usepackage{caption}
\usepackage[round]{natbib}
\usepackage{fancyhdr}
\usepackage{amsfonts}
\usepackage{times}
\usepackage{ifpdf}
\usepackage{latexsym}
\usepackage{graphicx}
\usepackage{enumerate}
\usepackage{pmboxdraw}
\usepackage{fancyvrb}
\usepackage[french,english]{babel}

% *** les environnements ***
%\theoremstyle{break}
\newtheorem{definition}{Definition}[section]
```

```
\newtheorem{proposition}[definition]{Proposition}
\newtheorem{theorem}[definition]{Theorem}
\newtheorem{lemma}[definition]{Lemma}
\newtheorem{corollary}[definition]{Corollary}
\newtheorem{remark}[definition]{Remark}
\newtheorem{openproblem}[definition]{Open problem}

\begin{document}

\author{
  @author_full_name@\\
  \url{@author_website@}\\
  \href{mailto:@author_email@}{@author_email@}
}
\title{@repository_name@}

\maketitle
\begin{abstract}
@abstract@
\end{abstract}

Current version: @current_date@

Current number of commits: @number_of_commits@

Current git SHA1: @current_git_SHA1@

Code lines: @number_of_lines@

\tableofcontents
\newpage

\section{Files tree}
\label{section:tree}

\begin{verbatim}
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

\section{Notes on licenses and dependencies}
\label{section:notes_on}

```

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

Here is a listing of the dependencies of the code in this repository:
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, pip, ruff;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
\item python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
\end{itemize}

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

@dependencies_notes@

\section{Listing of files}
```

```
\label{section:listing}

@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

3.41 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts library.
//
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
// either version 3 of the License,
// or (at your option) any later version.
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
// Blasé20240517Résultat.txt
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_dependencies_notes.exec.php
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
```

```
build_and_checks_dependencies/common_build_and_checks.exec.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
build_and_checks_dependencies/generate_from_template.libr.sh
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spelling_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/repository_data.libr.sh
build_and_checks_dependencies/shell_checks_complement.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/dependencies_data.json
build_and_checks_variables/dependencies_notes.txt
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpmd_baseline.xml
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
```

```
build_and_checks_variables/repository_data.txt.tpl
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
// build_and_checks_variables/upgrade_venvs_ts
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
// EncoreBlasé20240517-20240518.txt
git_helpers/git_checks.libr.sh
git_helpers/post-commit
git_helpers/post_commit_renaming_comment.libr.sh
git_helpers/pre-commit
IMAP_transfert.py
LiensUtiles.txt
// Uncomment some files in miscellaneous to do some testing.
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123456789012345
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345678901234\
56
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345678901234\
56\
\
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345678901234\
5\
\
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345678901234\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123\
\
// miscellaneous/\
a_long_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
e_that_it_works.txt
miscellaneous/funny.sh
miscellaneous/spampty_data.txt
miscellaneous/string_escaping.html
miscellaneous/string_escaping.xhtml
// miscellaneous/\
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_\
```

```
you_can_see_that_it_works.txt
// README.html
// README.md
README.md.tpl
string_escaping.libr.php
unicode.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

3.42 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD
license.

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
```

3.43 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
  />
  <violation rule="PHPMD\Rule\Design\NpathComplexity"
    file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
  />
  <violation rule="PHPMD\Rule\Design\LongMethod"
    file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
  />
  <violation rule="PHPMD\Rule\CleanCode\ElseExpression"
    file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
  />
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="build_and_checks_dependencies/build_dependencies_notes.exec.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="string_escaping.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="string_escaping.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\LongMethod"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="unicode.libr.php"
  />
```

```
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\CleanCode\MissingImport"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="unicode.libr.php"
  />
  <violation rule="PHPMD\Rule\Design\NpathComplexity"
    file="unicode.libr.php"
  />
</phpmd-baseline>
```

3.44 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
[tool.pylint.reports]
output-format = "colorized"
```

3.45 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+

author_email=Laurent.Lyaudet@gmail.com
author_full_name=Laurent Frédéric Bernard François Lyaudet
author_last_name=Lyaudet
author_name=Laurent Lyaudet
author_website=https://lyaudet.eu/laurent/

acknowledgments=Merci Dieu !
acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !

Great day! Now I allow myself lines of 75 characters instead of 70 :).
I checked how many characters fit in the PDF file generated by tex with
default paper size,
when using verbatim environment with line numbers for source code listing.
With a screen capture in GIMP,
I mesured 197px between two-digits line numbers and the left border of
the page.
(It is less than 197px for three-digits or more line numbers.)
The same distance from the right border yields 75 characters allowed.
Since when printing the HTML file, the number of allowed characters in
<pre> tags is way larger, 75 is the right limit with the tools I use.

max_line_length=75

latex_command=pdflatex

grammar_and_spelling_checker_command=


Why one venv for each Python dependency ?
```

```
Because no language of my knowledge did it right to handle all
possible dependency conflicts.
So in case of such a conflict, one specialized venv can get you out
of any problem.
The best solution would have been to allow install of different
versions of the same package, and have a mechanism:
- first intent, use the latest available version of a package among
those that satisfy the declared dependency constraints of the package
that has this dependency;
- second intent, check a dependency overload configuration file
with tree structure: if calling python package is A version 1.1.1
with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
you could have along that another rule saying that when called
directly B 2.2.2 should use C 4.4.4;
and even more complicated:
if calling python package is A version 1.1.1 with dependency B and D
both themselves with dependency C, use B 2.2.2, D 0.1.0,
and C 3.3.3 for B, and C 4.4.4 for D;
once you've seen that it can never be more complicated than that once
the same version number guarantees that the code is the same,
you create a config language "à la CSS" for cascading dependencies ;)
XD fit into JSON the dumbest possible way and you stop bothering
about dependency conflicts for the rest of history of informatics.
{
  "A":{
    "1.1.1": {
      "B": {
        "2.2.2": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "3.3.3" : {}
          }
        }
      },
      "D": {
        "0.1.0": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "4.4.4" : {}
          }
        }
      }
    }
  },
  "B": {
    "2.2.2": {
```

```
      "C": {
        // Dumb, you did not found it by yourself? Failed exam.
        "4.4.4" : {}
      }
    }
  }
}
Wonderful someone rediscovered nested structures XD
Seriously so many engineers or computer scientists are dumb to cry.

bandit_venv=../venv_bandit
black_venv=../venv_black
isort_venv=../venv_isort
mypy_venv=../venv_mypy
pylint_venv=../venv_pylint
ruff_venv=../venv_ruff


upgrade venvs can be auto or ask, any other value is interpreted
as ask.

upgrade_venvs=ask


time interval can be 0 to always upgrade or ask to upgrade,
or a number at most bash max int
https://superuser.com/questions/1030122/
what-is-the-maximum-value-of-a-numeric-bash-shell-variable
it should be enough to simulate never "upgrade or ask for upgrade"
unless you claim to live many billions years ;)
If it is wat, this is a random int XD (see RANDOM).
Note that the probability that it upgrades your dependencies is not
1/2, it increases over time, and is 1 after slightly
less than one day.
If it is watyouwant?, see SRANDOM XD.

upgrade_venvs_time_interval_in_seconds=wat


For the Nu HTML, CSS, and SVG checker.

vnu_jar_path=../jar_venv/vnu.jar

Not fully implemented yet, it will only warn that vnu jar was not
updated since some time.
"wat" works, but "watyouwant?" joke was not repeated ;).
```

```

upgrade_vnu_jar_time_interval_in_seconds=604800

use_vnu_for_xhtml=false
```

3.46 build_and_checks_variables/repository_data.txt.tpl

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.
The goal of this template is to list all possible variables.

repository_name=
repository_web_url=
repository_git_url=

abstract=
abstract=

license=
license2=

author_email=
author_full_name=
author_last_name=
author_name=
author_website=

acknowledgments=
acknowledgments=

max_line_length=

latex_command=

grammar_and_spelling_checker_command=

bandit_venv=
black_venv=
isort_venv=
mypy_venv=
pylint_venv=
ruff_venv=

upgrade_venvs=
```

```

upgrade_venvs_time_interval_in_seconds=

vnu_jar_path=

upgrade_vnu_jar_time_interval_in_seconds=

use_vnu_for_xhtml=

npm_lint_directories=
npm_lint_directories=
```

3.47 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
  "EM101",

  # commented-out-code Of course git log is always faster...
  # Note that if you're tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
  # debugged at some point I commented a part that was needed in
  # another case."...
  "ERA001",

  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
```

```
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
]
```

3.48 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
source-path="./git_helpers"
```

3.49 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck,
and also some of the warnings that were suppressed in some previous
state of the git repository (in that case, we use X suffix, see SC2162X):
- SC1090 : Can't follow non-constant source.
           Use a directive to specify location
           # shellcheck source=bla/bli.sh
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1091 : Not following (file not found most of the time).
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1111 : Use of slanted double quotes (valid inside single quotes)
           False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
           False positive when c contains \n.
- SC2034 : Unused variable
           False positive for globals that are declared in functions
           that are not currently called.
           Somehow it can be used to detect dead code.
- SC2059 : Don't use variables in the printf format string.
           Use printf '..%s..' "$foo".
           False positive when using a variable to build the format string.
```

```
- SC2086 : Double quote to prevent globbing and word splitting.
           False positive when returning integer variable
           via an indirection like return ${!foo}.
           See SC2248 below and the corresponding issue on GitHub.
- SC2119 : Use foo "$@" if function's $1 should mean script's $1.
           False positive when passing arguments would yield
           a wrong result.
- SC2120 : foo references arguments, but none are ever passed.
           False positives when some arguments are meant for debug in CLI.
- SC2125 : Brace expansions and globs are literal.
           False positive when assigning a variable declared with -i.
           https://github.com/koalaman/shellcheck/issues/914
- SC2145 : Argument mixes string and array.
           False positive when using "$@" line return and pipes.
           https://github.com/koalaman/shellcheck/issues/3192
           Example:
             # shellcheck disable=SC2145
             all_self_code_lines "$@"\
               | empty_lines_after_file_name
- SC2154 : var is referenced but not assigned.
           Very useful but false positive when I play with
           variable variable names.
- SC2162X: read without -r will mangle backslashes.
           False positive when read is used without variable:
           https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
           False positive on returning integer values.
           https://github.com/koalaman/shellcheck/issues/3405
- SC2309 : -eq treats this as a variable.
             Use = to compare as string (or expand explicitly with $var).
           Very useful but false positive when I play with
           variable variable names.
- SC2312 : Some command is not invoked separately and thus its return
           value is masked. Frequent with pipes.
- SCNNNNX: Use printf instead of echo.
           https://github.com/koalaman/shellcheck/issues/3419

The command to check if there is anything else:
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2034$" -e "=SC2059$" -e "=SC2086$" -e "=SC2119$" -e "=SC2120$"\
  -e "=SC2125$" -e "=SC2145$" -e "=SC2154$" -e "=SC2248$" -e "=SC2309$"\
  -e "=SC2312$" -e "=SC2154,SC2309$"
```

3.50 build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
```

```
45      LFBFL_verbose="--verbose"
46    fi
47    readonly LFBFL_verbose
48
49    local LFBFL_check_download=""
50    if [[ "$*" == *--check-download=keep* ]]; then
51      LFBFL_check_download="--check-download=keep"
52    elif [[ "$*" == *--check-download* ]]; then
53      LFBFL_check_download="--check-download"
54    fi
55    readonly LFBFL_check_download
56
57    source ./wget_sha512.libr.sh
58
59    LFBFL_subdir="build_and_checks_dependencies"
60    mkdir -p "${LFBFL_subdir}/licenses_templates"
61    LFBFL_variables_directory="${LFBFL_work_directory}/"
62    LFBFL_variables_directory+="build_and_checks_variables"
63    mkdir -p "${LFBFL_variables_directory}/temp"
64    if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
65      printf "temp/\nupgrade_venvs_ts\n"\
66        > "${LFBFL_variables_directory}/.gitignore"
67    fi
68
69    # LFBFL_dependencies_raw_content_download_URL
70    local LFBFL_dependencies_URL
71    LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
72    LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
73    readonly LFBFL_dependencies_URL
74
75    declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
76    # LFBFL_script_download_URL
77    declare -r\
78      LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
79    declare -r\
80      LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
81    local LFBFL_correct_sha512
82    LFBFL_correct_sha512='fb0dbd73d38e81588f1ab19d7f6624bcc0a18b2fae6e7'
83    LFBFL_correct_sha512+='536191f506b76fc84e25f0e794cdcb40cacd2d0ee8ca'
84    LFBFL_correct_sha512+='2ac16e671f039f8013afc3700e03dd9714a293a'
85    wget_sha512 "${LFBFL_file_path}"\
86      "${LFBFL_script}"\
87      "${LFBFL_correct_sha512}"\
88      "${LFBFL_verbose}"\
89      "${LFBFL_check_download}"
90    chmod +x "./${LFBFL_file_path}"
```

```

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

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```
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```
    <one line to give the program's name and a brief idea of what it does.>
    Copyright (C) <year>  <name of author>

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```

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```
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    This program comes with ABSOLUTELY NO WARRANTY; for details type ‘show w’.
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```

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3.52 COPYING.LESSER

```
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                        Version 3, 29 June 2007

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```

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```
a header file that is part of the Library.  You may convey such object
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       the Linked Version to produce a modified Combined Work, in the
       manner specified by section 6 of the GNU GPL for conveying
       Corresponding Source.

       1) Use a suitable shared library mechanism for linking with the
       Library.  A suitable mechanism is one that (a) uses at run time
       a copy of the Library already present on the user's computer
```

system, and (b) will operate properly with a modified version of the Library that is interface-compatible with the Linked Version.

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3.53 git_helpers/git_checks.libr.sh

```
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# This file is part of DevOrSysAdminScripts library.
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#
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# 2024-05-31T01:57+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
```

```
36 # 2024-05-31T01:49+02:00: This file was moved from
37 # "."
38 # to
39 # "./git_helpers".
40 # This file was renamed from "git_checks.sh" to "git_checks.libr.sh".
41
42 check_files(){
43   # Checks used on Django projects.
44   # Use `git commit --no-verify` after check.
45   declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
46   declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations"
47   local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
48   LFBFL_send_body_2+=" dès la création du modèle"
49   readonly LFBFL_send_body_2
50   declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
51   declare -r LFBFL_send_body_3=\
52 "Les nouveaux preloadings sont interdits -> #Prefetch()"
53   declare -i LFBFL_i_error=0
54   git diff --cached --name-only\
55     > /tmp/DOSAS_django_git_check1.temp
56   if grep models /tmp/DOSAS_django_git_check1.temp;
57   then
58     notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
59     notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
60     LFBFL_i_error=1
61   fi
62   if grep serializer /tmp/DOSAS_django_git_check1.temp;
63   then
64     notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
65     LFBFL_i_error=1
66   fi
67   rm /tmp/DOSAS_django_git_check1.temp
68   # shellcheck disable=SC2248
69   return ${LFBFL_i_error}
70 }
71
72
73 check_no_abusive_trailing_comma(){
74   # Useful for Python code, although it regularly yields
75   # false positives in Python code
76   # (but it is easy to distinguish by hand).
77   # Use `git commit --no-verify` after check.
78   declare -r LFBFL_send_summary_1="ATTENTION"
79   local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
80   LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
81   readonly LFBFL_send_body_1
```

```
  declare -i LFBFL_i_error=0
  git diff --cached -r\
    > /tmp/DOSAS_django_git_check2.temp
  if grep ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    LFBFL_i_error=1
  fi
  rm /tmp/DOSAS_django_git_check2.temp
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}


check_black_code_formatting(){
  # attention ça ne marche que sur les fichiers "stagés" avec git add
  declare -r LFBFL_s_files_paths=$(
    git diff --cached --name-only\
    | grep '\.py'
  )
  printf "%s\n" "${LFBFL_s_files_paths}"
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  declare -i LFBFL_i_error=0
  # Not very useful since LFBFL_file should be local to the loop by
  # default.
  local LFBFL_file_path
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    printf "Black will check formatting on file %s\n" "${LFBFL_file_path}"
    if ! black --check --diff "${LFBFL_file_path}";
    then
      LFBFL_i_error=1
    fi
  done
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

3.54 git_helpers/post-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "post-commit no such directory\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=post_commit_renaming_comment.libr.sh
source "./git_helpers/post_commit_renaming_comment.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./build_and_checks_dependencies/lines_filters.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "post-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}
```

```

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

3.55 git_helpers/post_commit_renaming_comment.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post_commit_renaming_comment.libr.sh"
# to
# "post_commit_renaming_comment.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
```

```
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
declare -i LFBFL_i_pushd_result2
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no such directory\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result2}
}
# shellcheck source=common_options.libr.sh
source "build_and_checks_dependencies/common_options.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "build_and_checks_dependencies/strings_functions.libr.sh"
declare -i LFBFL_i_popd_result2
popd || {
  LFBFL_i_popd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result2}
}

# Never forget that the worse security is the illusion of security
# (Kevin Mitnick). So, do not expect to have an accurate trackability
# of code history just because you use this post-commit hook or
# something else more evolved. This is useful but not a guarantee.
# For this reason, this function will only handle when all files
# impacted in the preceding commit are only renamed.
# It assumes that `diff` is the current default difference utility
# parametered with `git`. This point could be enhanced.
commit_a_file_renamed_comment(){
  # Options:
  #   --verbose
  #   --log-directory-change
  #   --max-comment-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_arg
  local LFBFL_s_format

```

```
  declare -i LFBFL_i_log_directory_change=0
  if [[ "$*" == *--log-directory-change* ]]; then
    LFBFL_i_log_directory_change=1
  fi
  readonly LFBFL_i_log_directory_change

  declare -i LFBFL_i_max_comment_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-comment-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-comment-line-length=}
      LFBFL_i_max_comment_line_length=$((LFBFL_arg))
      break
    fi
  done
  readonly LFBFL_i_max_comment_line_length

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    LFBFL_s_format="LFBFL_i_log_directory_change: %s\n"
    LFBFL_s_format+="LFBFL_i_max_comment_line_length: %s\n"
    # shellcheck disable=SC2059,SC2248
    printf "${LFBFL_s_format}"\
      "${LFBFL_i_log_directory_change}"\
      ${LFBFL_i_max_comment_line_length}
  fi
  #-----------------------------------------------------------------------

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -r LFBFL_renaming_lines=$(
    git log -p -1\
    | grep '^diff --git' -A 1
  )
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_renaming_lines: %s\n" "${LFBFL_renaming_lines}"
  fi

  declare -a LFBFL_renaming_lines_array
  mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
  readonly LFBFL_renaming_lines_array

  declare -r LFBFL_timestamp=$(date --iso-8601="minute")
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_timestamp: %s\n" "${LFBFL_timestamp}"
  fi
  local LFBFL_split_score_command1
```

```
122     declare -i LFBFL_i_split_score_command_properties1
123     # shellcheck disable=SC2248
124     get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} _
125     LFBFL_split_score_command1="${get_split_score_result}"
126     LFBFL_i_split_score_command_properties1=${i_get_split_score_result2}
127     readonly LFBFL_split_score_command1
128     readonly LFBFL_i_split_score_command_properties1
129     local LFBFL_split_score_command2
130     declare -i LFBFL_i_split_score_command_properties2
131     # shellcheck disable=SC2248
132     get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} /
133     LFBFL_split_score_command2="${get_split_score_result}"
134     LFBFL_i_split_score_command_properties2=${i_get_split_score_result2}
135     readonly LFBFL_split_score_command2
136     readonly LFBFL_i_split_score_command_properties2
137     declare -i LFBFL_i_renaming_happened=0
138     declare -i LFBFL_i
139     declare -i LFBFL_j
140     # declare -r LFBFL_special_file_name="files_names_listing.txt"
141     declare -r LFBFL_sfn="files_names_listing.txt"
142     local LFBFL_diff_line
143     local LFBFL_similarity_line
144     local LFBFL_old_file_path
145     local LFBFL_old_file_name
146     local LFBFL_old_file_directory
147     local LFBFL_new_file_path
148     local LFBFL_new_file_name
149     local LFBFL_new_file_directory
150     local LFBFL_useful_file_name
151     local LFBFL_extension
152     local LFBFL_comment_prefix
153     local LFBFL_comment_prefix2
154     local LFBFL_old_file_name2
155     local LFBFL_old_file_directory2
156     local LFBFL_new_file_name2
157     local LFBFL_new_file_directory2
158     local LFBFL_new_comment
159     declare -i LFBFL_i_copyright_line_number
160     declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
161     declare -i LFBFL_i_line_count
162     declare -i LFBFL_i_lines_after
163     local LFBFL_temp_file_path
164     for ((LFBFL_i=0; LFBFL_i<${#LFBFL_renaming_lines_array[@]};)); do
165       LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
166       LFBFL_i=$((LFBFL_i + 1))
167       LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
```

```
    LFBFL_i=$((LFBFL_i + 1))
    # We need to increment 3 times since grep adds "--" lines between.
    LFBFL_i=$((LFBFL_i + 1))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_diff_line: %s\nLFBFL_similarity_line: %s\n"\
        "${LFBFL_diff_line}"\
        "${LFBFL_similarity_line}"
    fi
    if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
      continue
    fi

    # Extract old file name from diff line. --------------------------
    LFBFL_old_file_path=$(
      printf "%s" "${LFBFL_diff_line}"\
      | cut -d ' ' -f 3
    )
    # It starts by "a/".
    LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
    LFBFL_old_file_name=$(basename "${LFBFL_old_file_path}")
    LFBFL_old_file_directory=$(dirname "${LFBFL_old_file_path}")
    # Extract new file name from diff line. --------------------------
    LFBFL_new_file_path=$(
      printf "%s" "${LFBFL_diff_line}"\
      | cut -d ' ' -f 4
    )
    # It starts by "b/".
    LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
    LFBFL_new_file_name=$(basename "${LFBFL_new_file_path}")
    LFBFL_new_file_directory=$(dirname "${LFBFL_new_file_path}")
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      LFBFL_s_format="LFBFL_old_file_path: %s\n"
      LFBFL_s_format+="LFBFL_old_file_name: %s\n"
      LFBFL_s_format+="LFBFL_old_file_directory: %s\n"
      LFBFL_s_format+="LFBFL_new_file_path: %s\n"
      LFBFL_s_format+="LFBFL_new_file_name: %s\n"
      LFBFL_s_format+="LFBFL_new_file_directory: %s\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_old_file_path}"\
        "${LFBFL_old_file_name}"\
        "${LFBFL_old_file_directory}"\
        "${LFBFL_new_file_path}"\
        "${LFBFL_new_file_name}"\
        "${LFBFL_new_file_directory}"
    fi
```

```
    # Extract "LFBFL_extension" with smart guessing for .tpl ---------
    # and set prefix for line with ©Copyright. ------------------------
    LFBFL_useful_file_name="${LFBFL_new_file_name}"
    LFBFL_extension="${LFBFL_useful_file_name##*.}"
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
        "${LFBFL_useful_file_name}"\
        "${LFBFL_extension}"
    fi
    for ((LFBFL_j=0; LFBFL_j<3; ++LFBFL_j)); do
      if [[ "${LFBFL_extension}" != "tpl" ]]; then
        break
      fi
      LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
      LFBFL_extension="${LFBFL_useful_file_name##*.}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
          "${LFBFL_useful_file_name}"\
          "${LFBFL_extension}"
      fi
    done
    LFBFL_comment_prefix=""
    if [[ "${LFBFL_extension}" == "sh" ]]; then
      LFBFL_comment_prefix="# "
    fi
    if [[ "${LFBFL_extension}" == "sql" ]]; then
      LFBFL_comment_prefix="-- "
    fi
    if [[ "${LFBFL_extension}" == "tex" ]]; then
      LFBFL_comment_prefix="% "
    fi
    if [[ "${LFBFL_useful_file_name}" == "${LFBFL_sfn}" ]]; then
      LFBFL_comment_prefix="// "
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_comment_prefix: %s\n" "${LFBFL_comment_prefix}"
    fi
    #-------------------------------------------------------------------
    # Split if necessary old_file_name, new_file_name, etc. to have
    # comment lines below max_comment_line_length.
    LFBFL_comment_prefix2="${LFBFL_comment_prefix}"'"'
    # -----
    LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_name2+="${LFBFL_old_file_name}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_old_file_name2}"\
```

```
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_old_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_old_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_old_file_directory2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_name2+="${LFBFL_new_file_name}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_name2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command1}"\
      ${LFBFL_i_split_score_command_properties1}
    LFBFL_new_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_new_file_directory2}"\
      "${LFBFL_comment_prefix2}"\
      ${LFBFL_i_max_comment_line_length}\
      '"'\
      "${LFBFL_split_score_command2}"\
      ${LFBFL_i_split_score_command_properties2}
    LFBFL_new_file_directory2="${repeated_split_last_line_result}"
    # Create new comments strings. -----------------------------------
    LFBFL_new_comment=""
    if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]];
    then
      LFBFL_new_comment="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was renamed from"
```

```
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_name2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[ LFBFL_i_log_directory_change -eq 1
      && "${LFBFL_new_file_directory}" != "${LFBFL_old_file_directory}"
      ]];
    then
      if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]];
      then
        LFBFL_new_comment+=$'\n'
      fi
      LFBFL_new_comment+="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was moved from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_directory2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_directory2}"'".'
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_new_comment: %s\n" "${LFBFL_new_comment}"
    fi
    # Find line with ©Copyright. -------------------------------------
    LFBFL_i_copyright_line_number=$(
      grep -n '©Copyright' "${LFBFL_new_file_path}"\
      | head --lines=1\
      | cut -d ':' -f 1
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_copyright_line_number: %s\n"\
        "${LFBFL_i_copyright_line_number}"
    fi
    if [[ LFBFL_i_copyright_line_number -eq 0 ]]; then
      notify-send "${LFBFL_send_summary_2}"\
        "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "No copyright line. Skipping file.\n"
      fi
      continue
    fi
```

```
    # Get total number of lines. ------------------------------------
    LFBFL_i_line_count=$(
      wc -l\
      < "${LFBFL_new_file_path}"
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_line_count: %s\n" "${LFBFL_i_line_count}"
    fi
    # Compute number of lines after. --------------------------------
    LFBFL_i_lines_after=$((
      LFBFL_i_line_count - LFBFL_i_copyright_line_number
    ))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_lines_after: %s\n" "${LFBFL_i_lines_after}"
    fi
    # Update file. --------------------------------------------------
    LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
    {
      head --lines="${LFBFL_i_copyright_line_number}"\
        "${LFBFL_new_file_path}"
      printf "%s\n" "${LFBFL_new_comment}"
      tail --lines="${LFBFL_i_lines_after}" "${LFBFL_new_file_path}"
    } >> "${LFBFL_temp_file_path}"
    mv "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
    # Add file to stage. --------------------------------------------
    git add "${LFBFL_new_file_path}"
    LFBFL_i_renaming_happened=1
  done
  readonly LFBFL_i_renaming_happened
  if [[ LFBFL_i_renaming_happened -eq 1 ]]; then
    # Commit --------------------------------------------------------
    local LFBFL_message="commit_a_file_renamed_comment()"
    LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
    readonly LFBFL_message
    printf "Commit message:\n%s\n" "${LFBFL_message}"
    git commit --no-verify --message="${LFBFL_message}"
  fi
}
```

3.56 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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# or (at your option) any later version.
#
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# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
# "pre-commit".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "pre-commit no such directory\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=git_checks.libr.sh
source "./git_helpers/git_checks.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "pre-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

declare -gi LFBFL_i_git_hook_error=0
```

```
if ! check_files; then
  LFBFL_i_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
  LFBFL_i_git_hook_error=1
fi
if ! check_black_code_formatting; then
  LFBFL_i_git_hook_error=1
fi
readonly LFBFL_i_git_hook_error
# shellcheck disable=SC2248
exit ${LFBFL_i_git_hook_error}
```

3.57 IMAP_transfert.py

```
"""
This file is part of DevOrSysAdminScripts library.

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Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""

import argparse
import email
import imaplib
```

```
32
33
34 class IMAPTransfertBadArgumentError(Exception):
35     """
36     A class to avoid a linter warning,
37     that needs an Error suffix to avoid another linter warning,
38     that needs a docstring to avoid another linter warning.
39     If you don’t see its intent, try badminton.
40     """
41
42
43 class SurpriseError(Exception):
44     """
45     A class to avoid a linter warning,
46     that needs an Error suffix to avoid another linter warning,
47     that needs a docstring to avoid another linter warning.
48     If you don’t see its intent after seeing its use, try volleyball.
49     But "Cheer up!", now you don’t need "pass" XD.
50     """
51
52
53 def IMAP_transfert(
54     from_login: str,
55     from_password: str,
56     from_host: str,
57     from_port: int,
58     to_login: str,
59     to_password: str,
60     to_host: str,
61     to_port: int,
62     *,
63     to_new_folder: str = "",
64     verbose: bool = False,
65 ) -> None:
66     """
67     Transfert all emails from some IMAP account
68     to another IMAP account.
69     """
70     empty = ""
71     if from_login == "":
72         raise IMAPTransfertBadArgumentError("From login is empty.")
73     # if from_password == "":
74     if from_password == empty:
75         # https://github.com/PyCQA/bandit/issues/714
76         # Dumb
77         raise IMAPTransfertBadArgumentError("From password is empty.")
```

```
    if from_host == "":
        raise IMAPTransfertBadArgumentError("From host is empty.")
    if from_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "From port is not positive.",
        )
    if to_login == "":
        raise IMAPTransfertBadArgumentError("To login is empty.")
    if to_password == empty:
        # And Dumber The infinite collection XD
        raise IMAPTransfertBadArgumentError("To password is empty.")
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
    https://www.php.net/manual/fr/function.filter-var.php
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
    """

    # pylint: disable=unreachable
    raise SurpriseError(
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print("Connecting to from account.")
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)
```

```

    if to_new_folder != "":
        if verbose:
            print("Creating new folder in to account.")
        to_account.create(to_new_folder)

    if verbose:
        print("Fetching all emails ids in from account.")
    from_account.select()
    result, ids = from_account.uid("search", "all")
    if verbose:
        print(f'from_account.uid("search") result:{result} ids:{ids}')
    ids = ids[0].decode().split()
    if verbose:
        print("Fetching and transfering each email in from account.")
    for my_id in ids:
        result, raw_message = from_account.uid(
            "fetch",
            my_id,
            "(RFC822)",
        )
        if verbose:
            print(f'from_account.uid("fetch", {my_id}) result:{result}')
        message = raw_message[0][1]
        my_email = email.message_from_string(message.decode("utf-8"))
        flags = imaplib.ParseFlags(raw_message[0][0])
        # flags = " ".join(flags) mypy is unhappy but IMAP flags are
        # ASCII strings, we give mypy what it wants.
        flags_as_strs = [flag.decode("ascii") for flag in flags]
        flags_str = " ".join(flags_as_strs)
        # date = imaplib.Internaldate2tuple(raw_message[0][0])
        date = '"' + my_email["date"] + '"'
        if verbose:
            print(
                f'from_account.uid("fetch", {my_id})'
                f" flags:{flags} date:{date}",
            )
        to_account.append(to_new_folder, flags_str, date, message)


def main() -> None:
    """
    Parse command-line arguments and
    transfert all emails from some IMAP account
    to another IMAP account.
    """
```

```
    parser = argparse.ArgumentParser(
        description=(
            "Transfert all emails from some IMAP account"
            " to another IMAP account"
        ),
    )
    parser.add_argument(
        "from_login",
        type=str,
        help="Email account login of user one.",
    )
    parser.add_argument(
        "from_password",
        type=str,
        help="Email account password of user one.",
    )
    parser.add_argument(
        "to_login",
        type=str,
        help="Email account login of user two.",
    )
    parser.add_argument(
        "to_password",
        type=str,
        help="Email account password of user two.",
    )
    parser.add_argument(
        "--to_new_folder",
        type=str,
        default="",
        help="Transfert emails into a new folder of this name.",
    )
    parser.add_argument(
        "--from_host",
        type=str,
        default="imap.gmail.com",
        help="Host of first email account",
    )
    parser.add_argument(
        "--from_port",
        type=int,
        default=993,
        help="Port on host of first email account",
    )
    parser.add_argument(
        "--to_host",
```

```
        type=str,
        default="imap.gmail.com",
        help="Host of second email account",
    )
    parser.add_argument(
        "--to_port",
        type=int,
        default=993,
        help="Port on host of second email account",
    )
    parser.add_argument(
        "-v",
        "--verbose",
        action="store_true",
        help="Print details of operations",
    )

    args = parser.parse_args()

    if args.verbose:
        print(
            "-----------------------------------------------------\n"
            "IMAP transfert:\n"
            "-----------------------------------------------------\n"
            "From account:\n"
            f" - Login: {args.from_login}\n"
            f" - Password: {args.from_password}\n"
            f" - Host: {args.from_host}\n"
            f" - Port: {args.from_port}\n"
            "To account:\n"
            f" - Login: {args.to_login}\n"
            f" - Password: {args.to_password}\n"
            f" - Host: {args.to_host}\n"
            f" - Port: {args.to_port}\n"
            "-----------------------------------------------------\n",
        )
        if args.to_new_folder is not None:
            print(
                f"Emails will go in folder {args.to_new_folder}"
                " after it is created.",
            )
    try:
        IMAP_transfert(
            args.from_login,
            args.from_password,
            args.from_host,
```

```
            args.from_port,
            args.to_login,
            args.to_password,
            args.to_host,
            args.to_port,
            to_new_folder=args.to_new_folder,
            verbose=args.verbose,
        )
    except (IMAPTransfertBadArgumentError, SurpriseError) as e:
        print(e)


if __name__ == "__main__":
    main()
```

3.58 LiensUtiles.txt

```
https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
-configurez-le-service-de-migration-de-donn%C3%A9es
----------------------------------------------------------------------
https://stackoverflow.com/a/78413334/5796086
https://stackoverflow.com/questions/21612980/\
why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
https://web.archive.org/web/20240501105801/https://stackoverflow.com\
/questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
78413334#78413334
----------------------------------------------------------------------
https://news.ycombinator.com/item?id=22214690
https://github.com/koalaman/shellcheck/issues/2005
https://opensource.stackexchange.com/questions/6271/\
calling-gpld-programs-from-a-shell-script
----------------------------------------------------------------------
```

3.59 miscellaneous/funny.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

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# along with DevOrSysAdminScripts.
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unprotected_funny(){
  # Works in a script but not in shell if job control is active.
  shopt -s lastpipe
  declare -i LFBFL_i_result=0
  local LFBFL_line
  printf "hahaha\n" | while read -r LFBFL_line; do
    if [[ "${LFBFL_line}" == "hahaha" ]]; then
      LFBFL_i_result=1
    fi
  done
  [[ LFBFL_i_result -eq 1 ]] && printf "I had a good laugh.\n"
}

protected_funny(){
  # Always works regarding the echoes.
  printf "hahaha\n" | while read -r LFBFL_line; do
    unprotected_funny
  done
}
```

3.60 miscellaneous/spampty_data.txt

```
a
 a
empty.txt:
one_space.txt:
all_space.txt:
normal.txt:a
normal_again.txt: a
```

3.61 miscellaneous/string_escaping.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>Truc</title>
</head>
<body>
<header id="title-block-header">
<h1 class="title">Truc</h1>
</header>
&lt;abc> &amp; &lt;de>
<p title="<&>’
 abcd‘\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>’&quot;
abcd‘\ machin"
This is some text
</pre>

<script>
let a = ’abcd & < > \\ \n  \"\’’;
alert(a);
let b = "abcd & < > \\ \n  \"\’";
alert(b);
</script>

</body>
</html>
```

3.62 miscellaneous/string_escaping.xhtml

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
          "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="fr" lang="fr">
<head>
<meta content="text/html; charset=UTF-8" http-equiv="Content-Type" />
  <title>Truc</title>
</head>
<body>
<h1 class="title">Truc</h1>
&lt;abc> &amp; &lt;de>
```

```

<p title="&lt;&amp;>'&#10;
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
This is some text
</pre>

<script type="text/javascript">
let a = 'abcd &amp; &lt; > \\ \n  \"\'';
alert(a);
let b = "abcd &amp; &lt; > \\ \n  \"\'";
alert(b);
</script>

</body>
</html>
```

3.63 README.md.tpl

```
# DevOrSysAdminScripts

[![CodeFactor-badge]][CodeFactor-package-page]
[![Codacy-badge]][Codacy-package-page]
![GitHub-top-language-badge]
![GitHub-code-size-in-bytes-badge]
[![Security:bandit:badge]][Bandit-GitHub.com]
[![Code-style:black:badge]][Black-GitHub.com]
[![Imports:isort:badge]][Isort-GitHub.io]
[![Typecheck:mypy:badge]][Typecheck-mypy-lang.org]
[![Linting:pylint:badge]][Pylint-GitHub.com]
[![Linting:ruff:badge]][Ruff-GitHub.com]

|                **Some "small" useful scripts**               |
|:-------------------------------------------------------------:|
| **for enhancing development or system-administrators tasks** |

Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.

I tried to follow some conventions for my bash code:
```

```

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- But the files "pre-commit" and "post-commit" were not renamed,
  since it is not possible to give them other names to use them as
  pre-commit/post-commit hooks in `git`.
- When some ".exec.sh" code is encapsulated into functions,
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.
- We use mapfile instead of read -r most of the time, but with the
  following exceptions: real use of read -r to read the command line of the
  shell, comments (of course), and a file funny.sh where it is kind of
  required to use read -r.
- We replaced all echo calls by printf calls to follow
  "The Open Group Base Specifications" since at least 2 decades:
  "New applications are encouraged to use printf instead of echo.",
  <https://pubs.opengroup.org/onlinepubs/000095399/>.
- We used `_?i_` or `_?j_`, etc. in integer variables names in Bash
  scripts.

[CodeFactor-badge]: https://www.codefactor.io/repository/github/\
llyaudet/DevOrSysAdminScripts/badge

[CodeFactor-package-page]: https://www.codefactor.io/repository/\
github/llyaudet/DevOrSysAdminScripts

[Codacy-badge]: https://app.codacy.com/project/badge/Grade/\
```

```
3dd6ef85e8db4460991d1ebc404b68dd

[Codacy-package-page]: https://app.codacy.com/gh/LLyaudet/\
DevOrSysAdminScripts/dashboard?utm_source=gh&utm_medium=referral\
&utm_content=&utm_campaign=Badge_grade

[GitHub-top-language-badge]: https://img.shields.io/github/\
languages/top/llyaudet/DevOrSysAdminScripts

[GitHub-code-size-in-bytes-badge]: https://img.shields.io/github/\
languages/code-size/llyaudet/DevOrSysAdminScripts

[Security:bandit:badge]: https://img.shields.io/badge/\
security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

[Code-style:black:badge]: https://img.shields.io/badge/\
code%20style-black-000000.svg

[Black-GitHub.com]: https://github.com/psf/black

[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/

[Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
mypy_badge.svg

[Typecheck-mypy-lang.org]: https://mypy-lang.org/

[Linting:pylint:badge]: https://img.shields.io/badge/\
linting-pylint-yellowgreen

[Pylint-GitHub.com]: https://github.com/pylint-dev/pylint

[Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
badge/v2.json

[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

3.64 string_escaping.libr.php

```php
<?php
/*
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*/

namespace string_escaping\HTML {
  function escape_text(string $s_string) : string {
    return str_replace(
      array(
        '&',
        '<',
      ),
      array(
        '&amp;',
        '&lt;',
      ),
      $s_string,
    );
  }



  function escape_attribute(string $s_string) : string {
```

```
    # in double quotes
    return str_replace(
      array('"'),
      array('&quot;'),
      $s_string,
    );
  }



  function escape_pre(string $s_string) : string {
    return str_replace(
      array(
        '&',
        '<',
      ),
      array(
        '&amp;',
        '&lt;',
      ),
      $s_string,
    );
  }
}




namespace string_escaping\XHTML {
  function escape_text(string $s_string) : string {
    return str_replace(
      array(
        '&',
        '<',
      ),
      array(
        '&amp;',
        '&lt;',
      ),
      $s_string,
    );
  }



  function escape_attribute(string $s_string) : string {
```

```
91      # in double quotes
92      return str_replace(
93        array(
94          '&',
95          '<',
96          '"',
97          "\n",
98        ),
99        array(
100         '&amp;',
101         '&lt;',
102         '&quot;',
103         '&#10;'
104       ),
105       $s_string,
106     );
107   }
108
109
110
111   function escape_pre(string $s_string) : string {
112     return str_replace(
113       array(
114         '&',
115         '<',
116       ),
117       array(
118         '&amp;',
119         '&lt;',
120       ),
121       $s_string,
122     );
123   }
124 }
125
126
127
128
129 namespace string_escaping\JS {
130   use function string_escaping\XHTML\escape_text as XHTML_escape_text;
131
132   function escape(string $s_string) : string {
133     return str_replace(
134       array(
135         '\\',
136         "\n",
```

```
        "'",
        '"',
      ),
      array(
        '\\\\',
        '\n',
        '\\\'',
        '\\"',
      ),
      $s_string,
    );
  }



  function escape_for_string_definition_inside_XHTML(string $s_string)
  : string {
    return XHTML_escape_text(escape($s_string));
  }
}

/*
$ php
<?php
require_once('string_escaping.libr.php');
var_dump(string_escaping\HTML\escape_text('<&"\'truc\\'));
var_dump(string_escaping\HTML\escape_attribute("<&\"\n'truc\\"));
var_dump(string_escaping\HTML\escape_pre("<&\"\n'truc\\"));
var_dump(string_escaping\XHTML\escape_text('<&"\'truc\\'));
var_dump(string_escaping\XHTML\escape_attribute("<&\"\n'truc\\"));
var_dump(string_escaping\XHTML\escape_pre("<&\"\n'truc\\"));
var_dump(string_escaping\JS\escape("abcd & < > \\ \n  \"'"));
var_dump(string_escaping\JS\escape_for_string_definition_inside_XHTML(
  "abcd & < > \\ \n  \"'"
));
*/
```

3.65 unicode.libr.php

```
<?php
/*
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of the GNU Lesser General Public License
```

```
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This file was renamed from "unicode.php" to "unicode.libr.php".
*/

function decimal_code_point_to_UTF8(
  int $i_code_point_in_decimal_notation
) : string {
  // var_dump($i_code_point_in_decimal_notation);
  // 0xxxxxxx ASCII
  if($i_code_point_in_decimal_notation < 128){
    return chr($i_code_point_in_decimal_notation);
  }
  $i_continuation_base_value = 128;
  // 110xxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 2048){  // 2**11
    $i_first_byte_base_value = 192;
    $i_last_byte_significant_bits = (
      $i_code_point_in_decimal_notation % 64  // 2**6
    );
    $i_first_byte_significant_bits = intdiv(
      (
        $i_code_point_in_decimal_notation
        - $i_last_byte_significant_bits
      ),
      64,  // 2**6
    );
    $s_result = chr(
      $i_first_byte_base_value
      + $i_first_byte_significant_bits
    );
```

```
54      $s_result .= chr(
55        $i_continuation_base_value
56        + $i_last_byte_significant_bits
57      );
58      return $s_result;
59    }
60    // 1110xxxx 10xxxxxx 10xxxxxx
61    if($i_code_point_in_decimal_notation < 65536){  // 2**16
62      $i_first_byte_base_value = 224;
63      $arr_arr_data_per_byte_reverse = [
64        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
65        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
66        ["i_bits" => 4, "i_base_value" => $i_first_byte_base_value],
67      ];
68      $i_rest = $i_code_point_in_decimal_notation;
69      $s_result = "";
70      foreach($arr_arr_data_per_byte_reverse as $arr_data){
71        // _Pragma unroll
72        // J'aurais pu "roller" les 4 cas des ifs.
73        // Mais pour les perfs, il faudrait tout unroller vu le nombre
74        // d'itérations. Et j'aimerais bien comparer du code unrollé
75        // aux 2 niveaux à la main avec celui unrollé par le compilateur
76        // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
77        // l'intérieur d'abord (pour ne pas refaire le même boulot)
78        // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
79        // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
80        $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
81        $i_rest = intdiv(
82          $i_rest - $i_significant_bits,
83          2**$arr_data["i_bits"],
84        );
85        $s_result = (  // .= or =.
86          chr($i_significant_bits + $arr_data["i_base_value"])
87          .$s_result
88        );
89      }
90      return $s_result;
91    }
92    // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
93    if($i_code_point_in_decimal_notation < 2097152){  // 2**21
94      $i_first_byte_base_value = 240;
95      $arr_arr_data_per_byte_reverse = [
96        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
97        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
98        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
99        ["i_bits" => 3, "i_base_value" => $i_first_byte_base_value],
```

```
100       ];
101       $i_rest = $i_code_point_in_decimal_notation;
102       $s_result = "";
103       foreach($arr_arr_data_per_byte_reverse as $arr_data){
104         // _Pragma unroll
105         $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
106         $i_rest = intdiv(
107           $i_rest - $i_significant_bits,
108           2**$arr_data["i_bits"],
109         );
110         $s_result = (  // .= or =.
111           chr($i_significant_bits + $arr_data["i_base_value"])
112           .$s_result
113         );
114       }
115       return $s_result;
116     }
117     throw new Exception(
118       "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
119       ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
120       ." 3 octets au lieu de 4 :face-palm:."
121       ." Du coup, ça peut donner quelques \"surprises\"."
122       ." Sinon, y'a aussi ASCII dans MySQL."
123       ." Et comme ça joue sur la mémoire et les performances,"
124       ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
125       ." :)"
126     );
127   }
128
129
130
131   function hexa_code_point_to_UTF8(
132     string $s_code_point_in_hexadecimal_notation
133   ) : string {
134     $s_code_point_in_hexadecimal_notation = strtoupper(
135       $s_code_point_in_hexadecimal_notation
136     );
137     $i_code_point_in_decimal_notation = 0;
138     $i_ord_0 = ord('0');
139     $i_ord_A = ord('A');
140     $i_digit_offset = -$i_ord_0;
141     $i_letter_offset = 10 - $i_ord_A;
142     for(
143       $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
144       $i < $i_max;
145       ++$i
```

```
146     ){
147       $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
148       $i_code_point_in_decimal_notation *= 16;
149       if(ctype_digit($s_hexa_digit)){
150         $i_code_point_in_decimal_notation += (
151           ord($s_hexa_digit) + $i_digit_offset
152         );
153         continue;
154       }
155       if(ctype_xdigit($s_hexa_digit)){
156         $i_code_point_in_decimal_notation += (
157           ord($s_hexa_digit) + $i_letter_offset
158         );
159         continue;
160       }
161       throw new Exception("Not an hexadecimal digit.");
162     }
163     return decimal_code_point_to_UTF8(
164       $i_code_point_in_decimal_notation
165     );
166   }
167
168
169
170   function check_string_is_valid_ASCII(string $s_string) : bool {
171     $i_offset_in_octets_from_string_start = 0;
172     $i_current_line_number = 1;
173     $i_offset_in_octets_from_line_start = -1;
174     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
175       $i_current_octet = ord($s_string[$i]);
176       $i_offset_in_octets_from_string_start = $i;
177       $i_offset_in_octets_from_line_start += 1;
178       if($i_current_octet > 127){
179         throw new Exception(
180           "Non-ASCII character found on line "
181           .$i_current_line_number
182           ."; the octet/character "
183           .($i_offset_in_octets_from_line_start + 1)
184           ." has value "
185           .$i_current_octet
186           ." which is not ASCII."
187           ." (Sequential position without line splitting:"
188           ." This is at octet/character "
189           .($i_offset_in_octets_from_string_start + 1)
190           ".)",
191         );
```

```
192       }
193       // if($s_string[$i] === "\n"){
194       if($i_current_octet === 10){
195         $i_current_line_number += 1;
196         $i_offset_in_octets_from_line_start = -1;
197       }
198     }
199     return true;
200   }
201
202
203
204   function check_file_is_valid_ASCII(string $s_file_name) : bool {
205     $s_string = file_get_contents($s_file_name);
206     if($s_string === false){
207       throw new Exception("File ".$s_file_name." not found.");
208     }
209     return check_string_is_valid_ASCII($s_string);
210   }
211
212
213
214   function check_string_is_valid_UTF8(string $s_string) : bool {
215     $i_continuation_octet_needed = 0;
216     $i_offset_in_octets_from_string_start = 0;
217     $i_offset_in_characters_from_string_start = -1;
218     $i_character_start_position_from_string_start = 0;
219     $i_current_line_number = 1;
220     $i_offset_in_octets_from_line_start = -1;
221     $i_offset_in_characters_from_line_start = -1;
222     $i_character_start_position_from_line_start = -1;
223     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
224       $i_current_octet = ord($s_string[$i]);
225       $i_offset_in_octets_from_string_start = $i;
226       $i_offset_in_octets_from_line_start += 1;
227       if($i_continuation_octet_needed > 0){
228         if($i_current_octet < 128 || $i_current_octet >= 192){
229           throw new Exception(
230             "Non-UTF8 character found on line "
231             .$i_current_line_number
232             ."; the octet "
233             .($i_offset_in_octets_from_line_start + 1)
234             ", part of the character "
235             .($i_offset_in_characters_from_line_start + 1)
236             ", has value "
237             .$i_current_octet
```

```
238           ." which is not a continuation octet."
239           ." This character starts at octet "
240           .($i_character_start_position_from_line_start + 1)
241           ." of the current line."
242           ." (Sequential positions without line splitting:"
243           ." This is at character "
244           .($i_offset_in_characters_from_string_start + 1)
245           ." and octet "
246           .($i_offset_in_octets_from_string_start + 1)
247           ."."
248           ." This character starts at octet "
249           .($i_character_start_position_from_string_start + 1)
250           ".)",
251         );
252       }
253       --$i_continuation_octet_needed;
254       continue;
255     }
256
257     $i_offset_in_characters_from_string_start += 1;
258     $i_offset_in_characters_from_line_start += 1;
259     $i_character_start_position_from_string_start = (
260       $i_offset_in_octets_from_string_start
261     );
262     $i_character_start_position_from_line_start = (
263       $i_offset_in_octets_from_line_start
264     );
265     if($i_current_octet < 128){ // 0xxxxxxx ASCII
266       // if($s_string[$i] === "\n"){
267       if($i_current_octet === 10){
268         $i_current_line_number += 1;
269         $i_offset_in_octets_from_line_start = -1;
270         $i_offset_in_characters_from_line_start = -1;
271       }
272       continue;
273     }
274     if($i_current_octet >= 128 && $i_current_octet < 192){
275       throw new Exception(
276         "Non-UTF8 character found on line "
277         .$i_current_line_number
278         ."; the octet "
279         .($i_offset_in_octets_from_line_start + 1)
280         ", part of the character "
281         .($i_offset_in_characters_from_line_start + 1)
282         ", has value "
283         .$i_current_octet
```

```
        ." which is a continuation octet."
        ." This character starts at octet "
        .($i_character_start_position_from_line_start + 1)
        ." of the current line."
        ." (Sequential positions without line splitting:"
        ." This is at character "
        .($i_offset_in_characters_from_string_start + 1)
        ." and octet "
        .($i_offset_in_octets_from_string_start + 1)
        ."."
        ." This character starts at octet "
        .($i_character_start_position_from_string_start + 1)
        ".)",
      );
    }
    if($i_current_octet >= 192 && $i_current_octet < 224){
      // 110xxxxx 10xxxxxx
      $i_continuation_octet_needed = 1;
      continue;
    }
    if($i_current_octet >= 224 && $i_current_octet < 240){
      // 1110xxxx 10xxxxxx 10xxxxxx
      $i_continuation_octet_needed = 2;
      continue;
    }
    if($i_current_octet >= 240 && $i_current_octet < 248){
      // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
      $i_continuation_octet_needed = 3;
      continue;
    }
    throw new Exception(
      "Non-UTF8 character found on line "
      .$i_current_line_number
      ."; the octet "
      .($i_offset_in_octets_from_line_start + 1)
      ., part of the character "
      .($i_offset_in_characters_from_line_start + 1)
      ., has value "
      .$i_current_octet
      ." which is invalid."
      ." This character starts at octet "
      .($i_character_start_position_from_line_start + 1)
      ." of the current line."
      ." (Sequential positions without line splitting:"
      ." This is at character "
      .($i_offset_in_characters_from_string_start + 1)
```

```
      ." and octet "
      .($i_offset_in_octets_from_string_start + 1)
      ."."
      ." This character starts at octet "
      .($i_character_start_position_from_string_start + 1)
      ".)",
    );
  }
  return true;
}



function check_file_is_valid_UTF8(string $s_file_name) : bool {
  $s_string = file_get_contents($s_file_name);
  if($s_string === false){
    throw new Exception("File ".$s_file_name." not found.");
  }
  return check_string_is_valid_UTF8($s_string);
}



/*
?>
<?php
require_once("./unicode.libr.php");
var_dump(hexa_code_point_to_UTF8("002B"));
var_dump(hexa_code_point_to_UTF8("00E6"));
var_dump(hexa_code_point_to_UTF8("1400"));
var_dump(hexa_code_point_to_UTF8("10111"));
// https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
// Some invisible unicode characters that are a problem:
// 2000-200F 2028-202F 205F 2060-206F
$arr = [
  '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
  '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
  '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
  '205F',
  '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
  '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
];
$arr_s = [];
$arr_s2 = [];
$arr_s3 = [];
foreach($arr as $s_hexa){
```

```
  $s = hexa_code_point_to_UTF8($s_hexa);
  $arr_s []= $s;
  if(strlen($s) === 2){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
  }
  if(strlen($s) === 3){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
      .'\x'.bin2hex($s[2]);
  }
  $arr_s2 []= $s2;
  $s3 = "  | sed -e 's/".$s2."/ /g'\\";
  $arr_s3 []= $s3;
  echo($s3."\n");
}
//*/
```

3.66 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
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#
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# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  # Options:
```

```
  #   --check-download (=keep to keep the downloaded files)
  #   --fixed-point-build
  #   --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  local LFBFL_fixed_point_build=""
  if [[ "$*" == *--fixed-point-build* ]]; then
    LFBFL_fixed_point_build="--fixed-point-build"
  fi
  readonly LFBFL_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download* ]]; then
    LFBFL_check_download="--check-download"
  fi
  readonly LFBFL_check_download
  (\
    cd build_and_checks_dependencies/\
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
  )
  ./build_and_checks.exec.sh "."\
    "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"\
    "${LFBFL_check_download}"
}

upgrade_build_and_checks "$@"
```

3.67 wget_sha512.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

wget_sha512(){
  # $1=file_path
  # $2=download_URL
  # $3=correct_sha512
  # Options:
  #   --check-download (=keep to keep the downloaded file)
  #   --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s wget_sha512 %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download* ]]; then
    LFBFL_check_download="--check-download"
  fi
  readonly LFBFL_check_download
  declare -i LFBFL_i_error=0

  if [[ ! -f "$1" ]]; then
    wget ${LFBFL_verbose:+"${LFBFL_verbose}"} -O "$1" "$2"
  fi
  declare -r LFBFL_present_sha512=$(
    sha512sum "$1"\
    | cut --fields=1 --delimiter=' '
  )
  if [[ "${LFBFL_present_sha512}" != "$3" ]]; then
    printf "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
      "$1" "$3" "${LFBFL_present_sha512}"
```

```
    LFBFL_i_error=1
  fi
  if [[ -n "${LFBFL_check_download}" ]]; then
    wget ${LFBFL_verbose:+"${LFBFL_verbose}"} -O "$1.wget_sha512.temp" "$2"
    declare -r LFBFL_present_sha512_2=$(
      sha512sum "$1.wget_sha512.temp"\
      | cut --fields=1 --delimiter=' '
    )
    if [[ "${LFBFL_check_download}" != "--check-download=keep" ]]; then
      rm "$1.wget_sha512.temp"
    fi
    if [[ "${LFBFL_present_sha512_2}" != "$3" ]]; then
      printf\
        "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
        "Online version of $1" "$3" "${LFBFL_present_sha512_2}"
      LFBFL_i_error=1
    fi
  fi
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !